Triple quantum K-theory of flag varieties

Matt Samuel

October 28, 2026

Abstract

We prove a triple quantum K-Pieri formula expanding the product of a quantum double Grothendieck polynomial and a quantized top block. Its coefficients are indexed by endpoints of admissible chains in the quantum Bruhat graph. A cyclic rotation of values turns these chains into the Pieri chains of Lenart and Sottile in the classical k-Bruhat order; this gives uniqueness of the chain to each endpoint and a closed formula for its quantum degree. The formula is first proved for $k = 1$ and $k = 2$, at $y = 0$ for all k, at $q = 0$ for all k (the classical equivariant case), and on the hyperplane $z = 1/\beta$ for all k, where it is an equivariant quantum K-Chevalley formula for the line bundle $\mathcal{O}(-\varpi_k)$ indexed by chains. For general k the equivariant dependence is shown to be rigid once a single identity, the backtrack identity, is known; that identity involves only chains and left edges and is reduced to a statement about permutations of k letters, which is proved when the prefix of the chain moves position $k-1$ and otherwise rewritten, by a braid bijection between $(k-1)$-chains and $(k-2)$-chains and the Yang–Baxter relations of the quantum Bruhat graph, as an identity between chains that is free of the equivariant parameter z. The latter identity is proved by an explicit bijection between two families of chains, which completes the proof of the formula.

1 Polynomial conventions

Let S_∞ denote the permutations of the positive integers with finite support. We use one-line notation, let $\ell(w)$ be the inversion number of w, and write t_{ab} for the transposition interchanging positions $a < b$. Thus wt_{ab} is obtained from w by interchanging $w(a)$ and $w(b)$.

Let β, z, and the entries of $x = (x_1, x_2, \ldots)$, $y = (y_1, y_2, \ldots)$, and $q = (q_1, q_2, \ldots)$ be independent indeterminates. Put

$$X_i = 1 + \beta x_i, \qquad Q_i = \beta^2 q_i, \qquad \overline{y}_a = -\frac{y_a}{1 + \beta y_a}.$$

Our formal group law is $a \oplus b = a + b + \beta ab$, so that $a \oplus \overline{a} = 0$. In particular, $z + \overline{y}_a$ below denotes an *ordinary* sum, not $z \oplus \overline{y}_a$.

We normalize the double Grothendieck polynomials by

$$\mathfrak{G}_{w_0^{(n)}}(x; y) = \prod_{\substack{i,j \geq 1 \\ i+j \leq n}} (x_i \oplus y_j),$$

and the isobaric divided-difference recursion

$$\mathfrak{G}_{ws_i}(x; y) = \pi_i^\beta \mathfrak{G}_w(x; y) \quad \text{if } \ell(ws_i) < \ell(w), \qquad \pi_i^\beta f = \frac{(1 + \beta x_{i+1}) f - (1 + \beta x_i) s_i f}{x_i - x_{i+1}}.$$

Here s_i interchanges x_i, x_{i+1} and fixes all other variables. In this convention $\mathfrak{G}_{21} = x_1 + y_1 + \beta x_1 y_1$, and $\mathfrak{G}_w|_{\beta=0}$ is the double Schubert polynomial with second alphabet $-y$.

For $0 \le r \le j$ define

$$F_r^j(X;q,\beta) = \sum_{\substack{J\subseteq\{1,\dots,j\}\\ |J|=r}} \left(\prod_{a\in J} X_a\right)\left(\prod_{\substack{a\in J\\ a+1\notin J}} (1-\beta^2 q_a)\right). \tag{1}$$

In particular $F_0^j = 1$. The Lenart–Maeno quantization map $\mathcal{Q}$ is linear over the coefficient parameters and is specified on standard elementary monomials by

$$\mathcal{Q}\left(\prod_{j=1}^{N-1} e_{r_j}(X_1,\dots,X_j)\right) = \prod_{j=1}^{N-1} F_{r_j}^j(X;q,\beta), \qquad 0\le r_j\le j. \tag{2}$$

One takes N sufficiently large for the polynomial being quantized. The standard elementary monomials form a basis of the space with $\deg_{X_i} \le N-i$, $1\le i<N$; this specifies the map on that space. The map is not asserted to be an algebra homomorphism. We set

$$\mathfrak{G}_u^q(x;y) = \mathcal{Q}\big(\mathfrak{G}_u(x;y)\big).$$

This is a definition, not an identification to be proved: at $\beta=-1$ one has $X_i = 1-x_i$ and $Q_i = q_i$, the polynomials F_r^j are the quantum elementary polynomials of Lenart–Maeno [4], and $\mathfrak{G}_u^q(x;y)$ is the quantum double Grothendieck polynomial obtained by applying their quantization map to the double Grothendieck polynomial normalized as above, which is the object studied in [9, 8]. The identifications with the conventions of those papers that the proofs actually use are collected in Remark 1 below.

The top block appearing in our formula is

$$B_k^q(x;z) := \mathcal{Q}\left(\prod_{i=1}^{k}(x_i+z)\right) = \beta^{-k}\sum_{r=0}^{k}\big(-(1-\beta z)\big)^{k-r} F_r^k(X;q,\beta). \tag{3}$$

The apparent negative powers of β in this expression cancel. At $\beta=0$ the block is understood by polynomial specialization, not by separately substituting into β^{-k}. We set $B_0^q = 1$. Notice that (3) quantizes the ordinary factors x_i+z, not the formal-group factors $x_i\oplus z$.

Remark 1 (Dictionary with the cited conventions)**.** *Results from the literature enter the proof at three points: the single-variable Chevalley formula (Lemma 16), the Pieri formula of Naito–Sagaki (Proposition 18), and the interpretation of Theorem 24 as a Chevalley formula for $\mathcal{O}(-\varpi_k)$. All three are stated in the sources at $\beta=-1$, with $X_i = 1-x_i$ and $Q_i = q_i$, and with $\mathfrak{G}_w^q(x;y)$ representing the opposite Schubert class $[\mathcal{O}^w]$ in $QK_T(Fl_N)$ under the identification $1-y_a = \mathbf{e}^{-\epsilon_a}$ of the equivariant parameters [8, Section 3.1]; we write $Y_a = (1-y_a)^{-1}$ for $\mathbf{e}^{\epsilon_a}$. The passage from $\beta=-1$ to general β is the scaling (14) below, under which $Y_a \mapsto (1+\beta y_a)^{-1}$, $(1-q_k)(1-x_k)\mapsto(1-\beta^2 q_k)(1+\beta x_k)$ and $\mathfrak{G}_w^q \mapsto c^{\ell(w)}\mathfrak{G}_w^q$ with $c=-\beta$; the powers of β that result are computed from the length–degree identity (4) at each use. Beyond linearity and this homogeneity, no property of the quantization map, in particular no multiplicativity, is used in the proof of Theorem 3.*

2 Chains and endpoint factors

Fix $u\in S_\infty$ and $k\ge 1$. Choose a finite rank $R>k$ with $u\in S_R$. The directed quantum Bruhat graph on S_R has an edge

$$v \xrightarrow{(a,b)} vt_{ab} \qquad (1\le a<b\le R)$$

when either

$$\begin{aligned} &\ell(vt_{ab}) = \ell(v) + 1 && \text{(a Bruhat edge)},\\ &\ell(vt_{ab}) = \ell(v) - 2(b-a) + 1 && \text{(a quantum edge)}. \end{aligned}$$

A Bruhat edge has weight 1; a quantum edge labeled (a, b) has weight $q_a q_{a+1} \cdots q_{b-1}$. Order labels by

$$(a, b) \prec (c, d) \quad \Longleftrightarrow \quad b > d \ \text{ or } \ (b = d \text{ and } a < c).$$

Definition 2. *An admissible k-chain from u in S_R is a directed path*

$$\gamma: \quad u = v_0 \xrightarrow{(a_1,b_1)} v_1 \xrightarrow{(a_2,b_2)} \cdots \xrightarrow{(a_s,b_s)} v_s = w$$

in the quantum Bruhat graph satisfying:

1. *$a_i \le k < b_i$ for every i;*
2. *all labels (a_i, b_i) are distinct;*
3. *$b_1 \ge b_2 \ge \cdots \ge b_s$;*
4. *if $i < s$ and $a_i = a_j$ for some $j < i$, then $(a_i, b_i) \prec (a_{i+1}, b_{i+1})$.*

The empty path, with endpoint u, is included. Let $E_R(u, k)$ be the set of endpoints of these paths. Write $s(\gamma)$ for the number of edges, and define $D(\gamma) = (D_1, \ldots, D_{R-1})$ by

$$D_j(\gamma) = \#\{\, i : \textit{the } i\textit{th edge is quantum and } a_i \le j < b_i \,\}.$$

Thus the weight of γ is $q^{D(\gamma)} = \prod_{j=1}^{R-1} q_j^{D_j(\gamma)}$.

Directly from the edge definitions, every such path satisfies

$$s(\gamma) = \ell(w) - \ell(u) + 2\sum_{j=1}^{R-1} D_j(\gamma). \tag{4}$$

For an endpoint w define three disjoint sets of *values* from the initial window $u(1), \ldots, u(k)$:

$$\begin{aligned} \mathrm{Fix}(u, w; k) &= \{u(i) : 1 \le i \le k,\ w(i) = u(i)\},\\ \mathrm{Left}(u, w; k) &= \{u(i) : 1 \le i \le k,\ w^{-1}(u(i)) < i\},\\ \mathrm{Out}(u, w; k) &= \{u(i) : 1 \le i \le k\} \setminus \big(\mathrm{Fix}(u, w; k) \cup \mathrm{Left}(u, w; k)\big). \end{aligned}$$

Set

$$m(u, w; k) = |\,\mathrm{Left}(u, w; k)| + |\,\mathrm{Out}(u, w; k)| = k - |\,\mathrm{Fix}(u, w; k)|.$$

The name Out includes values that move right while remaining in the window, as well as values that leave the window.

3 The triple quantum K-Pieri formula

We state the triple quantum K-Pieri formula, expanding a quantum double Grothendieck polynomial in the y-equivariant basis times a quantized top block carrying a second parameter z.

Theorem 3 (Triple quantum K-Pieri formula). *Let $k \geq 1$ and $u \in S_\infty$. Set*

$$R_0 = \max\{\max(\{2\} \cup \operatorname{supp}(u)),\, k+1\} + k, \qquad E = E_{R_0}(u,k).$$

For each endpoint $w \in E$, all admissible chains from u to w have the same quantum degree $D(u,w;k)$ and length $s(u,w;k)$. Moreover, $s(u,w;k) \geq m(u,w;k)$, and

$$B_k^q(x;z)\,\mathfrak{G}_u^q(x;y) = \sum_{w\in E} C_{u,w}^{(k)}(y,z,q,\beta)\,\mathfrak{G}_w^q(x;y), \tag{5}$$

where

$$\boxed{\begin{aligned} C_{u,w}^{(k)} = q^{D(u,w;k)}\beta^{s(u,w;k)-m(u,w;k)} \prod_{a\in \mathrm{Fix}(u,w;k)} \left(z - \frac{y_a}{1+\beta y_a}\right) \\ \times (1-\beta z)^{|\,\mathrm{Left}(u,w;k)|} \prod_{a\in \mathrm{Out}(u,w;k)} (1+\beta y_a)^{-1}. \end{aligned}} \tag{6}$$

Here $B_k^q(x;z) = \mathcal{Q}(\prod_{i=1}^k (x_i+z))$ is defined in (3); D records the quantum edges in a chain; s is its number of edges; and $m = |\,\mathrm{Left}\,| + |\,\mathrm{Out}\,| = k - |\,\mathrm{Fix}\,|$. The sets $\mathrm{Fix}, \mathrm{Left}, \mathrm{Out}$ record whether each value among $u(1), \ldots, u(k)$ stays fixed, moves to an earlier position, or otherwise moves (to a later position or out of the first k positions). The admissible chains and their quantum weights are defined above.

The formula is indexed by endpoints, so each endpoint appears once. The chain-independence of D and s is part of the theorem; it is proved in Corollary 11, which shows that there is exactly one admissible chain to each endpoint and gives D and s in closed form, see (7).

Remark 4 (Rank and polynomial setting). *The rank R_0 just makes the endpoint set finite; it does not impose relations from a finite flag variety. For $k = 0$ the formula reduces to $\mathfrak{G}_u^q = \mathfrak{G}_u^q$.*

Remark 5 (A useful degree check). *Let $|D| = \sum_j D_j$. By (4), the prefactor in (6) can also be written as*

$$q^D \beta^{s-m} = Q^D \beta^{\ell(w)-\ell(u)-m}.$$

The first expression is preferable for specialization at $\beta = 0$, since the inequality $s \geq m$ makes nonnegativity of the power explicit. With $\deg x_i = \deg y_i = \deg z = 1$, $\deg \beta = -1$, and $\deg q_i = 2$, the coefficient has degree $k + \ell(u) - \ell(w)$.

4 Proof calculations

The arguments below establish the rank bound, the uniqueness of the admissible chain to each endpoint together with closed formulas for s and D (by a rotation of values to the classical k-Bruhat order), polynomiality of the top block, the full formula for $k = 1$, the full formula at $y = 0$ for all k (Theorem 20, from the Naito–Sagaki marked-chain formula), the full equivariant formula at $z = 1/\beta$ for all k (Theorem 24, by the junction identity), the full equivariant formula at $q = 0$ for all k (Theorem 38, by the junction identity and a bootstrap from $y = 0$), and the full formula

for $k = 2$ (Proposition 46). For general k the same bootstrap reduces the junction identity to the backtrack identity (34), a statement about chains and left edges alone (Theorem 45); that identity is proved for $k = 2$, at $q = 0$ and at $z = 1/\beta$, and in Section 4.13 it is reduced to the block identity (36) on S_k, proved when position $k-1$ is moved by the prefix of the chain (Proposition 54) and otherwise reduced to the identity (37) (Theorem 56); Section 4.14 writes it as an operator identity, using the Yang–Baxter relations of the quantum Bruhat graph, eliminates the left paths from it (Proposition 62), proves it when no position below $k-1$ is fresh (Theorem 68), and then in general (Theorem 77), which completes the proof of Theorem 3 (Corollary 78).

Remark 6 (Computer checks)**.** *Several remarks below report that an intermediate identity has been verified by computer in a finite range of u and k. These checks were used to find the correct statements and are retained as consistency checks only: no step of the proof depends on them, and every statement used in the proof of Theorem 3 is either proved here for all u and k or quoted from the literature with a reference.*

4.1 Finiteness and the power of β

Proposition 7. *Put $n(u) = \max(\{1\} \cup \operatorname{supp}(u))$ and $M = \max\{n(u), k\} + 1$. Every label in an admissible k-chain from u has second entry at most M. Consequently its endpoint set stabilizes in every rank containing u and $\{1, \dots, M\}$. For every such chain γ with endpoint w,*

$$s(\gamma) \geq m(u, w; k).$$

Proof. The first label has maximal second entry, by the ordering condition. Suppose it were (a, b) with $b > M$. Then $a \leq k < b - 1$, $u(b) = b$, $u(b-1) = b-1$, and $u(a) < b - 1$. The intermediate value $u(b-1)$ prevents (a, b) from giving a Bruhat cover. Since $u(a) < u(b)$, the edge cannot be quantum either. This contradiction proves the rank bound. Since the upper indices decrease weakly, every label then has $b_i \leq b_1 \leq M$, so no position beyond M is ever touched by a chain. Whether (a, b) is a Bruhat edge, a quantum edge, or no edge from v depends only on the relative order of the values at the positions $a, \dots, b$ (see the length formulas in the proof of Lemma 8 below), which is unaffected by adjoining fixed points beyond M; and the argument just given excludes a first label with $b > M$ in every rank. Hence the admissible chains from u, their weights and their endpoints are the same in every rank containing u and $\{1, \dots, M\}$, proving stabilization.

Every position $i \leq k$ with $w(i) \neq u(i)$ must occur in at least one transposition of the path. Each transposition involves exactly one position in $\{1, \dots, k\}$. There are $k - |\operatorname{Fix}(u, w; k)| = m(u, w; k)$ changed positions, so at least this many edges are required. □

This proves that the displayed power β^{s-m} is nonnegative for each path, without presupposing independence of the path. Independence of s and D for a fixed endpoint, and in fact uniqueness of the path, is proved next.

4.2 Rotation to the classical k-Bruhat order

The quantum Bruhat graph has a cyclic symmetry, which we derive from the following reformulation of the edge condition. For $v \in S_R$ and $1 \leq a < b \leq R$ put

$$A(v; a, b) = \{v(r) : a < r < b\},$$

the set of values at the positions strictly between a and b. For distinct $x, y \in \{1, \dots, R\}$ let $\operatorname{arc}(x, y)$ be the set of values met strictly between x and y when moving upward cyclically from x to y:

$$\operatorname{arc}(x,y) = \begin{cases} \{x+1, \dots, y-1\} & x < y, \\ \{x+1, \dots, R\} \cup \{1, \dots, y-1\} & x > y. \end{cases}$$

Lemma 8 (Cyclic form of the edge condition). *Let $v \in S_R$ and $a < b$. Then $v \xrightarrow{(a,b)} vt_{ab}$ is an edge of the quantum Bruhat graph if and only if*

$$A(v; a, b) \cap \operatorname{arc}(v(a), v(b)) = \varnothing.$$

The edge is a Bruhat edge if $v(a) < v(b)$ and a quantum edge if $v(a) > v(b)$.

Proof. Passing from v to vt_{ab} changes the inversion status of the pair of positions (a, b) and of the pairs (a, r), (r, b) with $a < r < b$ exactly when $v(r)$ lies strictly between $v(a)$ and $v(b)$; no other pair is affected. Hence, if $v(a) < v(b)$,

$$\ell(vt_{ab}) = \ell(v) + 1 + 2\,\#\{r : a < r < b,\ v(a) < v(r) < v(b)\},$$

and if $v(a) > v(b)$,

$$\ell(vt_{ab}) = \ell(v) - 1 - 2\,\#\{r : a < r < b,\ v(b) < v(r) < v(a)\}.$$

A Bruhat edge therefore means $v(a) < v(b)$ and $A(v; a, b) \cap (v(a), v(b)) = \varnothing$. A quantum edge means $v(a) > v(b)$ and that all $b - a - 1$ intermediate values lie in $(v(b), v(a))$; since $A(v; a, b)$ does not contain $v(a)$ or $v(b)$, this says that $A(v; a, b)$ avoids the complement of $[v(b), v(a)]$, which is $\operatorname{arc}(v(a), v(b))$. □

For $0 \le \vartheta \le R$ let $\rho = \rho_\vartheta$ be the cyclic shift of $\{1, \dots, R\}$ given by

$$\rho(x) = x - \vartheta \quad (x > \vartheta), \qquad \rho(x) = x - \vartheta + R \quad (x \le \vartheta),$$

so that $\rho(\vartheta) = R$ is the largest value when $\vartheta \ge 1$, and the relabelled order of values is $\vartheta+1 < \vartheta+2 < \cdots < R < 1 < \cdots < \vartheta$. For $v \in S_R$ write $\rho v = \rho \circ v$ for the permutation obtained by relabelling the values of v. Since ρ is a cyclic shift, $\rho(\operatorname{arc}(x, y)) = \operatorname{arc}(\rho x, \rho y)$, and clearly $A(\rho v; a, b) = \rho(A(v; a, b))$. Lemma 8 gives at once:

Corollary 9 (Cyclic symmetry). *(a, b) is an edge from v if and only if it is an edge from ρv. The two edges have the same type unless exactly one of $v(a), v(b)$ is at most ϑ: the edge from v is quantum and the edge from ρv is Bruhat exactly when $v(b) \le \vartheta < v(a)$, and the edge from v is Bruhat and the edge from ρv is quantum exactly when $v(a) \le \vartheta < v(b)$.*

The point of the following lemma is that, for the paths we consider, a single rotation removes all quantum edges at once. It uses only the distinctness of the labels and the weak decrease of the upper indices, not condition 4 of Definition 2.

Lemma 10 (Rotation lemma). *Let $u \in S_R$, $1 \le k < R$, $\vartheta = u(k+1)$ and $\rho = \rho_\vartheta$. Let*

$$u = v_0 \xrightarrow{(a_1,b_1)} v_1 \xrightarrow{(a_2,b_2)} \cdots \xrightarrow{(a_s,b_s)} v_s$$

be a directed path in the quantum Bruhat graph whose labels satisfy $a_i \le k < b_i$, are pairwise distinct, and have $b_1 \ge b_2 \ge \cdots \ge b_s$. Then every edge of the rotated path $\rho v_0 \to \rho v_1 \to \cdots \to \rho v_s$ is a Bruhat edge. Consequently the ith edge of the original path is quantum if and only if $v_{i-1}(b_i) \le \vartheta < v_{i-1}(a_i)$.

Proof. Put $v'_i = \rho v_i$ and $N = R = \rho(\vartheta)$, the largest value; N sits at position $k+1$ in v'_0. By Corollary 9 the rotated path is a directed path with the same labels, and the last assertion follows from the first by the same corollary. We show that no rotated edge is quantum.

Position $k+1$ is involved in a label (a,b) only when $b = k+1$. Let $(c_1, k+1), \dots, (c_m, k+1)$ be the labels with upper index $k+1$, in order; by the weak decrease of upper indices they are the last m labels of the path, and by distinctness the c_l are pairwise distinct. Thus N occupies position $k+1$ until the label $(c_1, k+1)$ is applied, and position c_1 from then on.

Consider first an edge with label (a,b), $b > k+1$. Then $a < k+1 < b$, so $N \in A(v'; a, b)$ for the permutation v' at which it is applied. A quantum edge from v' would have $A(v'; a, b) \subseteq (v'(b), v'(a))$, which cannot contain the largest value. Hence the edge is Bruhat. The edge $(c_1, k+1)$ is Bruhat because the value N at position $k+1$ exceeds the value at c_1.

It remains to treat $(c_l, k+1)$ for $l \geq 2$. We argue by induction on l, assuming that $(c_1, k+1), \dots, (c_{l-1}, k+1)$ are Bruhat edges of the rotated path. Put $e_0 = N$ and let e_h be the value at position c_h immediately before the edge $(c_h, k+1)$, for $1 \leq h \leq l$. Immediately before $(c_h, k+1)$ the value at position $k+1$ is e_{h-1}, so the Bruhat property of these edges gives

$$N = e_0 > e_1 > \cdots > e_{l-1},$$

and after $(c_h, k+1)$ the value e_{h-1} sits at position c_h, where it stays, since c_h is not touched again. Position c_l is touched by none of the labels $(c_h, k+1)$, $h < l$, so $x := e_l$ has been the value at position c_l throughout. For $h < l$ with $c_h < c_l$ the position c_l lies strictly between c_h and $k+1$, and the Bruhat condition for the edge $(c_h, k+1)$ gives $x \notin (e_h, e_{h-1})$.

Suppose that the edge $(c_l, k+1)$ were quantum. Then $x > e_{l-1}$ and every value at a position strictly between c_l and $k+1$ lies in (e_{l-1}, x); in particular $e_{h-1} < x$ whenever $h < l$ and $c_h > c_l$. Since $e_{l-1} < x < e_0$ and x differs from every e_h, there is a unique $h \in \{1, \dots, l-1\}$ with $e_h < x < e_{h-1}$. If $c_h < c_l$ this contradicts $x \notin (e_h, e_{h-1})$; if $c_h > c_l$ it contradicts $e_{h-1} < x$. Hence the edge is Bruhat. □

A saturated chain in the k-Bruhat order is a path of Bruhat edges with labels (a,b), $a \leq k < b$. Lenart and Sottile [6] consider such chains subject to two conditions, (P1): $b_1 \geq \cdots \geq b_s$, and (P0): if $i < s$ and $a_i = a_j$ for some $j < i$, then $(a_i, b_i) \prec (a_{i+1}, b_{i+1})$, with the same order $\prec$ on labels as above; these are conditions 3 and 4 of Definition 2. Distinctness of labels is automatic for Bruhat chains satisfying (P1), since a repeated label would force equal upper indices in between and a return to a smaller permutation. Their uniqueness theorem [6, Theorem 2.2] states that for given v, w there is at most one saturated chain in the k-Bruhat order from v to w satisfying (P0) and (P1).

Corollary 11 (Classical reduction of the chains)**.** *Let $u \in S_R$, $1 \leq k < R$ with $R \geq \max\{n(u), k\} + 1$, $\vartheta = u(k+1)$ and $\rho = \rho_\vartheta$. Applying ρ to all vertices is a bijection from the admissible k-chains from u onto the saturated chains in the k-Bruhat order starting at ρu that satisfy (P0) and (P1); labels are preserved. Consequently:*

1. *For every $w \in E(u,k)$ there is exactly one admissible k-chain from u to w. In particular $s(u,w;k)$ and $D(u,w;k)$ are well defined, and the chain is a shortest directed path from u to w in the quantum Bruhat graph.*

2. *Along an admissible chain the value at a position $i \leq k$ changes exactly at the labels with lower index i, and in the rotated order it strictly increases at each of them; the value at a position $b > k$ changes exactly at the labels with upper index b and strictly decreases in the rotated order. In particular $w(i) \neq u(i)$ as soon as i is the lower index of some label, so $m(u,w;k)$ equals the number of distinct lower indices of the chain.*

3. Each position is involved in at most one quantum edge, and

$$D_j(u,w;k)=\begin{cases}\#\{\,i\le j:\ u(i)>u(k+1)\ge w(i)\,\} & j<k,\\ \#\{\,b>j:\ u(b)\le u(k+1)<w(b)\,\} & j\ge k,\end{cases}\qquad s(u,w;k)=\ell(w)-\ell(u)+2\sum_j D_j(u,w;k). \tag{7}$$

Proof. By Lemma 10 the rotation of an admissible k-chain is a path of Bruhat edges with the same labels, hence a saturated chain in the k-Bruhat order, and conditions 3 and 4 are (P1) and (P0). Conversely, by Corollary 9 a chain in the k-Bruhat order from ρu with (P0) and (P1) rotates back to a directed path from u with the same labels, which are distinct by the remark above; so it is an admissible k-chain. The inverse rotation is ρ^{-1}, so this is a bijection.

(1) Two admissible chains from u to w rotate to two chains from ρu to ρw satisfying (P0) and (P1), which coincide by the uniqueness theorem of Lenart and Sottile. Each edge of the quantum Bruhat graph raises the length by at most one, so a directed path from u to w has at least $\ell(\rho w)-\ell(\rho u)$ edges; the rotated chain has exactly this many.

(2) A Bruhat edge (a,b) replaces the value at a by the larger value at b and the value at b by the smaller value at a; no other position changes.

(3) By Lemma 10 the ith edge is quantum exactly when $v_{i-1}(b_i)\le\vartheta<v_{i-1}(a_i)$, that is, when the value entering position a_i is at most ϑ while the value leaving it exceeds ϑ, and the value entering position b_i exceeds ϑ while the value leaving it is at most ϑ. In the rotated order the values $>\vartheta$ precede the values $\le\vartheta$, so by (2) this transition happens at most once at each position, namely when the initial value at a_i exceeds ϑ and the final one does not, and when the initial value at b_i is at most ϑ and the final one exceeds it. A quantum edge (a,b) contributes 1 to D_j exactly when $a\le j<b$. For $j<k$ every quantum edge has $b>k>j$, so D_j counts the quantum edges with $a\le j$, which by the above is the first formula. For $j\ge k$ every quantum edge has $a\le k\le j$, so D_j counts the quantum edges with $b>j$, giving the second formula. The formula for s is (4). □

Remark 12. *The corollary identifies the indexing set of Theorem 3: $w\in E(u,k)$ if and only if ρw is reached from ρu by a Lenart–Sottile chain, that is, $\rho u\xrightarrow{c(k)}\rho w$ in the notation of [6], where ρ rotates the values so that $u(k+1)$ becomes the largest. Thus the quantum endpoint set for u is the classical endpoint set for ρu, and the quantum degree records which window values drop from above $u(k+1)$ to at most $u(k+1)$, and which outside values rise across it. We have checked (7) against the direct enumeration of chains for all $u\in S_n$, $n\le 7$, and all k. Lemma 10 also shows that the chains, but not the line paths of Section 4.5 below, can be analysed in the classical k-Bruhat order; backtracks such as $(k-1,k),(k-1,k)$ are inherently non-classical.*

4.3 An explicit polynomial top block

Write $a=1-\beta z$. For $S\subseteq\{1,\dots,k\}$ with no two consecutive elements, put

$$\epsilon(S)=\begin{cases}1 & k\in S,\\ 0 & k\notin S,\end{cases}\qquad U(S)=\{1,\dots,k\}\setminus\big(S\cup(S+1)\big),\quad S+1=\{j+1:j\in S\}.$$

Proposition 13. *The top block has the polynomial expression*

$$B_k^q(x;z)=\sum_{\substack{S\subseteq\{1,\dots,k\}\\ S\text{ has no consecutive elements}}}(-\beta)^{\epsilon(S)}a^{|S|-\epsilon(S)}\left(\prod_{j\in S}q_jX_j\right)\left(\prod_{i\in U(S)}(x_i+z)\right). \tag{8}$$

In particular, no inverse power of β is needed. For $k \geq 2$ it satisfies

$$\begin{aligned}(1-\beta^2 q_{k-1})B_k^q = \big(x_k + z + \beta(q_{k-1}-q_k) - \beta^2(q_k x_k + z q_{k-1})\big)B_{k-1}^q \\ + (1-\beta z)q_{k-1}(1-\beta^2 q_k)X_k B_{k-2}^q,\end{aligned} \tag{9}$$

with $B_0^q = 1$ and $B_1^q = x_1 + z - \beta q_1 X_1$.

Proof. Multiply (3) by β^k and substitute (1). In each product of factors $1 - Q_j$, choose the term $-Q_j$ for a set S of indices. Such a choice forces $j \in J$ and $j+1 \notin J$ for each $j \in S$, so two elements of S cannot be consecutive. Conversely these are the only restrictions on the remaining choices of J.

Each forced selected position contributes X_j. Each forced unselected position $j+1 \leq k$ contributes $-a$. At every other position, summing the selected and unselected possibilities gives $X_i - a$. Thus the contribution for S to $\beta^k B_k^q$ is

$$(-1)^{|S|} Q^S \left(\prod_{j\in S} X_j\right)(-a)^{|S|-\epsilon(S)} \prod_{i\in U(S)} (X_i - a), \qquad Q^S = \prod_{j\in S} Q_j.$$

Since $|U(S)| = k - 2|S| + \epsilon(S)$, $Q_j = \beta^2 q_j$, and $X_i - a = \beta(x_i + z)$, this expression is β^k times the summand in (8). This proves both that formula and polynomiality.

For the recurrence, introduce the auxiliary polynomials

$$H_0 = 1, \qquad H_1 = x_1 + z, \qquad H_j = (x_j + z)H_{j-1} + a q_{j-1} X_{j-1} H_{j-2} \quad (j \geq 2).$$

They enumerate tilings of $\{1, \ldots, j\}$ by single positions of weight $x_i + z$ and adjacent pairs $\{i, i+1\}$ of weight $a q_i X_i$. Separating the terms of (8) according to whether $k \in S$ gives

$$B_j^q = H_j - \beta q_j X_j H_{j-1} \qquad (j \geq 1). \tag{10}$$

Set $T_j = (1 - \beta^2 q_j)X_j$. The recurrence for H_j, together with $\beta(x_j + z - \beta q_j X_j) + a = T_j$, gives

$$B_k^q = (x_k + z - \beta q_k X_k)B_{k-1}^q + q_{k-1} X_{k-1} T_k H_{k-2}.$$

For $k \geq 3$, the same calculation at $k-1$ gives

$$\beta B_{k-1}^q = T_{k-1} H_{k-2} - a B_{k-2}^q.$$

For $k = 2$ this last identity follows directly from B_1^q. Multiplying the preceding expression for B_k^q by $1 - \beta^2 q_{k-1}$ and using this identity yields

$$(1-\beta^2 q_{k-1})B_k^q = \frac{T_k - a(1-\beta^2 q_{k-1})}{\beta} B_{k-1}^q + a q_{k-1} T_k B_{k-2}^q.$$

Expanding the numerator gives exactly (9); it is divisible by β. □

At $\beta = 0$, this becomes

$$B_k^q = (x_k + z)B_{k-1}^q + q_{k-1} B_{k-2}^q,$$

the elementary quantum cohomological recurrence. The additional terms in (9) are therefore essential K-theoretic corrections, not changes in notation.

4.4 Passing from finite flags to polynomial identities

We will use the equivariant quantum Chevalley formula, which is initially an identity of classes on a finite flag variety. The following observation explains the passage back to polynomials.

Proposition 14 (Stable detection). *At $\beta = -1$, let P be a polynomial in finitely many x_i and q_i, with coefficients rational in finitely many y_a. If its image in the localized equivariant quantum K-theory presentation is zero in every sufficiently large rank, then $P = 0$ as a polynomial.*

Proof. Suppose $P \neq 0$, and choose its least nonzero total quantum degree. In the quantum-adic expansion of its image, the coefficient of each monomial of that degree is the ordinary equivariant K-theory class of the corresponding coefficient polynomial $f(x; y)$. Indeed, all smaller quantum-degree coefficient polynomials of P are zero. The quantum K-theory module is free on the Schubert classes, so each such ordinary class must vanish.

Let $x_1, \ldots, x_r$ include the variables occurring in f, and let d bound its degree in each of these variables. Choose a rank $N > d + r$ containing all parameters in question. Restricting a vanishing ordinary class to every torus fixed point gives

$$f(\overline{y}_{\sigma(1)}, \ldots, \overline{y}_{\sigma(r)}; y) = 0 \qquad (\sigma \in S_N).$$

Work over the rational function field in the equivariant parameters. The N numbers $\overline{y}_a$ are distinct. After fixing distinct values for $r - 1$ variables, the remaining variable has $N - r + 1 > d$ possible values. Polynomial interpolation says that every coefficient in that variable vanishes. Repeating this argument in the other variables proves $f = 0$, a contradiction. □

We use here the presentation and Schubert representatives of [8]. Localizing the equivariant coefficient ring does not affect the conclusion, since clearing its denominators is injective.

4.5 The line Chevalley formula

The proof of Theorem 3 is an induction on k driven by multiplication by the single line class

$$T_k = (1 - \beta^2 q_k) X_k = (1 - \beta^2 q_k)(1 + \beta x_k),$$

exactly as the proof of the quantum Schubert Pieri formula is driven by multiplication by $x_k - z$. The following lemma is the K-theoretic single-variable Monk formula.

Definition 15 (Line paths). *Let $k \geq 1$ and $u \in S_\infty$, and put $M = \max\{n(u), k\} + 1$ as in Proposition 7. A* line path at k from u *is a directed path π in the quantum Bruhat graph with labels*

$$(a_1, k),\ (a_2, k),\ \ldots,\ (a_s, k), \qquad (k, b_1),\ (k, b_2),\ \ldots,\ (k, b_t),$$

where $s, t \geq 0$, $k > a_1 > a_2 > \cdots > a_s \geq 1$, and $M \geq b_1 > b_2 > \cdots > b_t > k$. Write $s(\pi)$ and $t(\pi)$ for the two lengths, $v(\pi)$ for the permutation reached after the first $s(\pi)$ edges, $\mathrm{end}(\pi)$ *for the final permutation, and $D(\pi)$ for the quantum degree, defined edgewise as in Definition 2. The empty path is allowed, with $v(\pi) = \mathrm{end}(\pi) = u$.*

Lemma 16 (Line Chevalley formula). *For every $k \geq 1$ and $u \in S_\infty$,*

$$T_k\, \mathfrak{G}^q_u(x; y) = \sum_\pi (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)} \frac{1}{1 + \beta y_{v(\pi)(k)}}\, \mathfrak{G}^q_{\mathrm{end}(\pi)}(x; y), \tag{11}$$

the sum running over all line paths at k from u.

Proof. Set $\beta=-1$ first and write $Y_a=(1-y_a)^{-1}$, so that the claimed coefficient is $(-1)^{t(\pi)}q^{D(\pi)}Y_{v(\pi)(k)}$. Work in $QK_T(Fl_N)$ for N large, with the presentation and Schubert representatives of [8]: the residue class of T_k is the line bundle class $[\mathcal{O}(-\epsilon_k)]$ with $\epsilon_k=\varpi_k-\varpi_{k-1}$, $\mathfrak{G}^q_w$ represents the opposite Schubert class $[\mathcal{O}^w]$, and the equivariant parameters are identified by $1-y_a=\mathbf{e}^{-\epsilon_a}$.

Kato's isomorphism Φ onto the equivariant K-group of the semi-infinite flag manifold sends $\mathbf{e}^{\mu}[\mathcal{O}^w]Q^{\xi}$ to $\mathbf{e}^{-\mu}[\mathcal{O}_{\mathbf{Q}}(wt_\xi)]$, and by [8, Section 3.1], quoting [7, Proposition 5.2], it intertwines quantum multiplication by $[\mathcal{O}(-\epsilon_k)]$ with $(1-\mathbf{t}_{k-1})\,[\mathcal{O}_{\mathbf{Q}}(w_0\epsilon_k)]\otimes(-)$, where $\mathbf{t}_{k-1}$ corresponds to Q_{k-1} and $\mathbf{t}_0=0$. The tensor product with $[\mathcal{O}_{\mathbf{Q}}(w_0\epsilon_k)]$ is computed by the general Chevalley formula of Lenart–Naito–Sagaki [5, Theorem 33], in the form recalled in [8, Section 5.1], for the weight $\lambda=-\epsilon_k$ and any reduced λ-chain. We use the chain of [8, Section 5.2]: the alcove path from $A_\circ$ to A_{ϵ_k} crosses exactly the hyperplanes

$$H_{\epsilon_a-\epsilon_k,\,0}\ (a=k-1,\dots,1),\qquad H_{\epsilon_k-\epsilon_b,\,1}\ (b=N,\dots,k+1),$$

in this order. The first group is crossed in the negative direction, so the corresponding roots of the chain are the positive roots $\epsilon_a-\epsilon_k$; the second group is crossed in the positive direction, so the corresponding roots are the negative roots $-(\epsilon_k-\epsilon_b)$. An admissible subset is therefore a line path π, and the statistic $n(A)$ counting negative roots is $t(\pi)$. The affine reflections of the first group are linear, while those of the second group fix ϵ_k, because $\epsilon_k\in H_{\epsilon_k-\epsilon_b,1}$; hence the weight statistic is $-v(\pi)\epsilon_k=-\epsilon_{v(\pi)(k)}$. Finally, $\langle-\epsilon_k,\alpha^\vee_{k-1}\rangle=1$ and all other pairings of $-\epsilon_k$ with simple coroots are nonpositive, so the auxiliary sum over partitions in the general formula is a geometric series contributing $(1-\mathbf{t}_{k-1})^{-1}$, which cancels the factor $1-\mathbf{t}_{k-1}$ above. Transporting back through Φ, which negates the exponent of $\mathbf{e}$ and converts t_ξ into Q^ξ, gives

$$T_k\star[\mathcal{O}^u]=\sum_{\pi}(-1)^{t(\pi)}Q^{D(\pi)}\mathbf{e}^{\epsilon_{v(\pi)(k)}}[\mathcal{O}^{\mathrm{end}(\pi)}]=\sum_{\pi}(-1)^{t(\pi)}q^{D(\pi)}Y_{v(\pi)(k)}[\mathcal{O}^{\mathrm{end}(\pi)}]$$

in $QK_T(Fl_N)$. For $k=1$ there are no left edges and this is the first fundamental weight case of the Chevalley formula in [5, Section 5], after the substitution $[\mathcal{O}^{s_1}]=1-\mathbf{e}^{-\varpi_1}T_1$.

The bound $b_1\le M$ is proved as in Proposition 7: $v(\pi)$ agrees with u outside $\{1,\dots,k\}$, so if $b_1>M$ then $v(\pi)(b_1)=b_1$, $v(\pi)(b_1-1)=b_1-1$ and $v(\pi)(k)<b_1-1$, and (k,b_1) is neither a Bruhat nor a quantum edge. Thus the right side of (11) is independent of N, and Proposition 14 upgrades the identity in $QK_T(Fl_N)$ to a polynomial identity at $\beta=-1$. The homogenization to general β is the scaling used in Proposition 17 below: with $c=-\beta$ and $(x,y,q)\mapsto(cx,cy,\beta^2q)$ one has $T_k\mapsto T_k$, $Y_a\mapsto(1+\beta y_a)^{-1}$, $\mathfrak{G}^q_w\mapsto c^{\ell(w)}\mathfrak{G}^q_w$, and

$$c^{\ell(\mathrm{end}(\pi))-\ell(u)}\beta^{2|D(\pi)|}(-1)^{t(\pi)}=(-\beta)^{s(\pi)}\beta^{t(\pi)}$$

by the length–degree identity $\ell(\mathrm{end}(\pi))-\ell(u)=s(\pi)+t(\pi)-2|D(\pi)|$. □

We record the operator form in which the lemma is used. For a formal combination $\Theta=\sum_v\theta_v\,\mathfrak{G}^q_v$ whose coefficients are free of x, put

$$\mathcal{T}_k\Theta:=\sum_v\theta_v\sum_{\pi\text{ from }v}(-\beta)^{s(\pi)}\beta^{t(\pi)}q^{D(\pi)}\,\frac{\mathfrak{G}^q_{\mathrm{end}(\pi)}}{1+\beta y_{v(\pi)(k)}},\tag{12}$$

so that Lemma 16 reads $T_k\Theta=\mathcal{T}_k\Theta$ as polynomials.

4.6 Proof of the triple formula for $k = 1$

Proposition 17. *Theorem 3 holds for $k = 1$ and arbitrary $u \in S_\infty$.*

Proof. Take $k = 1$ in Lemma 16. There are no left edges, so $v(\pi) = u$ for every line path, the equivariant factor is the constant $(1 + \beta y_{u(1)})^{-1}$, and

$$(1 - \beta^2 q_1) X_1 \mathfrak{G}_u^q = \frac{1}{1 + \beta y_{u(1)}} \sum_\gamma \beta^{s(\gamma)} q^{D(\gamma)} \mathfrak{G}^q_{\mathrm{end}(\gamma)}, \tag{13}$$

where γ runs over directed paths with labels forming a subsequence of $(1, M), (1, M-1), \ldots, (1, 2)$. These are exactly the admissible chains of Definition 2 for $k = 1$: all lower indices are 1, and distinctness and weak decrease of the upper indices make them strictly decreasing. There is at most one such path to a given endpoint. In fact, a nonempty product $t_{1b_1} \cdots t_{1b_s}$ with distinct b_i is a single cycle on $\{1, b_1, \ldots, b_s\}$. The cycle of $u^{-1}w$ determines the ordered list of these indices. It also shows that a nonempty path cannot have endpoint u, and that the value $u(1)$ has moved out of position 1. Consequently a nonempty path has $m = 1$, $\mathrm{Fix} = \mathrm{Left} = \varnothing$, and $\mathrm{Out} = \{u(1)\}$.

Finally, $\beta B_1^q = (1 - \beta^2 q_1) X_1 - (1 - \beta z)$. Subtract the second term from (13). The empty-path coefficient, after division by β, is

$$\frac{(1 + \beta y_{u(1)})^{-1} - (1 - \beta z)}{\beta} = z - \frac{y_{u(1)}}{1 + \beta y_{u(1)}}.$$

Every nonempty path has coefficient

$$q^{D(\gamma)} \frac{\beta^{s(\gamma)-1}}{1 + \beta y_{u(1)}}.$$

These are precisely (6) in the two cases. All exponents are nonnegative, so clearing equivariant denominators extends the result to $\beta = 0$ as well. □

For later use we make the scaling in the proof of Lemma 16 explicit. Put $c = -\beta$ and replace (x, y, z, q) at $\beta = -1$ by $(cx, cy, cz, \beta^2 q)$. Then

$$\mathfrak{G}_w^{\beta^2 q}(cx; cy)\big|_{\beta=-1} = c^{\ell(w)} \mathfrak{G}_w^q(x; y), \tag{14}$$

where on the left the quantum arguments are $\beta^2 q$ and the Grothendieck parameter is fixed at -1. Indeed, the normalization of $\mathfrak{G}_{w_0}$ and the operators π_i^β make $\mathfrak{G}_w(x; y)$ homogeneous of degree $\ell(w)$ for $\deg x_i = \deg y_a = 1$, $\deg \beta = -1$, which gives (14) for the classical polynomials; the substitution fixes every $X_i = 1 + \beta x_i$, hence every standard elementary monomial, so by uniqueness of the expansion in that basis it multiplies each coefficient by $c^{\ell(w)}$; and F_r^j depends on (q, β) only through $\beta^2 q$, so applying $\mathcal{Q}$ to both sides gives (14). The scaling takes place with β inverted; whenever both sides of a resulting identity have no negative powers of β, it holds before that inversion as well.

4.7 What the published marked-chain formula proves for all k

There is also an unconditional all-k calculation, but it is non-equivariant and initially indexed by paths rather than endpoints. For a nonempty admissible chain γ, let $r(\gamma)$ be the number of distinct lower indices a_i. Let $F(\gamma)$ be the following set of edge indices:

- every i in the maximal initial segment with $b_1 = \cdots = b_i$ and $a_1 > \cdots > a_i$;
- every $i < s(\gamma)$ for which $(a_i, b_i) \succ (a_{i+1}, b_{i+1})$.

Put $h(\gamma) = |F(\gamma)|$. These are exactly the indices forced to be marked in the Naito–Sagaki marking rule; in particular $h(\gamma) \geq 1$.

Proposition 18. *For every $k \geq 1$ and $u \in S_\infty$,*

$$B_k^q(x;z)\mathfrak{G}_u^q(x;0) = z^k \mathfrak{G}_u^q(x;0) + \sum_{\substack{\gamma \text{ admissible} \\ s(\gamma)>0}} q^{D(\gamma)} \beta^{s(\gamma)-r(\gamma)} z^{k-r(\gamma)} (1-\beta z)^{h(\gamma)-1} \mathfrak{G}^q_{\mathrm{end}(\gamma)}(x;0). \tag{15}$$

This statement does not require uniqueness of a path to its endpoint.

Proof. For $1 \leq p \leq k$, let $c[k,p] = (k-p+1, k-p+2, \ldots, k+1)$, in cycle notation. The single Grothendieck polynomial of this cycle is

$$\mathfrak{G}_{c[k,p]}(x;0) = \sum_{j=p}^{k} \binom{j-1}{p-1} \beta^{j-p} e_j(x_1, \ldots, x_k).$$

The binomial theorem, coefficient by coefficient in e_j, gives

$$\prod_{i=1}^{k}(x_i + z) = z^k + \sum_{p=1}^{k} z^{k-p}(1-\beta z)^{p-1} \mathfrak{G}_{c[k,p]}(x;0).$$

Indeed the coefficient of e_j on the right is $z^{k-j}\big((1-\beta z)+\beta z\big)^{j-1} = z^{k-j}$. Applying the *linear* map $\mathcal{Q}$ gives

$$B_k^q = z^k + \sum_{p=1}^{k} z^{k-p}(1-\beta z)^{p-1} \mathfrak{G}^q_{c[k,p]}(x;0). \tag{16}$$

No multiplicativity of $\mathcal{Q}$ is used.

Naito–Sagaki's Pieri theorem [9, Theorem 2.13], homogenized from $\beta = -1$, says that the contribution of γ to $\mathfrak{G}_u^q \mathfrak{G}^q_{c[k,p]}$ is

$$q^{D(\gamma)} \beta^{s(\gamma)-p} \# \operatorname{Mark}_p(\gamma).$$

A mark may occur only at a first occurrence of a lower index. All indices in $F(\gamma)$ must be marked; the remaining first occurrences can be chosen freely. The admissibility conditions ensure that forced indices are first occurrences. Thus, as also recorded in [9, Remark 2.14],

$$\# \operatorname{Mark}_p(\gamma) = \binom{r(\gamma)-h(\gamma)}{p-h(\gamma)}.$$

Here $1 \leq h(\gamma) \leq r(\gamma) \leq k$.

Fix a nonempty chain, and abbreviate its statistics by s, r, h, D. Its coefficient after multiplying (16) by $\mathfrak{G}_u^q$ is therefore

$$\begin{aligned} & q^D \sum_{p=h}^{r} \binom{r-h}{p-h} \beta^{s-p} z^{k-p} (1-\beta z)^{p-1} \\ &= q^D \beta^{s-r} z^{k-r} (1-\beta z)^{h-1} \sum_{p=h}^{r} \binom{r-h}{p-h} (\beta z)^{r-p} (1-\beta z)^{p-h} \\ &= q^D \beta^{s-r} z^{k-r} (1-\beta z)^{h-1}. \end{aligned}$$

The last sum is 1 by the binomial theorem. The separate term $z^k \mathfrak{G}_u^q$ supplies the empty path. This proves (15). Homogenization is justified by the same scaling and length–degree identity used in Proposition 17; the exponents $s - p$ are nonnegative. □

With Corollary 11 in hand, the path-indexed sum (15) is in fact indexed by endpoints, and its exponents can be read off from the endpoint. Write γ_w for the unique admissible k-chain from u to $w \in E(u,k)$.

Lemma 19 (Forced marks count left moves)**.** *Let $w \in E(u,k)$, $w \neq u$. Then*

$$r(\gamma_w) = m(u,w;k) \qquad \textit{and} \qquad h(\gamma_w) = |\operatorname{Left}(u,w;k)| + 1.$$

Proof. The first equality is Corollary 11(2). For the second, note first that $(a_i, b_i) \succ (a_{i+1}, b_{i+1})$ means $b_i = b_{i+1}$ and $a_i > a_{i+1}$; call such an index i a *descent* of γ_w. If i_0 is the last index of the maximal initial segment with equal upper indices and decreasing lower indices, then $1, \dots, i_0 - 1$ are descents and i_0 is not, so $F(\gamma_w)$ is the disjoint union of $\{i_0\}$ and the set of descents, and $h(\gamma_w) - 1$ is the number of descents. We show that the descents are in bijection with $\operatorname{Left}(u,w;k)$.

By Corollary 11 we may rotate the values and assume that γ_w is a saturated chain in the k-Bruhat order satisfying (P0) and (P1); the sets Fix, Left, Out and the descents only involve positions and are unchanged. Consider a maximal run $(a_1, b), \dots, (a_p, b)$ of labels with the same upper index b, and let α_j be the value at position a_j just before the label (a_j, b) is applied, α_0 the value at b before the run. Each label is a Bruhat cover, so $\alpha_0 > \alpha_1 > \cdots > \alpha_p$, and the net effect of the run is that α_0 moves to a_1, α_j moves from a_j to a_{j+1} for $1 \le j < p$, and α_p moves from a_p to b. Position b is touched by no other label, so α_p stays at b for ever, and every value entering the window from outside does so at the start of a run. In particular a value moves within the window only in steps $a_j \to a_{j+1}$, which is a move to the left exactly at a descent.

Let i be a descent, so that the labels i and $i+1$ are consecutive labels $(a_j, b), (a_{j+1}, b)$ of a run with $a_{j+1} < a_j$, and the value α_j moves from a_j to a_{j+1}. By (P0) the lower index a_j is a first occurrence, so position a_j was untouched and $\alpha_j = u(a_j)$ is a window value making its first move. By the prohibition lemma of Lenart and Sottile [6, Lemma 2.9], a chain satisfying (P0) and (P1) contains no subsequence $(j', m), \dots, (i', m), \dots, (i', l)$ with $i' < j' \le k < l < m$; applied with $(j', m) = (a_j, b)$ and $(i', m) = (a_{j+1}, b)$, this says that position a_{j+1} is never touched again (a later label (a_{j+1}, l) has $l < b$ by (P1) and distinctness). Hence $u(a_j)$ ends at position $a_{j+1} < a_j$ and lies in $\operatorname{Left}(u,w;k)$. Distinct descents move distinct values, so this gives an injection from descents into $\operatorname{Left}(u,w;k)$.

Conversely let $c = u(i) \in \operatorname{Left}(u,w;k)$. It moves at least once; its first move is by a label (i, b). If this label ends its run, c stays at b and is in Out; otherwise c moves to the lower index a' of the next label. If $a' < i$ this is a descent, and c is the value attached to it. If $a' > i$, then every later move of c starts from a position that has already been touched, so by (P0) it is a step to a larger lower index or the end of a run; thus c only moves right or leaves the window, and ends in Out, a contradiction. Hence every value in $\operatorname{Left}(u,w;k)$ is attached to a descent, and the correspondence is a bijection. □

Theorem 20 (The non-equivariant case)**.** *Theorem 3 holds at $y = 0$: for every $k \ge 1$ and $u \in S_\infty$,*

$$B_k^q(x;z)\, \mathfrak{G}_u^q(x;0) = \sum_{w \in E(u,k)} q^{D(u,w;k)} \beta^{s(u,w;k) - m(u,w;k)} z^{|\operatorname{Fix}(u,w;k)|} (1-\beta z)^{|\operatorname{Left}(u,w;k)|}\, \mathfrak{G}_w^q(x;0).$$

Proof. By Corollary 11(1) the sum in (15) has exactly one term for each $w \in E(u,k) \setminus \{u\}$, namely $\gamma = \gamma_w$, and the term $z^k \mathfrak{G}_u^q$ is the term $w = u$ with $\operatorname{Fix} = \{u(1), \dots, u(k)\}$. By Lemma 19, $k - r(\gamma_w) = k - m = |\operatorname{Fix}|$, $s(\gamma_w) - r(\gamma_w) = s - m$ and $h(\gamma_w) - 1 = |\operatorname{Left}|$. This is (6) at $y = 0$. □

Thus the only part of Theorem 3 that remains after this point is the dependence on the equivariant parameters y. At $z = 1/\beta$ this dependence is settled in Theorem 24 below, at $q = 0$ in Theorem 38, whose proof uses Theorem 20 to fix an integer multiplicity, and for $k = 2$ in Proposition 46; in general it is reduced, by the same use of Theorem 20, to the backtrack identity of Section 4.12, which is proved in Sections 4.13 and 4.14.

4.8 Reduction of level k to levels $k-1$ and $k-2$

We now set up the induction on k. Throughout, $a = 1 - \beta z$ and H_j denotes the tiling polynomial introduced in the proof of Proposition 13:

$$H_0 = 1, \qquad H_1 = x_1 + z, \qquad H_j = (x_j + z)H_{j-1} + a\, q_{j-1} X_{j-1} H_{j-2} \quad (j \geq 2),$$

and $B_j^q = H_j - \beta q_j X_j H_{j-1}$ for $j \geq 1$ by (10). We use the convention $q_0 = 0$ and $B_{-1}^q = 0$.

Lemma 21 (Two-step block identities)**.** *For every $k \geq 1$,*

$$\beta B_k^q + a B_{k-1}^q = T_k H_{k-1}, \tag{17}$$

$$(1 - \beta^2 q_{k-1}) H_{k-1} = B_{k-1}^q + a\beta q_{k-1} B_{k-2}^q. \tag{18}$$

Consequently

$$(1 - \beta^2 q_{k-1})\big(\beta B_k^q + a B_{k-1}^q\big) = T_k\big(B_{k-1}^q + a\beta q_{k-1} B_{k-2}^q\big). \tag{19}$$

Proof. Substituting $B_k^q = H_k - \beta q_k X_k H_{k-1}$, $B_{k-1}^q = H_{k-1} - \beta q_{k-1} X_{k-1} H_{k-2}$ and the recurrence for H_k,

$$\beta B_k^q + a B_{k-1}^q = \big(\beta(x_k + z) + a - \beta^2 q_k X_k\big) H_{k-1} = (1 - \beta^2 q_k) X_k H_{k-1},$$

since $\beta(x_k + z) + a = 1 + \beta x_k = X_k$; the two terms involving $a\beta q_{k-1} X_{k-1} H_{k-2}$ cancel. This is (17). For (18) with $k \geq 3$, expand both sides using the recurrence for H_{k-1} and the expressions of B_{k-1}^q, B_{k-2}^q through H: the right side is

$$H_{k-1} + \beta q_{k-1}(a - X_{k-1}) H_{k-2} - a\beta^2 q_{k-1} q_{k-2} X_{k-2} H_{k-3},$$

and $\beta(a - X_{k-1}) = -\beta^2(x_{k-1} + z)$, so this equals $H_{k-1} - \beta^2 q_{k-1}\big((x_{k-1}+z)H_{k-2} + a q_{k-2} X_{k-2} H_{k-3}\big) = (1 - \beta^2 q_{k-1}) H_{k-1}$. For $k = 2$ the same computation applies without the last term, and for $k = 1$ both sides equal 1. Multiplying (17) by $1 - \beta^2 q_{k-1}$ and substituting (18) gives (19). □

At $\beta = 0$, (19) reduces to $B_k = (x_k + z) B_{k-1} + q_{k-1} B_{k-2}$, the recurrence used for the quantum Schubert Pieri formula. At $\beta = -1$ it reads $(1 - q_{k-1})\big(A B_{k-1} - B_k\big) = T_k\big(B_{k-1} - A q_{k-1} B_{k-2}\big)$ with $A = 1 + z$.

The target identity. For $j \geq 0$ and $u \in S_\infty$ write

$$P_j(u) := \sum_{w \in E(u,j)} C_{u,w}^{(j)}\, \mathfrak{G}_w^q, \qquad P_0(u) = \mathfrak{G}_u^q, \qquad P_{-1}(u) = 0,$$

for the right side of (5) at level j, with $C_{u,w}^{(j)}$ given by (6). The endpoint data (s, D) entering $C_{u,w}^{(j)}$ are well defined by Corollary 11.

Proposition 22 (Reduction). *Let $k \geq 1$ and suppose Theorem 3 holds at levels $k-1$ and $k-2$ for the permutation u, that is, $B^q_{k-1}\mathfrak{G}^q_u = P_{k-1}(u)$ and $B^q_{k-2}\mathfrak{G}^q_u = P_{k-2}(u)$. Then Theorem 3 holds at level k for u if and only if*

$$\beta(1-\beta^2 q_{k-1})\,P_k(u) = \mathcal{T}_k\big(P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)\big) - a(1-\beta^2 q_{k-1})\,P_{k-1}(u), \tag{20}$$

with $\mathcal{T}_k$ the line operator (12).

Proof. By (19) and the hypotheses,

$$(1-\beta^2 q_{k-1})\big(\beta B^q_k \mathfrak{G}^q_u + aP_{k-1}(u)\big) = T_k\big(P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)\big),$$

and the right side equals $\mathcal{T}_k\big(P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)\big)$ by Lemma 16. Hence $\beta(1-\beta^2 q_{k-1})B^q_k\mathfrak{G}^q_u$ equals the right side of (20). Since the $\mathfrak{G}^q_w$ are linearly independent over the coefficient field and $\beta(1-\beta^2 q_{k-1})$ is a nonzero element of the integral domain of coefficients, $B^q_k\mathfrak{G}^q_u = P_k(u)$ holds if and only if (20) does. □

For $k = 1$ the identity (20) is $\beta P_1(u) = \mathcal{T}_1\mathfrak{G}^q_u - a\mathfrak{G}^q_u$, which is exactly the computation in the proof of Proposition 17. Thus Theorem 3 follows, by induction on k, from (20) for all $k \geq 2$ and all u. We call (20) the *junction identity*; it is the K-theoretic counterpart of the recurrence $\mathcal{E}_{k,k} = (x_k - z_1)\mathcal{E}_{k-1,k-1} + q_{k-1}\mathcal{E}_{k-2,k-2}$ for quantum factorial elementary polynomials expanded by the single-variable Monk formula in the quantum Schubert case.

Coefficient form. Extracting the coefficient of $\mathfrak{G}^q_w$ in (20), and writing $\pi\colon v \to w$ for a line path at k from v with endpoint w, the junction identity reads

$$\begin{aligned}\beta(1-\beta^2 q_{k-1})\,C^{(k)}_{u,w} = \sum_v \sum_{\pi\colon v\to w} \frac{(-\beta)^{s(\pi)}\beta^{t(\pi)}q^{D(\pi)}}{1+\beta y_{v(\pi)(k)}}\Big(C^{(k-1)}_{u,v} + a\beta q_{k-1}C^{(k-2)}_{u,v}\Big)\\ - a(1-\beta^2 q_{k-1})\,C^{(k-1)}_{u,w},\end{aligned} \tag{21}$$

where $C^{(j)}_{u,v} = 0$ for $v \notin E(u,j)$ and $C^{(-1)} = 0$. A line path $\pi\colon v \to w$ is determined by its label sequence, and $v = w\,t_{kb_t}\cdots t_{kb_1}t_{a_sk}\cdots t_{a_1k}$; the path must be directed from v in the quantum Bruhat graph. The term $\pi = \varnothing$, $v = w$, is the diagonal term, with equivariant factor $(1+\beta y_{w(k)})^{-1}$; the paths with $s(\pi) = 0$ are the up-neighbours $v = wt_{kb_t}\cdots t_{kb_1}$, and the paths with $t(\pi) = 0$ are the down-neighbours $v = wt_{a_sk}\cdots t_{a_1k}$. In the quantum Schubert case only single edges occur and (21) becomes the four-term expansion of that proof; here a path may combine several down edges followed by several up edges, and each term carries a K-theoretic sign and an equivariant denominator determined by the intermediate permutation $v(\pi)$.

Remark 23 (Well-definedness). *The identity* (21) *is to be proved for each endpoint w. By Corollary 11 the data (s, D) entering $C^{(k)}_{u,w}$ are already known to be well defined, and D is given by* (7)*; so the left side is unambiguous, and a proof of* (21) *need not establish path-independence separately. It does, however, have to show that the right side collapses to the single monomial $q^{D(u,w;k)}$ times the expected endpoint factors, and that it vanishes for $w \notin E(u,k)$. In the quantum Schubert proof the analogous path-independence was proved through the cycle structure of $u^{-1}w$; here it comes from the rotation to the classical k-Bruhat order.*

The diagonal case. Suppose $u(k) = w(k)$. Position k is touched by a k-chain only through labels (k, b), and by Corollary 11(2) a touched position never regains its original value. Hence the admissible k-chains from u to w have no label with lower index k; they are exactly the admissible $(k-1)$-chains from u to w all of whose labels have second entry at least $k+1$, and for these chains the conditions of Definition 2 at levels k and $k-1$ coincide. Corollary 11(2) at level $k-1$, where k is an outside position, shows likewise that the $(k-1)$-chains to w with some label (a, k) have $w(k) \neq u(k)$. Therefore

$$\mathrm{Fix}(u, w; k) = \mathrm{Fix}(u, w; k-1) \sqcup \{u(k)\}, \qquad \text{Left, Out, } s, \ D \text{ unchanged},$$

and

$$C^{(k)}_{u,w} = \Big(z - \frac{y_{u(k)}}{1+\beta y_{u(k)}}\Big) C^{(k-1)}_{u,w} = \frac{(1+\beta y_{u(k)})^{-1} - a}{\beta}\, C^{(k-1)}_{u,w}. \tag{22}$$

Inserting (22) into (21), the diagonal term $\pi = \varnothing$ together with the last term produce exactly $\beta C^{(k)}_{u,w}$ plus $\frac{a\beta q_{k-1}}{1+\beta y_{u(k)}} C^{(k-2)}_{u,w}$, and what remains to be shown in the case $u(k) = w(k)$ is the cancellation

$$\sum_{v} \sum_{\substack{\pi:\ v \to w \\ \pi \neq \varnothing}} \frac{(-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)}}{1+\beta y_{v(\pi)(k)}} \Big(C^{(k-1)}_{u,v} + a\beta q_{k-1} C^{(k-2)}_{u,v} \Big) + \frac{\beta q_{k-1}}{1+\beta y_{u(k)}} \Big(\beta C^{(k-1)}_{u,w} + a C^{(k-2)}_{u,w} \Big) = 0, \tag{23}$$

where we used $\beta(z + \overline{y}_{u(k)}) + a = (1+\beta y_{u(k)})^{-1}$ to combine the two multiples of $q_{k-1} C^{(k-1)}_{u,w}$. This is the K-theoretic form of the reflection–drop cancellation of the quantum Schubert proof: there the nonempty paths are single edges t_{ak}, t_{kb} and the drop term is $q_{k-1} c_{k-2}(u, w)$. Note the combination $\beta C^{(k-1)}_{u,w} + a C^{(k-2)}_{u,w}$, the coefficient analogue of $\beta B^q_{k-1} + a B^q_{k-2} = T_{k-1} H_{k-2}$.

4.9 The case $z = 1/\beta$: the Chevalley formula for ϖ_k

At $z = 1/\beta$, that is $a = 0$, the junction identity loses its level-$(k-2)$ part and its last term, and we can prove it completely. This gives the full equivariant statement of Theorem 3 on the hyperplane $a = 0$, for all k and u. Since $x_i + 1/\beta = X_i/\beta$, the block is $B^q_k|_{z=1/\beta} = \beta^{-k} \mathcal{Q}(X_1 \cdots X_k) = \beta^{-k} F^k_k = \beta^{-k}(1-\beta^2 q_k) X_1 \cdots X_k$, and the coefficient (6) specializes by $(1-\beta z)^{|\,\mathrm{Left}\,|} = 0^{|\,\mathrm{Left}\,|}$ and $z + \overline{y}_c = (1+\beta y_c)^{-1}/\beta$ to

$$C^{(k)}_{u,w}\big|_{z=1/\beta} = \begin{cases} q^D \beta^{s-k} \prod_{i=1}^{k} (1+\beta y_{u(i)})^{-1} & \mathrm{Left}(u, w; k) = \varnothing, \\ 0 & \text{otherwise}, \end{cases}$$

using $m + |\,\mathrm{Fix}\,| = k$ and $\mathrm{Fix} \sqcup \mathrm{Out} = \{u(1), \ldots, u(k)\}$ when $\mathrm{Left} = \varnothing$. Thus the statement to be proved is the following.

Theorem 24 (Theorem 3 at $z = 1/\beta$)**.** *For every $k \geq 1$ and $u \in S_\infty$,*

$$(1-\beta^2 q_k)\, X_1 X_2 \cdots X_k\, \mathfrak{G}^q_u(x; y) = \prod_{i=1}^{k} \frac{1}{1+\beta y_{u(i)}} \sum_{\substack{w \in E(u,k) \\ \mathrm{Left}(u,w;k) = \varnothing}} \beta^{s(u,w;k)} q^{D(u,w;k)}\, \mathfrak{G}^q_w(x; y). \tag{24}$$

The polynomial $(1-\beta^2 q_k) X_1 \cdots X_k = F^k_k$ represents, in $QK_T(Fl_N)$ with the identifications used in the proof of Lemma 16, the class of the line bundle $\mathcal{O}(-\varpi_k)$: by [8, Section 3.1], $[\mathcal{O}(-\varpi_k)]$ is

the quantum product of $[\mathcal{O}(-\epsilon_1)]$, $\frac{1}{1-Q_1}[\mathcal{O}(-\epsilon_2)], \ldots, \frac{1}{1-Q_{k-1}}[\mathcal{O}(-\epsilon_k)]$, and $\frac{1}{1-Q_{j-1}}[\mathcal{O}(-\epsilon_j)]$ is represented by $\frac{1-\beta^2 q_j}{1-\beta^2 q_{j-1}} X_j$. So (24) is a chain-indexed form of the equivariant quantum K-Chevalley formula for $\mathcal{O}(-\varpi_k)$, with the equivariant weight $\prod_{i\le k}(1+\beta y_{u(i)})^{-1}$ independent of the endpoint. We prove it from the single-variable formula (Lemma 16) by the junction identity, and we record the combinatorial lemmas separately, since they are also the ingredients of the general case.

Left-free chains. Call an admissible k-chain, or its endpoint, *Left-free* if $\mathrm{Left}(u, w; k) = \varnothing$. Recall that $\prec$ is the total order on labels with $(a, b) \prec (c, d)$ if $b > d$, or $b = d$ and $a < c$.

Lemma 25 (Left-free chains)**.** *An admissible k-chain is Left-free if and only if its label sequence is strictly increasing for $\prec$. Conversely, every directed path in the quantum Bruhat graph from u whose labels (a_i, b_i) satisfy $a_i \le k < b_i$ and are strictly increasing for $\prec$ is an admissible k-chain, and it is Left-free.*

Proof. For an admissible chain the upper indices are weakly decreasing, so the sequence is strictly $\prec$-increasing exactly when there is no index i with $b_i = b_{i+1}$ and $a_i > a_{i+1}$, that is, no descent in the sense of the proof of Lemma 19. That proof shows that the descents are in bijection with $\mathrm{Left}(u, w; k)$. For the converse, a strictly $\prec$-increasing sequence has distinct labels and weakly decreasing upper indices, and condition 4 of Definition 2 holds for every index, repeated lower index or not. Hence the path is an admissible chain, and it is Left-free by the first part. □

Left-free chains are therefore the paths of the quantum alcove model for the reflection order $\prec$ on the roots $\epsilon_a - \epsilon_b$, $a \le k < b$, which is consistent with the Chevalley interpretation of (24).

Three lemmas on rotated paths. Fix $u \in S_R$ and k, and let $\vartheta = u(k+1)$ and $\rho = \rho_\vartheta$ be as in Lemma 10. For $v \in S_R$ and $1 \le j < R$ put $\sigma_j(v) = \#\{i \le j : v(i) \le \vartheta\}$.

Lemma 26 (Degree along rotated paths)**.** *Let $u = v_0 \to v_1 \to \cdots \to v_s = w$ be a directed path in the quantum Bruhat graph with labels (a_i, b_i), $a_i \le k < b_i$, such that every edge of the rotated path $\rho v_0 \to \cdots \to \rho v_s$ is a Bruhat edge. Then $D_j = \sigma_j(w) - \sigma_j(u)$ for every j, where D_j is the number of quantum edges of the path with $a_i \le j < b_i$.*

Proof. An edge (a, b) with $a \le j < b$ applied to v changes σ_j by $+1$ if $v(b) \le \vartheta < v(a)$, by -1 if $v(a) \le \vartheta < v(b)$, and not at all otherwise. By Corollary 9, since the rotated edge is Bruhat, the second case does not occur, and the first case occurs exactly when the edge is quantum. Edges with both positions at most j or both greater than j do not change σ_j and are not counted by D_j. Summing along the path gives the claim. □

Lemma 27 (Commuting a right edge)**.** *Let $x \in S_R$ and $a < k < b' < b$. Then $x \lessdot xt_{kb} \lessdot xt_{kb}t_{ab'}$ is a chain of Bruhat covers if and only if $x \lessdot xt_{ab'} \lessdot xt_{ab'}t_{kb}$ is. The two chains have the same endpoint.*

Proof. The transpositions commute, so the endpoints agree. Write $A = x(a)$, $K = x(k)$, $B' = x(b')$ and $B = x(b)$. Recall that $y \lessdot yt_{ij}$ means $y(i) < y(j)$ and that no value at a position strictly between i and j lies in the interval $(y(i), y(j))$. The positions strictly between k and b other than b', and the positions strictly between a and b' other than k, carry the same values in all four permutations involved, so only the four values A, K, B', B matter. The first chain is a chain of covers if and only if

$$K < B, \quad B' \notin (K, B), \quad A < B', \quad B \notin (A, B')$$

together with the conditions on the other positions, and the second if and only if

$$A < B', \quad K \notin (A, B'), \quad K < B, \quad A \notin (K, B)$$

together with the same conditions on the other positions. Assume the first set. If $B' < K$, then $K \notin (A, B')$ and $A < B' < K$ give $A \notin (K, B)$. If $B' > B$, then $B \in (K, B')$ and $B \notin (A, B')$ force $A > B > K$, so again $K \notin (A, B')$ and $A \notin (K, B)$. Conversely assume the second set. If $K < A$, then $A \notin (K, B)$ forces $A > B$, so $B < A < B'$, whence $B \notin (A, B')$ and $B' \notin (K, B)$. If $K > B'$, then $B' < K$ gives $B' \notin (K, B)$ and $B > K > B'$ gives $B \notin (A, B')$. □

For a sequence γ of labels (a, b) with $a \leq k < b$ let $\mathrm{sort}(\gamma)$ be the sequence obtained by moving all labels with lower index k to the end, preserving the relative order of these labels and of the remaining labels.

Lemma 28 (Sorting). *Let γ be a strictly $\prec$-increasing sequence of labels (a, b) with $a \leq k < b$, and let $x \in S_R$. Then γ is a chain of Bruhat covers from x if and only if* $\mathrm{sort}(\gamma)$ *is, and the two chains have the same endpoint.*

Proof. If a label (k, b) precedes a label (a, b') with $a < k$ in γ, then $(k, b) \prec (a, b')$ forces $b' < b$, since $b' = b$ would require $k < a$. The sorted sequence is obtained from γ by repeatedly interchanging an adjacent pair $(k, b), (a, b')$ with $a < k$ into $(a, b'), (k, b)$; the pairs interchanged are always in their original relative order, so $b' < b$ holds at each step. Each step preserves the endpoint and, by Lemma 27 applied to the permutation reached before the pair, the property of being a chain of covers. □

Lemma 29 (Rotation of sorted words). *Let $\omega = \gamma'\,(k, b_1) \cdots (k, b_r)$ be a directed path in the quantum Bruhat graph from u, where the labels of γ' have lower index at most $k-1$ and upper index at least $k+1$, are pairwise distinct with weakly decreasing upper indices, and $b_1 > \cdots > b_r > k$. Then every edge of the rotated path $\rho u \to \cdots$ is a Bruhat edge.*

Proof. The labels of γ' satisfy the hypotheses of Lemma 10, so the rotated edges of γ' are Bruhat. Work in the rotated order, with $N = \rho(\vartheta)$ the largest value, initially at position $k+1$. Let $(c_1, k+1), \ldots, (c_m, k+1)$ be the labels of γ' with upper index $k+1$; they are its last m labels. As in the proof of Lemma 10, put $e_0 = N$ and let e_h be the value at position c_h just before $(c_h, k+1)$; then $e_0 > e_1 > \cdots > e_m$, after γ' the value at position $k+1$ is e_m, and e_{h-1} sits at position c_h. Position k is touched by no label of γ'; let x be its value. Since $c_h < k < k+1$, the Bruhat condition for $(c_h, k+1)$ gives $x \notin (e_h, e_{h-1})$ for each h, and x differs from all e_h. Hence $x \notin (e_m, N)$, and as $x < N$ we get $x < e_m$.

Let $y_0 = x$ and let y_i be the value at position k after the edge (k, b_i). We show by induction on i that (k, b_i) is Bruhat and that $e_m > y_i$ whenever $b_i \neq k+1$; the induction hypothesis is $e_m > y_{i-1}$, true for $i = 1$. Position $k+1$ still carries e_m when (k, b_i) is applied, because $b_1, \ldots, b_{i-1} > b_i \geq k+1$. If (k, b_i) were quantum, every value at a position strictly between k and b_i would lie in $(v(b_i), y_{i-1})$, and $v(b_i) < y_{i-1}$; for $b_i > k+1$ this puts e_m below y_{i-1}, and for $b_i = k+1$ it says $e_m = v(b_i) < y_{i-1}$; both contradict $e_m > y_{i-1}$. So the edge is Bruhat, $y_i = v(b_i) > y_{i-1}$, and if $b_i > k+1$ the Bruhat condition gives $e_m \notin (y_{i-1}, y_i)$, whence $e_m > y_i$. If $b_i = k+1$ it is the last right edge. □

Corollary 30 (Sorting bijection). *Let $w \in S_R$. The map $\gamma \mapsto \mathrm{sort}(\gamma)$ is a bijection from the Left-free admissible k-chains from u to w onto the directed paths from u to w of the form $\omega = \gamma'\,(k, b_1) \cdots (k, b_r)$, where γ' is a Left-free admissible $(k-1)$-chain from u none of whose labels has upper index k, and $b_1 > \cdots > b_r > k$. It preserves the number of edges and the quantum degree D.*

Proof. Let γ be a Left-free k-chain from u to w. By Lemma 10 the rotated path is a chain of Bruhat covers from ρu to ρw; by Lemma 25 its labels are strictly $\prec$-increasing, so by Lemma 28 $\mathrm{sort}(\gamma)$ is a chain of Bruhat covers from ρu to ρw, and by Corollary 9 it is a directed path from u to w in the quantum Bruhat graph. Its labels with lower index less than k form a strictly $\prec$-increasing directed path γ' from u with upper indices greater than k, hence a Left-free $(k-1)$-chain by Lemma 25, and its labels (k,b) appear in $\prec$-order, that is with decreasing b. Both γ and $\mathrm{sort}(\gamma)$ have all rotated edges Bruhat, so by Lemma 26 they have the same D, and they have the same number of edges.

Conversely let ω be a path of the stated form. By Lemma 29 its rotated edges are Bruhat. Let γ be the $\prec$-increasing rearrangement of its labels, so that $\mathrm{sort}(\gamma) = \omega$; the labels are distinct since γ' has distinct labels and the (k, b_i) are distinct from them. By Lemma 28 γ is a chain of Bruhat covers from ρu to ρw, hence a directed path from u to w, and by Lemma 25 a Left-free k-chain. The two constructions are inverse to each other. □

Proof of Theorem 24. We induct on k, reading the right side of (24) as $\mathfrak{G}^q_u$ for $k = 0$, where the statement is trivial. Let $k \geq 1$ and assume (24) at level $k-1$. Since $(1-\beta^2 q_{k-1})F^k_k = T_k F^{k-1}_{k-1}$, the induction hypothesis and Lemma 16 give

$$(1-\beta^2 q_{k-1})F^k_k\,\mathfrak{G}^q_u = \prod_{i=1}^{k-1} \frac{1}{1+\beta y_{u(i)}} \sum_{\gamma} \beta^{s(\gamma)} q^{D(\gamma)}\, \mathcal{T}_k\, \mathfrak{G}^q_{\mathrm{end}(\gamma)},$$

the sum over Left-free $(k-1)$-chains γ from u (one for each Left-free endpoint, by Corollary 11). Expanding $\mathcal{T}_k$ by (12), the coefficient of $\mathfrak{G}^q_w$ on the right is a sum over *composites* (γ, π), with π a line path at k from $\mathrm{end}(\gamma)$ to w, of the weights

$$\beta^{s(\gamma)} q^{D(\gamma)}\, (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)}\, \frac{1}{1+\beta y_{v(\pi)(k)}}.$$

Since the $\mathfrak{G}^q_w$ are linearly independent and $1-\beta^2 q_{k-1}$ is a nonzero polynomial, (24) at level k is equivalent to the identity, for every w,

$$\begin{aligned} \sum_{(\gamma,\pi)\colon u\to w} & \beta^{s(\gamma)} q^{D(\gamma)} (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)} \frac{1}{1+\beta y_{v(\pi)(k)}} \\ &= \frac{1-\beta^2 q_{k-1}}{1+\beta y_{u(k)}}\, \beta^{s(u,w;k)} q^{D(u,w;k)}\, [\, w \in E(u,k) \text{ Left-free}\,], \end{aligned} \tag{25}$$

which is the junction identity (21) at $a = 0$. We prove (25) by grouping composites by their underlying path.

Words and cuts. A composite (γ, π) determines the concatenated label sequence $\omega = \gamma\pi$, which we call its *word*; it is a directed path from u to w. By Lemma 25 the labels of γ are $\prec$-increasing, so its labels with upper index k, which have the largest possible $\prec$-value, form a terminal segment of γ with increasing lower indices; the left edges of π have upper index k and decreasing lower indices. Hence

$$\omega = \gamma_{>k}\ (a'_1, k)\,(a'_2, k) \cdots (a'_m, k)\ (k, b_1) \cdots (k, b_r),$$

where $\gamma_{>k}$ consists of the labels of γ with upper index greater than k, the middle part is the *block* of ω, and $b_1 > \cdots > b_r > k$. The composite is recovered from ω by the *cut* $j \in \{0, \ldots, m\}$, the number of block labels belonging to γ. Conversely, given a directed path ω of this shape from u to w, the cut j yields a composite exactly when $\gamma_{>k}(a'_1, k) \cdots (a'_j, k)$ is a Left-free $(k-1)$-chain and

$(a'_{j+1},k)\cdots(a'_m,k)(k,b_1)\cdots(k,b_r)$ is a line path, that is, by Lemma 25 and Definition 15, exactly when

$$a'_1 < a'_2 < \cdots < a'_j \qquad \text{and} \qquad a'_{j+1} > a'_{j+2} > \cdots > a'_m.$$

All composites with word ω have the same endpoint, the same quantum degree $D(\omega)$, the same total number $|\omega|$ of labels, and the same value $v(\pi)(k)$, namely the value $c(\omega)$ at position k after the block, since position k is touched only by the block and the right edges. The weight of the composite with cut j is therefore

$$(-1)^{m-j}\,\beta^{|\omega|}q^{D(\omega)}\,\frac{1}{1+\beta y_{c(\omega)}}.$$

Words with two valid cuts cancel. Let $m \geq 1$. If j is a valid cut with $1 \leq j < m$, the largest block entry is a'_j or a'_{j+1}; if j and $j' > j$ are both valid, then $a'_{j+1},\ldots,a'_{j'}$ lie both in an increasing prefix and in a decreasing suffix, so $j' = j+1$. Thus a word has at most two valid cuts, and if it has two, they are consecutive and their weights cancel. If j is valid, then $j+1$ is valid exactly when $a'_j < a'_{j+1}$, and $j-1$ is valid exactly when $a'_j > a'_{j+1}$ (for $1 \leq j < m$; for $j = 0$ or $j = m$ the block is monotone and the cut 1, respectively $m-1$, is valid as well). Hence a word with exactly one valid cut j has $1 \leq j < m$ and $a'_j = a'_{j+1}$: the same label (a'_j,k) is traversed twice in succession. Two consecutive traversals of a label (a,k) form a Bruhat edge followed by a quantum edge or vice versa, which forces $1 = 2(k-a)-1$, that is $a = k-1$. So a word with a single valid cut has block

$$a'_1 < \cdots < a'_{j-1} < k-1, \qquad a'_j = a'_{j+1} = k-1, \qquad k-1 > a'_{j+2} > \cdots > a'_m. \tag{26}$$

Moving the backtrack. From any permutation, $(k-1,k)$ is an edge (Bruhat if the value at $k-1$ is smaller, quantum otherwise), and the pair $(k-1,k)(k-1,k)$ returns to the starting permutation with quantum degree q_{k-1}. Hence the pair can be inserted into or removed from a word at any point without affecting the other edges. For a word with block (26) and $m \geq 3$, let p be the larger of the block entries adjacent to the pair (a'_{j-1} or a'_{j+2}, or the only one present), and move the pair to the other side of p. The result is again a directed path to w with a block of the form (26), with the same $|\omega|$, $D(\omega)$ and $c(\omega)$, and its unique cut differs from j by one, so the two weights cancel. This is an involution without fixed points on the words with a single valid cut and $m \geq 3$.

Surviving words. What remains are the words with empty block, $\omega = \gamma_{>k}(k,b_1)\cdots(k,b_r)$, and the words $\omega = \gamma_{>k}(k-1,k)(k-1,k)(k,b_1)\cdots(k,b_r)$. Inserting or removing the pair is a bijection between the two kinds, the second kind having weight $-\beta^2 q_{k-1}$ times the first (one more chain label, one more left edge, degree q_{k-1}). In both kinds position k is untouched before the right edges, so $c(\omega) = u(k)$. The left side of (25) is therefore

$$\frac{1-\beta^2 q_{k-1}}{1+\beta y_{u(k)}}\sum_{\omega}\beta^{|\omega|}q^{D(\omega)},$$

the sum over directed paths $\omega = \gamma'(k,b_1)\cdots(k,b_r)$ from u to w with γ' a Left-free $(k-1)$-chain without upper index k and $b_1 > \cdots > b_r > k$. By Corollary 30 this sum equals $\sum_\gamma \beta^{s(\gamma)}q^{D(\gamma)}$ over the Left-free k-chains from u to w, which by Corollary 11 is $\beta^{s(u,w;k)}q^{D(u,w;k)}$ if $w \in E(u,k)$ is Left-free and 0 otherwise. This proves (25), hence (24) at level k, and Theorem 24. □

Remark 31. *For $k = 1$ the argument has no block and recovers Proposition 17 at $z = 1/\beta$. The proof shows more precisely that in the junction identity at $a = 0$ the only composites that do not cancel are the sorted composite $\gamma_{>k}\,\big|\,(k,b_1)\cdots(k,b_r)$ of each Left-free endpoint and its backtrack $\gamma_{>k}(k-1,k)\,\big|\,(k-1,k)(k,b_1)\cdots(k,b_r)$. This is the pattern observed numerically for the moved case in Section 4.11 below; the general case differs in that cut-sliding leaves residues proportional to a (Lemma 33) and in that chains with* $\mathrm{Left} \neq \varnothing$ *contribute.*

Remark 32 (The specialization of Maeno–Naito–Sagaki). *A second one-point specialization of Theorem 3 is in the literature. Let*

$$u = (n{-}k{+}1,\, n{-}k{+}2,\, \ldots,\, n{+}1,\;\, n{-}k,\, \ldots,\, 2,\, 1) \in S_{n+1},$$

so that positions $1, \ldots, k+1$ carry the $k+1$ largest values in increasing order and the remaining values follow in decreasing order. The admissible k-chains from u are exactly

$$\gamma_j = (k, k{+}1),\, (k{-}1, k{+}1),\, \ldots,\, (j, k{+}1), \qquad 1 \le j \le k+1,$$

with γ_{k+1} the empty chain. Indeed, a first label (a, b) with $b \ge k{+}2$ is neither Bruhat, as $u(a) > u(b)$, nor quantum, as the value $n{+}1$ at position $k{+}1$ is not in $(u(b), u(a))$; so all labels have upper index $k+1$, and after γ_j the only label $(a, k+1)$ with $a < j$ that is an edge is $(j{-}1, k{+}1)$, because the value at a is smaller than the value $n-k+j$ at $k+1$, excluding a quantum edge, while the values at positions $a+1, \ldots, j-1$ lie between them unless $a = j-1$. For $j \ge 2$ position 1 is untouched, so $u(1) \in \mathrm{Fix}$ and the coefficient carries the factor $z + \overline{y}_{u(1)} = (1 - a\, t_{u(1)})/(\beta t_{u(1)})$, $t_c = 1 + \beta y_c$, which vanishes at $a = t_{u(1)}^{-1}$. For $j = 1$ the endpoint is

$$w = (n{-}k{+}2,\, \ldots,\, n{+}1,\;\, n{-}k{+}1,\;\, n{-}k,\, \ldots,\, 1),$$

with Left $= \{u(2), \ldots, u(k)\}$, Out $= \{u(1)\}$, $s = k$, $D = 0$, *and coefficient $a^{k-1} t_{u(1)}^{-1} = t_{u(1)}^{-k}$ at $a = t_{u(1)}^{-1}$. Since $\beta^k B_k^q = \sum_p (-a)^{k-p} F_p^k$, Theorem 3 predicts*

$$\sum_{p=0}^{k} (-1)^p\, t_{u(1)}^p\, F_p^k\, \mathfrak{G}_u^q = \mathfrak{G}_w^q \qquad (\beta = -1),$$

which is [8, Proposition 2.3] transported to $QK_T(Fl_{n+1})$ as in their Section 3.1 (with $1 - y_c = \mathbf{e}^{-\epsilon_c}$, $u = w_{k+1}$ and $w = w_k$ in their notation), the key step in their proof that quantum double Grothendieck polynomials represent Schubert classes. Their proof of this special case expands each term of F_p^k by the general Chevalley formula for the weight $w_0 \epsilon_J$, $J \subseteq \{1, \ldots, k\}$.

4.10 The cancellation mechanism: cut composites and local moves

Every term of (21) is a *cut composite*: a directed path in the quantum Bruhat graph

$$u = v_0 \to v_1 \to \cdots \to v_r = v \;\big|\; v \to \cdots \to w,$$

whose first part is an admissible chain at level $k-1$ or $k-2$ and whose second part is a line path at k; the bar marks the cut. The term attached to it is the product of the chain coefficient $C_{u,v}^{(j)}$ (times $a\beta q_{k-1}$ when $j = k-2$) and the line weight $(-\beta)^s \beta^t q^D (1 + \beta y_{v(\pi)(k)})^{-1}$. The whole edge sequence from u to w determines q^D for the chain and the path together, and it determines $v(\pi)$; only the position of the cut and the level distinguish the bookkeeping. The cancellation in (21) is realized by two moves on cut composites, which we now make precise, together with the identity $\beta(z + \overline{y}_c) + a = (1 + \beta y_c)^{-1}$ used in (22).

Lemma 33 (Cut sliding). *Let γ be an admissible $(k-1)$-chain from u to v, let $\pi = (k-1, k) \cdot \pi'$ be a line path at k from v whose first edge is $(k-1, k)$, and put $v' = v t_{k-1,k}$ and $\kappa \in \{1, q_{k-1}\}$ for the weight of the edge $v \to v'$. Then:*

1. *$\gamma' = \gamma \cdot (k-1,k)$ is an admissible $(k-1)$-chain from u to v' if and only if $(k-1,k) \notin \gamma$, and π' is a line path at k from v' with $v(\pi') = v(\pi)$, $s(\pi') = s(\pi) - 1$, $t(\pi') = t(\pi)$ and $q^{D(\pi)} = \kappa\, q^{D(\pi')}$.*

2. *If $v(k-1) \neq u(k-1)$ and $v(k) \neq u(k-1)$, then* $\mathrm{Fix}, \mathrm{Left}, \mathrm{Out}$ *computed for v and for v' agree, so that $C^{(k-1)}_{u,v'} = \beta\kappa\, C^{(k-1)}_{u,v}$, and the two terms of (21) indexed by (v,π) and (v',π') cancel exactly.*

3. *If $v(k-1) = u(k-1)$, then γ is also an admissible $(k-2)$-chain to v, $C^{(k-1)}_{u,v} = (z + \overline{y}_{u(k-1)})C^{(k-2)}_{u,v}$, and the two terms indexed by (v,π) and (v',π') sum to*

$$a\kappa\,(-\beta)^{s(\pi')}\beta^{t(\pi')}q^{D(\pi')}\,\frac{C^{(k-2)}_{u,v}}{1+\beta y_{v(\pi)(k)}}. \tag{27}$$

Proof. (1) The edge $(k-1,k)$ joins adjacent positions, so it is always an edge of the quantum Bruhat graph, Bruhat if $v(k-1) < v(k)$ and quantum with weight q_{k-1} otherwise; it is the same edge in γ' and in π, whence the statement about weights. In γ' the second entries remain weakly decreasing since k is the least possible value, and the only condition of Definition 2 that involves the new last label is distinctness; condition 4 for the previous label (a_r, b_r) is automatic, because $b_r > k$, or $b_r = k$ and $a_r \neq k-1$. Removing the first edge of a line path leaves a line path with the same left-part endpoint $v(\pi)$.

(2) Passing from v to v' exchanges the values $c = v(k-1)$ and $d = v(k)$. The value c leaves the window $\{1,\dots,k-1\}$: it cannot lie in Left (a value in position $k-1$ cannot have moved left), and if it lies in Out or is not a window value of u at all, its status is unchanged. The value d enters position $k-1$: if $d = u(i)$ with $i \leq k-2$ it was in Out and remains there, and otherwise it is not a window value. Hence m is unchanged while the chain length grows by one and the quantum weight is multiplied by κ, giving $C^{(k-1)}_{u,v'} = \beta\kappa C^{(k-1)}_{u,v}$. The line weights of π and π' differ by the factor $(-\beta)\kappa$, so the two terms are negatives of each other.

(3) Here $v(k-1) = u(k-1)$, so by Corollary 11(2) position $k-1$ is untouched by γ: no label of γ has first entry $k-1$, and γ is admissible at level $k-2$ with the same data; this gives the factorization of $C^{(k-1)}_{u,v}$ as in (22). In v' the value $c = u(k-1)$ has moved to position k, so it passes from Fix to Out while d behaves as in (2); thus $C^{(k-1)}_{u,v'} = \kappa\,(1+\beta y_c)^{-1}(z+\overline{y}_c)^{-1}C^{(k-1)}_{u,v}$. Writing $X = (-\beta)^{s(\pi')}\beta^{t(\pi')}q^{D(\pi')}(1+\beta y_{v(\pi)(k)})^{-1}$, the term of (v,π) is $X(-\beta)\kappa C^{(k-1)}_{u,v}$ and the term of (v',π') is $X\kappa(1+\beta y_c)^{-1}(z+\overline{y}_c)^{-1}C^{(k-1)}_{u,v}$. Their sum is

$$X\kappa\,\frac{(1+\beta y_c)^{-1} - \beta(z+\overline{y}_c)}{z+\overline{y}_c}\,C^{(k-1)}_{u,v} = X\kappa a\, C^{(k-2)}_{u,v},$$

which is (27). □

Lemma 33 makes no assertion in the case $v(k) = u(k-1)$, in which a fixed window value would enter position $k-1$. We do not need it: wherever parts (2) and (3) are applied below (Proposition 41, Proposition 46 and Lemma 48), the hypothesis $v(k) \neq u(k-1)$ holds, as is noted there. The residue (27) has exactly the shape of a level-$(k-2)$ term of (21), up to the factor βq_{k-1}: the missing q_{k-1} appears either through $\kappa = q_{k-1}$ or through a factor q_{k-1} inside $q^{D(\pi')}$, and the missing β through the line path π' being one edge shorter. This is why the junction input is $P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)$ and not $P_{k-1}(u)$ alone.

Lemma 34 (Braid move)**.** *Let $a < k-1$. For any permutation v, the two-edge sequences $(k-1,k),(a,k)$ and $(a,k-1),(k-1,k)$ from v have the same endpoint $vt_{k-1,k}t_{ak} = vt_{a,k-1}t_{k-1,k}$, and if both are directed paths in the quantum Bruhat graph they have the same quantum weight. Consequently, if γ is an admissible $(k-2)$-chain to v with $\gamma\cdot(a,k-1)$ admissible, and $\pi = (k-1,k)\cdot(a,k)\cdot\pi''$ is a line path from v, then $(k-1,k)\cdot\pi''$ is a line path from $vt_{a,k-1}$ with the same $v(\pi)$, and the terms indexed by (v,π) and $(vt_{a,k-1},(k-1,k)\cdot\pi'')$ at level $k-2$ cancel exactly whenever the window statistics of v and $vt_{a,k-1}$ agree.*

Proof. The identity $t_{k-1,k}t_{ak} = t_{a,k-1}t_{k-1,k}$ holds in S_∞. For the weights write $x = v(a)$, $y = v(k-1)$, $z = v(k)$. The adjacent transposition $(k-1,k)$ is always an edge, quantum exactly when the larger value is on the left. If $x<y$, the edge (a,k) from $vt_{k-1,k}$ is Bruhat, and its cover condition applied to the intermediate position $k-1$, which holds z, gives $z\notin(x,y)$; hence $[y>z]=[x>z]$, and both paths have weight $q_{k-1}^{[y>z]}$. If $x>y$, the edge (a,k) from $vt_{k-1,k}$ is quantum, so $y<z<x$, and both paths have weight $q_aq_{a+1}\cdots q_{k-1}$. The label $(a,k-1)$ is admissible at level $k-2$ since $a\le k-2<k-1$, and second entries remain weakly decreasing because $k-1$ is the least possible value. The chain gains one edge (a factor β when m is unchanged) while the line path loses one left edge (a factor $-\beta$), and the quantum weights agree, so the two terms cancel. □

When the window statistics change in Lemma 34 (the value $v(a)$ or $v(k-1)$ is a fixed window value), a residue of the same kind as (27) is produced instead.

The drop term. The diagonal term of (21), $\pi=\varnothing$ and $v=w$, contributes $(1+\beta y_{w(k)})^{-1}\big(C_{u,w}^{(k-1)}+a\beta q_{k-1}C_{u,w}^{(k-2)}\big)$, and after (22) what is left of it is the drop term in (23). In the language of cut composites it is the composite with an empty line path, and Lemma 33 pairs it with the composite $u\xrightarrow{\gamma_w}w\xrightarrow{(k-1,k)}wt_{k-1,k}\;\big|\;(k-1,k)$, for which the move of (1) is blocked because the label $(k-1,k)$ would be repeated. Its term is $-\beta\kappa'\,(1+\beta y_{u(k)})^{-1}C_{u,wt_{k-1,k}}^{(k-1)}$ with $\kappa\kappa'=q_{k-1}$, and by the computation in the proof of Lemma 33 it equals $-\beta^2q_{k-1}(1+\beta y_{u(k)})^{-1}C_{u,w}^{(k-1)}$ when $w(k-1)\neq u(k-1)$, and $-\beta q_{k-1}(1+\beta y_{u(k)})^{-1}(1+\beta y_{u(k-1)})^{-1}C_{u,w}^{(k-2)} = -\beta q_{k-1}(1+\beta y_{u(k)})^{-1}\big(\beta C_{u,w}^{(k-1)}+aC_{u,w}^{(k-2)}\big)$ when $w(k-1)=u(k-1)$, using (22) at level $k-1$. In both cases it is the negative of the drop term, provided $C_{u,w}^{(k-2)}=0$ in the first case.

A worked class. Take $u=21345$, $k=3$ and the endpoint $w=12345$, which is not in $E(u,3)$, so the right side of (21) must vanish. The nonzero terms are indexed by three cut composites of the single edge sequence $(1,3),(2,3),(1,3)$ from u to w at level 2, its braid image at level 1, and the drop:

$$\begin{array}{lll}
\text{level 2}: & (1,3)\,\big|\,(2,3),(1,3) & \beta^2q_1q_2\,(z+\overline{y}_1)\,t_2^{-1}t_3^{-1},\\
\text{level 2}: & (1,3),(2,3)\,\big|\,(1,3) & -\beta q_1q_2\,t_1^{-1}t_2^{-1}t_3^{-1},\\
\text{level 1}: & (1,3)\,\big|\,(2,3),(1,3) & a\beta^3q_1q_2^2\,t_2^{-1}t_3^{-1},\\
\text{level 1}: & (1,3),(1,2)\,\big|\,(2,3) & -a\beta^3q_1q_2^2\,t_2^{-1}t_3^{-1},\\
\text{drop}: & & a\beta q_1q_2\,t_2^{-1}t_3^{-1},
\end{array}$$

with $t_c=1+\beta y_c$. The two level-1 terms are a braid pair (Lemma 34) and cancel. The two level-2 terms are a cut sliding pair of type (3) in Lemma 33, since the slid value $v(2)=1=u(2)$ is fixed; their sum is the residue $-a\beta q_1q_2t_2^{-1}t_3^{-1}$, which cancels the drop term. The residue is absorbed by the drop rather than by a level-1 path term because the relevant level-1 endpoint is w itself.

Status of the local analysis. Lemmas 33 and 34 are proved. To complete the diagonal case in this language one would still need a global statement: that the cut composites with a given endpoint w, $w(k) = u(k)$, decompose into classes generated by the two moves, and that within each class the exact cancellations and the residues of type (27) sum to zero against the level-$(k-2)$ terms and the drop. We do not prove this global statement in this form, and the proof of Theorem 3 does not use it. Instead, Section 4.12 reorganizes the junction identity: the right edges are discarded (Lemma 42), the y-dependence is shown to be rigid (Proposition 43), and what remains is the backtrack identity (34), proved in Sections 4.13 and 4.14; the diagonal case $w(k) = u(k)$ is then handled together with the moved case. (As a consistency check in the sense of Remark 6, we have verified (23) termwise, with symbolic coefficients, for all $u \in S_5$ and all k, and for a sample of $u \in S_6$ with $k \leq 5$, including endpoints $w \notin E(u,k)$, where it asserts the vanishing of the right side.)

The moved cases. Suppose $u(k) \neq w(k)$. Then $C_{u,w}^{(k-1)}$ and $C_{u,w}^{(k-2)}$ need not vanish, in contrast to the quantum Schubert case: a $(k-1)$-chain may move position k through a label (a,k). The identity (21) must then be established by matching each k-chain to w with a $(k-1)$- or $(k-2)$-chain to some v followed by a line path $v \to w$, and by cancelling all remaining terms in pairs. The $k = 1$ case shows the pattern: there $P_{k-2} = 0$, every nonempty line path is a single up-path, and the matching is the identity. In this language the case analysis for $k \geq 2$ is carried out at $a = 0$ above (Theorem 24) and at $q = 0$ below (Theorem 38), where the matching is obtained by counting rather than by an explicit bijection; the general case is treated, after the reorganization just described, in Sections 4.12–4.14.

4.11 Special cases

Two specializations isolate the two difficulties of the junction identity from each other. The first, $q = 0$, is settled completely below (Theorem 38); Section 4.12 then shows that its proof extends to arbitrary q modulo one identity, proves that identity for $k = 2$, and Sections 4.13 and 4.14 prove it in general.

The classical case $q = 0$. Set all $q_j = 0$. Then all edges are Bruhat covers, the level-$(k-2)$ part of the junction input and the drop term vanish (both carry q_{k-1}), and (20) reduces to

$$\beta\, P_k(u) = \mathcal{T}_k P_{k-1}(u) - a\, P_{k-1}(u), \tag{28}$$

the expansion of $\beta B_k + a B_{k-1} = (1 + \beta x_k) B_{k-1}$ by the classical equivariant K-Chevalley formula. The diagonal case is then vacuous:

Lemma 35. *Let $q = 0$ and $w(k) = u(k)$. There is no admissible $(k-1)$-chain γ from u to some v together with a nonempty line path at k from v to w. Consequently* (28) *holds at every such w, by* (22).

Proof. Follow the value in position k along the composite path. Every label (a,k), whether in γ or in the left part of the line path, is a Bruhat cover, so it replaces the value in position k by a strictly smaller one; every right edge (k,b) replaces it by a strictly larger value coming from a position $b > k$. Suppose first that some label (a,k) occurs, and let (α,k) be the first. It moves $u(k)$ to position α, which is touched afterwards only by a left edge (α,k) of the line path; that edge would have to be a Bruhat cover with $u(k) = v(\alpha) < v(k)$, whereas every value occupying position k after (α,k) is smaller than $u(k)$. Hence $u(k)$ never returns to position k, and the right edges, which only import values from positions beyond k, cannot return it either. If no label (a,k) occurs, the line

path consists of right edges only, and the first of them moves $u(k)$ to a position $b > k$ that is never touched again. In all cases $w(k) \neq u(k)$. □

For the moved case at $q = 0$ the computations of Section 4.10 show that only cut sliding is needed, together with the end braid $t_{k-1,k}t_{kb} = t_{k-1,b}t_{k-1,k}$, which converts a right edge following a junction edge $(k-1,k)$ into two $(k-1)$-chain labels; the residues (27) cancel the term $-aC_{u,w}^{(k-1)}$ of (21) exactly as in the example $u = 21435$, $k = 3$, $w = 24315$, where the pair $u \,\big|\, (2,3),(3,4)$ and $ut_{23} \,\big|\, (3,4)$ leaves the residue $a\beta(z+\overline{y}_2)t_1^{-1} = aC_{u,w}^{(2)}$.

The word-and-cut bookkeeping of Section 4.9 organizes the classical case completely at the level of individual words. We normalize the coefficients by

$$\widehat{C}_{u,v}^{(j)} := \beta^j C_{u,v}^{(j)} = \beta^s \prod_{i=1}^{j} \phi_i(u,v;j), \qquad \phi_i = \begin{cases} t_{u(i)}^{-1} - a & u(i) \in \mathrm{Fix}, \\ a & u(i) \in \mathrm{Left}, \\ t_{u(i)}^{-1} & u(i) \in \mathrm{Out}, \end{cases}$$

with $t_c = 1 + \beta y_c$; this is (6) at $q = 0$ written with $z + \overline{y}_c = (t_c^{-1} - a)/\beta$ and $m + |\,\mathrm{Fix}\,| = j$. In this notation the classical junction identity (28) at the endpoint w reads

$$\widehat{C}_{u,w}^{(k)} + a\,\widehat{C}_{u,w}^{(k-1)} = \sum_{(\gamma,\pi)\colon u \to w} \beta^{s(\gamma)}(-\beta)^{s(\pi)}\beta^{t(\pi)}\, t_{v(\pi)(k)}^{-1} \prod_{i<k} \phi_i(u, \mathrm{end}\,\gamma; k-1), \tag{29}$$

the sum over composites of an admissible $(k-1)$-chain γ and a line path π of Bruhat edges. As in Section 4.9, a composite has a word $\omega = \gamma_{>k}\,(a'_1,k)\cdots(a'_m,k)\,(k,b_1)\cdots(k,b_r)$ and a cut j; now the chain part may contain descents, so the valid cuts are no longer determined by monotonicity alone. Call the block entry a'_r *fresh* if position a'_r is touched neither by $\gamma_{>k}$ nor by $a'_1,\dots,a'_{r-1}$, and write $\Sigma(\omega)$ for the sum of the weights of all composites with word ω.

Proposition 36 (Classical word lemma)**.** *Let $q = 0$ and let ω be the word of a composite, with block $(a'_1,\dots,a'_m)$, and let S be the length of the longest strictly decreasing suffix $a'_{m-S+1} > \cdots > a'_m$ of the block ($S = 0$ if $m = 0$). Then:*

1. *The block entries are pairwise distinct.*
2. *The valid cuts form the interval $m - S \le j \le J$, where J is the least index $r < m$ such that a'_r is not fresh and $a'_r > a'_{r+1}$, or $J = m$ if there is no such r.*
3. *$\Sigma(\omega) \neq 0$ if and only if $J = m$ and all entries $a'_{m-S+1},\dots,a'_m$ of the decreasing suffix are fresh. In that case*

$$\Sigma(\omega) = \beta^{|\omega|}\, t_{c(\omega)}^{-1}\, a^S \prod_{i \notin \{a'_1,\dots,a'_m\}} \phi_i^0 \prod_{r \le m-S} \psi_r,$$

where $c(\omega)$ is the value at position k after the block, $\phi_i^0 = \phi_i(u, u\gamma_{>k}; k-1)$, and for $r \le m-S$: $\psi_r = \phi_{a'_r}^0$ if a'_r is not fresh, $\psi_r = a$ if a'_r is fresh and $a'_r > a'_{r+1}$, and $\psi_r = t_{u(a'_r)}^{-1}$ if a'_r is fresh and either $a'_r < a'_{r+1}$ or $r = m - S$.

Proof. (1) A Bruhat edge (a,k) requires the value at a to be smaller than the value at k, and it moves the smaller value to k; so along the block the value at position k strictly decreases. If (a,k) occurred at two block indices $r < r'$, the value placed at a by the first occurrence would still be there at the second, and it exceeds every later value at k; so the second occurrence is not a Bruhat edge.

(2) By (1), an entry a'_r is fresh exactly when position a'_r is not touched by $\gamma_{>k}$. The line part $(a'_{j+1},k)\cdots(a'_m,k)(k,b_1)\cdots(k,b_r)$ is a line path exactly when $a'_{j+1}>\cdots>a'_m$, that is $j\geq m-S$. The chain part $\gamma_{>k}(a'_1,k)\cdots(a'_j,k)$ has distinct labels by (1) and weakly decreasing upper indices, and condition 4 of Definition 2 is automatic inside $\gamma_{>k}$ and at its last label. At a block index $r<j$ the lower index a'_r is repeated exactly when a'_r is not fresh, since at level $k-1$ a window position is touched only by labels with that lower index, and then condition 4 requires $(a'_r,k)\prec(a'_{r+1},k)$, that is $a'_r<a'_{r+1}$. Hence the chain part is admissible exactly when $j\leq J$.

(3) Fix a valid cut j and let v_j be the endpoint of the chain part. We determine the status of each window position $i<k$ in v_j, starting from its status in $v_0=u\gamma_{>k}$. Let x_r be the value at a'_r just before (a'_r,k); after this label x_r is at k, and after (a'_{r+1},k) (if $r<j$) it is at a'_{r+1}, where it stays, since by (1) position a'_{r+1} is not touched again by the block. If a'_r is fresh then $x_r=u(a'_r)$, so position a'_r has status Out if $r=j$, status Left if $r<j$ and $a'_{r+1}<a'_r$, and status Out if $r<j$ and $a'_{r+1}>a'_r$; before the label, for $r>j$, its status is Fix. If a'_r is not fresh, then $x_r\neq u(a'_r)$, and we claim that the status of every value is unchanged by the block labels of the chain part other than the fresh ones just described. Indeed, a value $c=u(i)$ with status Left in v_0 sits at a position $p<i$ reached by a descent $(i,b),(p,b)$ of $\gamma_{>k}$, and a chain part containing (p,k) would contain the subsequence $(i,b),(p,b),(p,k)$ with $p<i\leq k-1<k<b$, which the prohibition lemma [6, Lemma 2.9] excludes for an admissible chain. A value with status Out at a window position p is moved by (p,k) to position k, and by the next label, if any, to a'_{r+1} with $a'_{r+1}>p$ by condition 4, since p is a repeated lower index; so it keeps status Out. Values at positions greater than k are not moved by the block. This proves that the weight of the composite with cut j is

$$\beta^{|\omega|}(-1)^{m-j}\,t^{-1}_{c(\omega)}\prod_{i\notin\{a'_r\}}\phi^0_i\prod_{a'_r\text{ not fresh}}\phi^0_{a'_r}\prod_{a'_r\text{ fresh}}\phi^{(j)}_r,$$

where

$$\phi^{(j)}_r=\begin{cases}a & r<j,\ a'_{r+1}<a'_r,\\ t^{-1}_{u(a'_r)} & r<j,\ a'_{r+1}>a'_r,\ \text{or } r=j,\\ t^{-1}_{u(a'_r)}-a & r>j,\end{cases}$$

the sign coming from the $m-j$ left edges of the line part.

Now sum over $j\in[m-S,J]$. If $J<m-S$ there is no valid cut and $\Sigma(\omega)=0$; note that $J\neq m-S$ when $m-S\geq 1$, since a'_{m-S} is an ascent by the maximality of S. For $r<m-S$ we have $r<j$ throughout, so $\phi^{(j)}_r=\psi_r$ is independent of j; for $r=m-S\geq 1$, the entry a'_{m-S} is an ascent, so $\phi^{(j)}_{m-S}=t^{-1}_{u(a'_{m-S})}$ both for $j=m-S$ and for $j>m-S$. If $m-S<J<m$, the entries a'_r with $m-S<r<J$ are fresh descents, a'_J is not fresh, and the factors for $r>J$ do not depend on j. Writing $\tau_i=t^{-1}_{u(a'_{m-S+i})}$ and $T_l=a^l\prod_{l<i<L+1}(\tau_i-a)$ for $0\leq l\leq L:=J-(m-S)-1$, the remaining sum is, up to a common factor,

$$\sum_{j=m-S}^{J}(-1)^{m-j}\prod_{m-S<r<J}\phi^{(j)}_r$$
$$=\pm\Big(T_0-(T_0+T_1)+(T_1+T_2)-\cdots+(-1)^L(T_{L-1}+T_L)+(-1)^{L+1}T_L\Big)=0,$$

because the cut $j=m-S+l$ with $1\leq l\leq L$ contributes $a^{l-1}\tau_l\prod_{i>l}(\tau_i-a)=T_{l-1}+T_l$, the cut $j=m-S$ contributes T_0, and the cut $j=J$ contributes $a^L=T_L$. If $J=m$ and some suffix entry is not fresh, it must be a'_m (the others are descents), its factor $\phi^0_{a'_m}$ is constant, and the same

telescoping with $L = S-1$ gives 0. If $J = m$ and all suffix entries are fresh, the same computation with $L = S$, where now the cut $j = m$ contributes $a^{S-1}\tau_S = T_{S-1} + T_S$, gives $T_S = a^S$. Collecting the constant factors yields the stated formula. □

Thus at $q = 0$ a word contributes either nothing or a single product. We now show that this product depends only on the endpoint w, not on the word. Call a word with $\Sigma(\omega) \neq 0$ *surviving*. For $j \geq 0$ and arbitrary w put

$$\Phi^{(j)}(u, w) := \beta^{\ell(w)-\ell(u)} \prod_{i=1}^{j} \phi_i(u, w; j),$$

the endpoint factors of (6) at $q = 0$, evaluated at w whether or not $w \in E(u, j)$. Since $s(u, w; j) = \ell(w) - \ell(u)$ at $q = 0$, we have $\Phi^{(j)}(u, w) = \widehat{C}^{(j)}_{u,w}$ for $w \in E(u, j)$. As $\phi_i(u, w; k) = \phi_i(u, w; k-1)$ for $i < k$,

$$\Phi^{(k)}(u, w) = \phi_k(u, w; k)\, \Phi^{(k-1)}(u, w), \qquad \Phi^{(k)}(u, w)\big|_{y=0} = \beta^{\ell(w)-\ell(u)}(1-a)^{|\operatorname{Fix}(u,w;k)|} a^{|\operatorname{Left}(u,w;k)|}. \tag{30}$$

Proposition 37 (Surviving words carry the endpoint weight)**.** *Let $q = 0$ and let ω be a surviving word from u to w. If no label of ω involves position k, then $w(k) = u(k)$ and $\Sigma(\omega) = \Phi^{(k)}(u, w) + a\,\Phi^{(k-1)}(u, w)$. Otherwise $w(k) \neq u(k)$ and $\Sigma(\omega) = \Phi^{(k)}(u, w)$.*

Proof. Write $\omega = \gamma_{>k}(a'_1, k) \cdots (a'_m, k)(k, b_1) \cdots (k, b_r)$, let $v_0 = u\gamma_{>k}$, and let v be the permutation reached after the block, so that $|\omega| = \ell(w) - \ell(u)$. By Proposition 36, $J = m$, so the cut $j = m$ is valid: $\gamma_{>k}(a'_1, k) \cdots (a'_m, k)$ is an admissible $(k-1)$-chain from u to v; and a'_m is fresh when $m \geq 1$. We compare the factors in Proposition 36(3) with the statuses of $u(1), \ldots, u(k)$ in w at level k.

The right edges touch only the positions $k, b_1, \ldots, b_r$, all greater than every $i < k$. A value $u(i)$, $i < k$, sitting at one of these positions in v is in Out in v at level $k-1$ and in w at level k; any other $u(i)$ does not move. Hence $\phi_i(u, v; k-1) = \phi_i(u, w; k)$ for all $i < k$.

Suppose $m = 0$. Then $v(k) = u(k)$, so $c(\omega) = u(k)$ and $\Sigma(\omega) = \beta^{|\omega|} t^{-1}_{u(k)} \prod_{i<k} \phi_i(u, w; k)$. If $r = 0$ then $w = v$, $u(k) \in \operatorname{Fix}$, and $t^{-1}_{u(k)} = \phi_k(u, w; k) + a$, which gives the first assertion by (30). If $r \geq 1$, the edge (k, b_1) moves $u(k)$ to position b_1, which is not touched again; so $u(k) \in \operatorname{Out}$ and $t^{-1}_{u(k)} = \phi_k(u, w; k)$.

Suppose $m \geq 1$. Position k is first touched by (a'_1, k), which moves $u(k)$ to $a'_1 < k$, and position a'_1 is not touched again, because the block entries are distinct and the right edges touch only positions $\geq k$. So $u(k) \in \operatorname{Left}(u, w; k)$ and $\phi_k(u, w; k) = a$. Since a'_m is fresh, (a'_m, k) moves $u(a'_m)$ to position k; thus $c(\omega) = u(a'_m)$, and $u(a'_m)$ ends at k or at b_1, so $\phi_{a'_m}(u, w; k) = t^{-1}_{u(a'_m)}$. Hence the factor $t^{-1}_{c(\omega)}$ together with the factor a that the suffix entry a'_m contributes to a^S equals $\phi_{a'_m}(u, w; k)\, \phi_k(u, w; k)$. For the remaining positions $i < k$, $i \neq a'_m$, we show that the factor assigned to i in Proposition 36(3) is $\phi_i(u, v; k-1)$, which is $\phi_i(u, w; k)$. If $i = a'_r$ is fresh with $r < m$, then $u(i)$ is moved by (a'_r, k) to k and by (a'_{r+1}, k) to a'_{r+1}, where it stays; the assigned factor is a when $a'_{r+1} < a'_r$ (whether or not r lies in the suffix) and $t^{-1}_{u(i)}$ when $a'_{r+1} > a'_r$, in agreement with the status Left, respectively Out, of $u(i)$ in v. If i is not a fresh block entry, the assigned factor is $\phi^0_i = \phi_i(u, v_0; k-1)$, and the proof of Proposition 36(3), applied to the valid cut $j = m$, shows that the status of $u(i)$ is the same in v_0 and in v. Multiplying all factors gives $\Sigma(\omega) = \beta^{|\omega|} \prod_{i \leq k} \phi_i(u, w; k) = \Phi^{(k)}(u, w)$. □

The proposition reduces the classical junction identity to counting surviving words, and the count can be read off from the non-equivariant case.

Theorem 38 (The classical case)**.** *Theorem 3 holds at $q = 0$: for all $k \geq 1$ and $u \in S_\infty$,*

$$\prod_{i=1}^{k}(x_i + z)\,\mathfrak{G}_u(x;y) = \sum_{w \in E(u,k)} \beta^{\ell(w)-\ell(u)-m} \prod_{c \in \mathrm{Fix}} \Big(z - \frac{y_c}{1+\beta y_c}\Big)(1-\beta z)^{|\,\mathrm{Left}\,|} \prod_{c \in \mathrm{Out}} \frac{1}{1+\beta y_c}\,\mathfrak{G}_w(x;y),$$

where $E(u,k)$ is the set of endpoints of the Lenart–Sottile chains from u in the k-Bruhat order and Fix, Left, Out, m *are taken with respect to $(u, w; k)$.*

Proof. At $q = 0$ only chains without quantum edges contribute, since by Corollary 11(1) no endpoint is reached by two admissible chains; the contributing chains are the saturated chains in the k-Bruhat order satisfying (P0) and (P1), with $D = 0$ and $s = \ell(w) - \ell(u)$. So the displayed formula is (5) at $q = 0$, and as in the rest of this paragraph we take the quantum Bruhat graph to consist of Bruhat edges only. We induct on k. The case $k = 1$ is Proposition 17. For $k \geq 2$, by Proposition 22 at $q = 0$ it suffices to prove (28), that is (29) for every w. Fix w. If $w(k) = u(k)$, this is Lemma 35. Let $w(k) \neq u(k)$ and let N_w be the number of surviving words from u to w, a nonnegative integer that does not depend on y or z, since by Proposition 36(3) survival is a combinatorial condition on the word. Every surviving word to w has a label involving position k, by Proposition 37, so the right side of (29) equals $N_w\,\Phi^{(k)}(u,w)$.

We rewrite the left side. The sets $E(u,k)$ and $E(u,k-1)$ do not both contain w: along a k-chain the value in the window position k can only increase, while along a $(k-1)$-chain the value in the outside position k can only decrease, so $w \in E(u,k) \cap E(u,k-1)$ forces $w(k) = u(k)$. If $w \in E(u,k-1)$, its chain contains a label with upper index k, because $w(k) \neq u(k)$ and no other label touches position k; the first one, (α, k), moves $u(k)$ to position $\alpha < k$, and $u(k)$ stays there, since afterwards only labels with upper index k occur by condition 3, position α is touched only by the label (α, k), and labels are not repeated. Thus $u(k) \in \mathrm{Left}(u, w; k)$, and by (30) $a\,\widehat{C}_{u,w}^{(k-1)} = a\,\Phi^{(k-1)}(u,w) = \Phi^{(k)}(u,w)$. Therefore the left side of (29) equals $c_w\,\Phi^{(k)}(u,w)$ with $c_w = [w \in E(u,k)] + [w \in E(u,k-1)] \in \{0,1\}$, and we must show $N_w = c_w$.

For this we specialize to $y = 0$. Theorem 20 at $q = 0$ gives $B_j \mathfrak{G}_u(x;0) = P_j(u)|_{y=0}$ for $j = k-1$ and $j = k$, where $B_j = \prod_{i \leq j}(x_i + z)$. Together with $\beta B_k + a B_{k-1} = X_k B_{k-1}$ and Lemma 16 at $y = 0$, this yields (28) at $y = 0$, hence (29) at $y = 0$ for our w: $N_w\,\Phi^{(k)}(u,w)|_{y=0} = c_w\,\Phi^{(k)}(u,w)|_{y=0}$. By (30) the common factor is the nonzero polynomial $\beta^{\ell(w)-\ell(u)}(1-a)^{|\,\mathrm{Fix}\,|}a^{|\,\mathrm{Left}\,|}$, so $N_w = c_w$. Hence (29) holds for all y, which completes the induction. □

Corollary 39 (The matching)**.** *Let $q = 0$ and $w(k) \neq u(k)$. There is exactly one surviving word from u to w if $w \in E(u,k) \cup E(u,k-1)$, and none otherwise.*

Remark 40. *The proof counts the surviving words without constructing them. When $u(k) \in \mathrm{Out}(u, w; k)$ the surviving word is easy to describe: it is γ_w with its labels (k, b) moved to the end. Indeed the first label (k, b) of γ_w ends its run, as $u(k)$ would otherwise move left; every later label (k, b') ends its run by condition 4, since k is then a repeated lower index and $(k, b') \prec (a, b')$ is impossible for $a < k$. So each (k, b) commutes, as a transposition, with all labels after it; the rearranged word has the same product and the same length, hence is again a chain of covers, and its chain part $\gamma_{>k}$ is admissible, since a label with a repeated lower index that was followed by a removed (k, b) is now followed by a label with smaller upper index. When $u(k) \in \mathrm{Left}(u, w; k)$ the surviving word is related to γ_w, or to the $(k-1)$-chain when $w \in E(u, k-1)$, by braid relations such as $t_{kb}t_{ab} = t_{ak}t_{kb}$ and $t_{ab}t_{ak} = t_{ak}t_{kb}$ as well; for $u = 21435$, $k = 3$, $w = 43215$ the $(k-1)$-chain is $(1,4), (2,4), (1,3), (2,3)$ and the surviving word is $(1,4)\,|\,(1,3),(2,3)\,|\,(3,4)$. We have verified Corollary 39 and Proposition 37 directly for all $u \in S_5$ and $k \leq 5$.*

Theorem 38 is the classical equivariant case of Theorem 3; together with Theorems 20 and 24 it leaves open only the simultaneous presence of q, y, and a. The argument has a feature worth isolating: the word lemma shows that the y-dependence of every term that survives cancellation is the same, namely that of the endpoint weight, after which the integer multiplicity is fixed by the non-equivariant formula. The quantum junction identity has the same shape, with backtracks and the level-$(k-2)$ terms as additional ingredients, and a quantum word lemma of this kind would complete the proof in the same way.

Theorem 38 has a consequence in the x-variables, which has no quantum analogue. Since $\prod_{j\le k}(x_j+z)$ is symmetric in x_i, x_{i+1} for $i\neq k$, the isobaric divided difference π_i^x commutes with multiplication by it, and $\pi_i^x\mathfrak{G}_u=\mathfrak{G}_{us_i}$ when $us_i<u$, $\pi_i^x\mathfrak{G}_w=-\beta\mathfrak{G}_w$ when $ws_i>w$. Hence the classical coefficients satisfy

$$C_{us_i,w'}=\begin{cases} C_{u,w's_i}-\beta\, C_{u,w'} & w's_i>w',\\ 0 & w's_i<w', \end{cases}\qquad (us_i<u,\ i\neq k,\ q=0), \tag{31}$$

a statement about the chain combinatorics that we had verified directly for all $u\in S_4$ before Theorem 38 was available. The identity shows where the factor $(1-\beta z)^{|\,\mathrm{Left}\,|}$ comes from: when a window value c moves from position $i+1$ to position i, the two terms on the right combine as $(1+\beta y_c)^{-1}-\beta(z+\overline{y}_c)=1-\beta z$. Conversely, (31) would reduce the classical theorem by descending induction in the right weak order to the k-Grassmannian permutations u (no descents outside position k), for which all endpoints are again k-Grassmannian and the statement is an equivariant K-theoretic Pieri rule for the Grassmannian, multiplying by $\prod_{j\le k}(x_j+z)=(1-\beta z)^k\prod_j(x_j\oplus z')$ with $z'=z/(1-\beta z)$; we do not pursue this.

The case $k=2$. For $k=2$ a line path has at most one left edge, namely $(1,2)$, and the level-0 part of the junction input is $a\beta q_1\mathfrak{G}_u$. The cut composites are the sequences

$$(1,b_1),\dots,(1,b_r)\ \big|\ [(1,2)]\ (2,c_1),\dots,(2,c_t),$$

with the junction edge $(1,2)$ possibly absent, present on one side of the cut, or present on both sides. The last possibility is a *backtrack*: a chain ending with a quantum edge $(1,2)$ followed by the same edge reversed as a Bruhat line edge, or vice versa. Its contribution relative to the composite without the two edges is $\beta\cdot(-\beta)\cdot q_1=-\beta^2q_1$, and this is exactly how the factor $1-\beta^2q_1$ on the left of (20) is produced. For instance, for $u=21345$ and $w=32145\in E(u,2)$, the only terms are $ut_{13}\,\big|\,(2,3)$, with value $\beta t_1^{-1}t_2^{-1}$, and the backtrack $ut_{13}t_{12}\,\big|\,(1,2),(2,3)$, with value $-\beta^3q_1t_1^{-1}t_2^{-1}$, whose sum is $\beta(1-\beta^2q_1)C^{(2)}_{u,w}$. For $w=23145$ the backtrack $ut_{12}\,\big|\,(1,2),(2,3)$ has value $-\beta^2q_1t_1^{-1}t_2^{-1}$ and the level-0 composite $u\,\big|\,(2,3)$ has value $a\beta^2q_1t_1^{-1}$; their sum is $-\beta^3q_1(z+\overline{y}_2)t_1^{-1}=-\beta^2q_1\,\beta C^{(2)}_{u,w}$, again by $(1+\beta y_2)^{-1}-a=\beta(z+\overline{y}_2)$. Thus for $k=2$ all three mechanisms—cut sliding, backtracks, and the absorption of residues by the level-0 terms—are visible, while the braid move of Lemma 34 cannot occur. We have verified (21) for $k=2$ termwise for all $u\in S_5$. The diagonal case can be settled completely.

Proposition 41 (Diagonal case for $k=2$)**.** *Let $k=2$ and $w(2)=u(2)$. Then (23) holds; hence (21) holds at w.*

Proof. We classify the nonempty cut composites ending at w. Position 2 is touched by every edge of a line path at 2. A right edge $(2,c)$ sends the value in position 2 to the position $c>2$, which is never touched again, so if the line path contains a right edge, the final value in position 2 is not $u(2)$ unless $u(2)$ was brought to position c earlier—impossible, as c is touched only once. Hence the

line path is $\pi = (1,2)$, and the value $u(2)$ sits in position 1 of v, so the chain γ to v contains the label $(1,2)$. For a level-1 chain this label is the last one, as 2 is the least admissible second entry, and $(1,2)$ moves $u(2)$ to position 1 only if it is the first label whose second entry is 2; since labels are distinct, $\gamma = \gamma_0 \cdot (1,2)$ with $(1,2) \notin \gamma_0$, and $w = vt_{12} = u\gamma_0$. Thus γ_0 is the unique 1-chain from u to w (Proposition 17), and there is exactly one level-1 composite, the backtrack $\gamma_w \cdot (1,2) \,\big|\, (1,2)$. There is no level-0 composite, because a line path from u cannot restore $u(2)$ either. Write κ, κ' for the weights of the two traversals of $(1,2)$, so $\kappa\kappa' = q_1$.

If $w \neq u$, the window value $u(1)$ is in Out for both w and wt_{12} (it has left position 1 and $w(1) \neq u(1)$; also $w(2) = u(2) \neq u(1)$, so the hypotheses of the lemma hold), so by Lemma 33(2) $C^{(1)}_{u,wt_{12}} = \beta\kappa C^{(1)}_{u,w}$, and the backtrack contributes $(-\beta)\kappa'(1+\beta y_{u(2)})^{-1}\beta\kappa C^{(1)}_{u,w} = -\beta^2 q_1 (1+\beta y_{u(2)})^{-1} C^{(1)}_{u,w}$, while the drop term is $\beta q_1 (1+\beta y_{u(2)})^{-1}\beta C^{(1)}_{u,w}$ because $C^{(0)}_{u,w} = 0$. They cancel.

If $w = u$, then γ_0 is empty, $C^{(1)}_{u,u} = z + \overline{y}_{u(1)}$, and $u(1)$ passes from Fix to Out in ut_{12}, so $C^{(1)}_{u,ut_{12}} = \kappa(1+\beta y_{u(1)})^{-1}$ and the backtrack contributes $-\beta q_1 (1+\beta y_{u(1)})^{-1}(1+\beta y_{u(2)})^{-1}$. The drop term is $\beta q_1 (1+\beta y_{u(2)})^{-1}\big(\beta(z+\overline{y}_{u(1)}) + a\big) = \beta q_1 (1+\beta y_{u(1)})^{-1}(1+\beta y_{u(2)})^{-1}$. They cancel. □

The moved case for $k = 2$ requires in addition the classification of 2-chains: their labels form segments $(1,b)$, $(2,b)$, $(1,b)(2,b)$ or $(2,b)(1,b)$ for strictly decreasing b, the last form being allowed only at the first occurrence of the lower index 2. The computations through S_5 exhibit the following pattern, which we record because it illustrates the three mechanisms; the case $k = 2$ is proved in full, by a different route, in Proposition 46. If $w \in E(u,2)$ and its 2-chain has no segment of the form $(2,b)(1,b)$, then sorting the chain, all labels $(1,b)$ first and all labels $(2,c)$ afterwards, gives a cut composite $\gamma^* \,\big|\, \pi^*$ whose term is $\beta C^{(2)}_{u,w}$, and its backtrack $\gamma^*(1,2) \,\big|\, (1,2)\pi^*$ contributes $-\beta^3 q_1 C^{(2)}_{u,w}$; for instance $u = 24135$, $w = 31524$, with chain $(2,5),(1,4),(2,3)$ and composite $(1,4) \,\big|\, (2,5),(2,3)$. A segment $(2,b)(1,b)$ is where a window value moves left, and it is resolved by $t_{2b}t_{1b} = t_{1b}t_{12}$: for $u = e$ and $w = 231$, with chain $(2,3),(1,3)$, the composites are $(1,2) \,\big|\, (2,3)$ with term βt_1^{-2} and $\varnothing \,\big|\, (1,2),(2,3)$ with term $-\beta^2(z+\overline{y}_1)t_1^{-1}$, a cut sliding pair of type (3) whose residue $\beta a t_1^{-1}$ is exactly $\beta C^{(2)}_{e,231}$: the factor $1-\beta z$ attached to a value in Left is produced by $(1+\beta y_c)^{-1} - \beta(z+\overline{y}_c) = 1-\beta z$. The level-0 composite $e \,\big|\, (1,2),(2,3)$, with term $-a\beta^3 q_1 t_1^{-1}$, then cancels the part $\beta^3 q_1 C^{(2)}_{e,231}$ of the left side. When the 1-chain to w ends with $(1,2)$, the term $-a(1-\beta^2 q_1)C^{(1)}_{u,w}$ is cancelled by the residue of the pair $u \,\big|\, (1,2)\pi'$, $ut_{12} \,\big|\, \pi'$ together with the level-0 composite $u \,\big|\, (1,2)\pi'$, using $t_{12}t_{2c_1}\cdots t_{2c_t} = t_{1c_1}\cdots t_{1c_t}t_{12}$ and Postnikov's lemma for the equality of weights. The outstanding point is the validity in the quantum Bruhat graph of the sorted and braided sequences; Lemma 28 settles the sorting for chains without descents, and Lemma 27 is the local step for moving a right edge past a label with smaller upper index in general, but the braided segments $(2,b)(1,b)$ are not covered. This case analysis is superseded by Section 4.12, where the case $k = 2$ is proved in full (Proposition 46) by discarding the right edges first.

The y-recursion. Finally, there is a reduction that is valid for all q and does not use the Chevalley formula at all. The quantization map commutes with the isobaric divided differences π_i^y in the second alphabet, and $\pi_i^y \mathfrak{G}^q_w = \mathfrak{G}^q_{s_i w}$ if $s_i w < w$, $\pi_i^y \mathfrak{G}^q_w = -\beta \mathfrak{G}^q_w$ otherwise. Applying π_i^y to (5) for $s_i u > u$ and using $\pi_i(fg) = (1+\beta y_{i+1})\partial_i(f)g + s_i(f)\pi_i(g)$ shows that the theorem for $s_i u$ implies the theorem for u provided the coefficients satisfy

$$C_{u,w} = \pi_i^y\big(C_{s_i u,w}\big) + \begin{cases} s_i\big(C_{s_i u,\, s_i w}\big) & s_i w > w, \\ \beta\, s_i\big(C_{s_i u,w}\big) & s_i w < w. \end{cases} \tag{32}$$

We have verified (32) for the closed form, with q arbitrary, for all $u \in S_5$ and $k \le 3$. It reduces the theorem for all $u \in S_n$ to the single permutation $u = w_0^{(n)}$, for which the chains are indexed by subsets of $\{1, \dots, k\}$ in the classical case; the quantum case of this base is not simpler than the general one, which is why we rely on the Chevalley induction instead.

4.12 The general case: reduction to the backtrack identity

We now return to the junction identity with q, y and a all present, and show that the bootstrap of Theorem 38 goes through verbatim once a single identity is available. That identity involves no right edges, no endpoint set $E(u,k)$ and no equivariant bootstrap; it is proved below for $k = 2$, which completes the proof of Theorem 3 for $k = 2$, and at $a = 0$ and at $q = 0$ (Proposition 46). Section 4.13 reduces it further, to an identity on permutations of k letters, and proves it in one of the two resulting cases; Section 4.14 proves the other case, first when the prefix of the chain moves every position below $k-1$ and then in general.

Throughout, $k \ge 2$ and u are fixed. We extend the normalization of Section 4.11 to arbitrary q: for $j \ge 0$ and $v \in E(u,j)$,

$$\widehat{C}^{(j)}_{u,v} := \beta^j C^{(j)}_{u,v} = q^{D(u,v;j)} \beta^{s(u,v;j)} \prod_{i=1}^{j} \phi_i(u,v;j),$$

again by $\beta(z + \overline{y}_c) = t_c^{-1} - a$ and $m + |\operatorname{Fix}| = j$, and $\widehat{C}^{(j)}_{u,v} = 0$ for $v \notin E(u,j)$.

Left and right paths. A line path π at k from x (Definition 15) is the concatenation of its *left path* $\lambda = (a_1,k)\cdots(a_s,k)$ and its *right path* $(k,b_1)\cdots(k,b_t)$, and $v(\pi)$ is the endpoint of λ; conversely a left path from x followed by a right path from its endpoint is a line path. A *left composite* at level $j \in \{k-1, k-2\}$ is a pair (γ, λ) consisting of an admissible j-chain γ from u to some x and a left path λ at k from x; its *endpoint* is the endpoint v of λ, its *word* is the label sequence $\gamma\lambda$, and its weight is

$$\operatorname{wt}(\gamma,\lambda) := (a\beta^2 q_{k-1})^{[j=k-2]}\, (-\beta)^{|\lambda|} q^{D(\lambda)}\, t^{-1}_{v(k)}\, \widehat{C}^{(j)}_{u,x}.$$

Let $M_{u,v}$ be the sum of the weights of all left composites with endpoint v, and for a right path ϱ from v to w let $r_{v,w}$ be the sum of $\beta^{t(\varrho)} q^{D(\varrho)}$ over such paths. Multiplying (21) by β^{k-1} and splitting every line path into its left and right paths gives

$$(1 - \beta^2 q_{k-1})\, \widehat{C}^{(k)}_{u,w} = \sum_{v} M_{u,v}\, r_{v,w} - a(1 - \beta^2 q_{k-1})\, \widehat{C}^{(k-1)}_{u,w}. \tag{33}$$

The endpoint weight. For arbitrary v put

$$\Psi(u,v) := \prod_{i=1}^{k} \phi_i(u,v;k) + [\, v(k) = u(k) \,]\, a \prod_{i=1}^{k-1} \phi_i(u,v;k-1),$$

so that $\Psi(u,v) = \prod_{i \le k} \phi_i(u,v;k)$ if $v(k) \ne u(k)$ and $\Psi(u,v) = t^{-1}_{u(k)} \prod_{i<k} \phi_i(u,v;k-1)$ if $v(k) = u(k)$, because then $\phi_k(u,v;k) = t^{-1}_{u(k)} - a$. We say that a quantity is *rigid at* v if it equals $N\,\Psi(u,v)$ with N independent of y. Note that $\Psi(u,v)|_{y=0} = (1-a)^{|\operatorname{Fix}|} a^{|\operatorname{Left}|} \ne 0$, where the sets are taken at level k if $v(k) \ne u(k)$ and at level $k-1$ otherwise.

Lemma 42 (Right paths preserve the endpoint weight)**.** *Let v be the endpoint of a left composite and let ϱ be a right path at k from v to w. Then $\Psi(u,w) = \Psi(u,v)$, and $w(k) = u(k)$ only if ϱ is empty.*

Proof. In v the value $u(k)$ sits at a position $\le k$: a chain at level $k-1$ or $k-2$ and a left path touch position k only through labels (a,k), which move $u(k)$ to $a < k$, and from then on $u(k)$ is moved only by labels whose two positions are at most k, since in a chain the labels with upper index greater than k precede those with upper index at most k. The right path touches only the positions $k, b_1, \dots, b_t$ with $b_i > k$, and sends the value at k to b_1 and the value at b_i to b_{i+1} or, for $i = t$, to k. Hence if $v(k) = u(k)$ and $\varrho \neq \varnothing$ then $w(k) \neq u(k)$, and if $v(k) \neq u(k)$ then $u(k)$ is at a position $< k$ in both v and w. This proves the second assertion.

For $i < k$, let p be the position of $u(i)$ in v. If $p < k$ it is untouched by ϱ; if $p \ge k$ then $u(i)$ ends at a position $\ge k > i$, so its status is Out in v and in w. Thus $\phi_i(u,v;k-1) = \phi_i(u,v;k) = \phi_i(u,w;k)$ for all $i < k$. If $v(k) \neq u(k)$, then $u(k)$ is in Left for both v and w and both weights equal $\prod_{i \le k} \phi_i$. If $v(k) = u(k)$ and $\varrho = \varnothing$, then $w = v$. If $v(k) = u(k)$ and $\varrho \neq \varnothing$, then $u(k)$ ends at a position $> k$, so $\phi_k(u,w;k) = t_{u(k)}^{-1}$ and $\Psi(u,w) = t_{u(k)}^{-1} \prod_{i<k} \phi_i(u,w;k) = \Psi(u,v)$. □

Proposition 43 (Reduction to left rigidity)**.** *Let $k \ge 2$ and suppose that Theorem 3 holds at levels $k-1$ and $k-2$ for u. If $M_{u,v}$ is rigid at every v, then the junction identity (21) holds for u at every w, and hence Theorem 3 holds at level k for u.*

Proof. Write $M_{u,v} = N_v \Psi(u,v)$ with N_v independent of y. By Lemma 42,

$$\sum_v M_{u,v}\, r_{v,w} = \widetilde{N}_w\, \Psi(u,w), \qquad \widetilde{N}_w = \sum_v N_v\, r_{v,w},$$

with $\widetilde{N}_w$ independent of y. We show that the difference of the two sides of (33) is a y-free multiple of $\Psi(u,w)$. Since (33) holds at $y = 0$—Theorem 20 at levels $k, k-1, k-2$ together with Lemma 21 and Lemma 16 at $y = 0$ give (20) at $y = 0$, exactly as in the proof of Proposition 22—and $\Psi(u,w)|_{y=0} \neq 0$, the multiple is zero and (33) holds identically. By Proposition 22 this is Theorem 3 at level k.

Suppose $w(k) \neq u(k)$. Put $c_j = [w \in E(u,j)]\, q^{D(u,w;j)} \beta^{s(u,w;j)}$ for $j = k, k-1$. Then $\widehat{C}_{u,w}^{(k)} = c_k \prod_{i \le k} \phi_i(u,w;k)$. If $w \in E(u,k-1)$, the $(k-1)$-chain to w contains a label with upper index k, as nothing else touches position k; the first one, (α,k), moves $u(k)$ to position $\alpha < k$, and $u(k)$ stays there, because the later labels all have upper index k and the only one touching position α would be a repetition of (α,k). Thus $u(k) \in \mathrm{Left}(u,w;k)$, $\phi_k(u,w;k) = a$, and $a\, \widehat{C}_{u,w}^{(k-1)} = c_{k-1} \prod_{i \le k} \phi_i(u,w;k)$. Hence the difference of the two sides of (33) equals

$$\big((1-\beta^2 q_{k-1})(c_k + c_{k-1}) - \widetilde{N}_w\big)\, \Psi(u,w).$$

Suppose $w(k) = u(k)$. By (22), $\widehat{C}_{u,w}^{(k)} = (t_{u(k)}^{-1} - a)\widehat{C}_{u,w}^{(k-1)}$ and $\widehat{C}_{u,w}^{(k-1)} = c_{k-1} \prod_{i<k} \phi_i(u,w;k-1)$, so the difference of the two sides of (33) is $\big((1-\beta^2 q_{k-1})c_{k-1} - \widetilde{N}_w\big)\, t_{u(k)}^{-1} \prod_{i<k} \phi_i(u,w;k-1) = \big((1-\beta^2 q_{k-1})c_{k-1} - \widetilde{N}_w\big)\Psi(u,w)$. □

The proposition absorbs the diagonal case: (23) and Proposition 41 are no longer needed separately, since the bootstrap handles $w(k) = u(k)$ and $w(k) \neq u(k)$ alike. It also shows why the right edges can be ignored: they never change the y-dependence.

Distinct left composites. A level-$(k-1)$ left composite (γ, λ) is *distinct* if no label of λ occurs in γ, that is, if the labels of its word are pairwise distinct. Let $A_{u,v}$ be the sum of the weights of the distinct left composites with endpoint v.

Lemma 44 (Distinct left composites are classical)**.** *Let $\vartheta = u(k)$, $\rho = \rho_\vartheta$, and $\sigma_j(x) = \#\{i \le j : x(i) \le \vartheta\}$.*

1. *For a distinct left composite with word ω and endpoint v, every edge of the rotated word is a Bruhat edge, and $q^{D(\omega)} = q^{\sigma(v)-\sigma(u)}$.*

2. *Rotation is a bijection from the distinct left composites from u onto the left composites from ρu in the classical setting $q = 0$, and*

$$A_{u,v} = q^{\sigma(v)-\sigma(u)}\, \beta^{\ell(\rho v)-\ell(\rho u)}\, n_{\rho u,\rho v}\, \Psi(u,v),$$

where $n_{u',v'}$ is the number of surviving words from u' to v' with empty right part, in the sense of Proposition 36. In particular $A_{u,v}$ is rigid at every v.

Proof. The labels of $\omega = \gamma\lambda$ satisfy $a \le k-1 < b$, are pairwise distinct, and have weakly decreasing upper indices, since those of γ are at least k and those of λ equal k. Lemma 10 with $k-1$ in place of k shows that every rotated edge is Bruhat, and Lemma 26, with the same substitution, gives the degree. For (2), the rotated chain is an admissible $(k-1)$-chain of Bruhat covers from ρu by Corollary 11, the rotated left path is a left path of Bruhat covers, and conversely every classical left composite from ρu rotates back to a distinct left composite from u by Corollary 9 and the distinctness of the labels. The statuses $\mathrm{Fix}, \mathrm{Left}, \mathrm{Out}$ are determined by positions and are unchanged by rotation, and t_c becomes $t'_{\rho(c)}$ in the relabelled alphabet $y'_{\rho(c)} = y_c$. Hence the weight of a distinct left composite is $q^{\sigma(v)-\sigma(u)}$ times the weight, at $q = 0$ and in the alphabet y', of its rotation. At $q = 0$, the left composites with a given word ω are the cuts of ω in Proposition 36 with $r = 0$ right edges, their total is $\Sigma(\omega)$, and by Proposition 37 it equals $\beta^{|\omega|}\Psi$ for surviving words and 0 otherwise—both cases of that proposition give Ψ by its definition, and the proofs of both propositions use only the chain and block labels, never the right edges. Transporting back, using $\Psi(\rho u, \rho v)(y') = \Psi(u,v)(y)$ and $|\omega| = \ell(\rho v) - \ell(\rho u)$, gives the formula. □

The backtrack identity. By Lemma 44, $M_{u,v}$ is rigid if and only if $M_{u,v} - A_{u,v}$ is, and $M_{u,v} - A_{u,v}$ is the contribution of the level-$(k-1)$ left composites with a repeated label together with all level-$(k-2)$ left composites. We shall prove, in Sections 4.13 and 4.14, that this contribution is exactly $-\beta^2 q_{k-1} A_{u,v}$:

$$M_{u,v} = (1 - \beta^2 q_{k-1})\, A_{u,v} \qquad \text{for every } v. \tag{34}$$

We call (34) the *backtrack identity*. It is a statement about $(k-1)$-chains, $(k-2)$-chains and left edges only.

Theorem 45. *Let $k \ge 2$ and suppose Theorem 3 holds at levels $k-1$ and $k-2$ for u. If the backtrack identity* (34) *holds for (u,k), then Theorem 3 holds at level k for u. Consequently, if* (34) *holds for all u and all $k \ge 2$, then Theorem 3 holds in general.*

Proof. By (34) and Lemma 44, $M_{u,v}$ is rigid at every v, so Proposition 43 applies. The second statement is the induction on k, with Proposition 17 as the base and $P_{-1} = 0$, $P_0(u) = \mathfrak{G}_u^q$. □

Proposition 46. *The backtrack identity* (34) *holds (i) for $k = 2$ and every u; (ii) at $a = 0$; (iii) at $q = 0$. Consequently Theorem 3 holds for $k = 2$, for all u and all q, y, z.*

Proof. (iii) At $q = 0$ the level-$(k-2)$ composites vanish, both sides of (34) are sums over level-$(k-1)$ left composites, and no word has a repeated label: all block edges are Bruhat covers, so the value at position k strictly decreases along the block, and a repeated label (a, k) would return a larger value to position k.

(ii) At $a = 0$ only Left-free chains contribute and the level-$(k-2)$ composites vanish. The analysis in the proof of Theorem 24 applies to words without right edges: a distinct word has zero or two valid cuts, and in the latter case the two composites cancel, so $A_{u,v}$ is the sum over the composites $(\gamma, \varnothing)$ with γ free of upper index k; a word with a repeated label has a single valid cut and the block (26); those with $m \geq 3$ cancel in pairs under moving the backtrack; and the remaining ones, $(\gamma\,(k-1,k), (k-1,k))$, have weight $-\beta^2 q_{k-1}$ times the weight of $(\gamma, \varnothing)$ by Lemma 33(2), where at $a = 0$ the correction of Lemma 33(3) is absent.

(i) Let $k = 2$. The left paths at 2 are $\varnothing$ and $(1,2)$; a 1-chain is $(1,b_1)\cdots(1,b_s)$ with $b_1 > \cdots > b_s \geq 2$, so $(1,2)$ can only be its last label; and the only 0-chain is empty. Every v is the endpoint of at most one 1-chain γ_v. Fix v and let $v' = vt_{12}$, with κ and κ' the weights of the edges $v' \to v$ and $v \to v'$, so that $\kappa\kappa' = q_1$. The left composites with endpoint v are $(\gamma_v, \varnothing)$, $(\gamma_{v'}, (1,2))$, and, if $v \in \{u, ut_{12}\}$, the level-0 composites $(\varnothing, \varnothing)$ or $(\varnothing, (1,2))$. If neither v nor v' is in $E(u,1)$ there is nothing to prove, so by Lemma 33(1) one of the following holds.

If γ_v exists, does not contain $(1,2)$, and is nonempty, then $\gamma_{v'} = \gamma_v(1,2)$, and $(\gamma_{v'}, (1,2))$ is the only composite with a repeated label. Here $u(1)$ sits at a position ≥ 3 in v and $v(2) = u(2)$, so the hypotheses of Lemma 33(2) hold, and it gives $\widehat{C}^{(1)}_{u,v'} = \beta\kappa'\widehat{C}^{(1)}_{u,v}$, so $M_{u,v} = t^{-1}_{v(2)}\widehat{C}^{(1)}_{u,v}(1-\beta^2 q_1) = (1-\beta^2 q_1)A_{u,v}$.

If $v = u$, then $\gamma_u = \varnothing$, $\widehat{C}^{(1)}_{u,u} = t^{-1}_{u(1)} - a$, $\widehat{C}^{(1)}_{u,ut_{12}} = \beta\kappa' t^{-1}_{u(1)}$ by Lemma 33(3), and the three composites give

$$M_{u,u} = t^{-1}_{u(2)}\big((t^{-1}_{u(1)} - a) - \beta^2 q_1 t^{-1}_{u(1)} + a\beta^2 q_1\big) = (1-\beta^2 q_1)\, t^{-1}_{u(2)}(t^{-1}_{u(1)} - a) = (1-\beta^2 q_1)A_{u,u}.$$

If $\gamma_v = \gamma'(1,2)$ with $\gamma' \neq \varnothing$, then $\gamma_{v'} = \gamma'$, both composites are distinct, and $\widehat{C}^{(1)}_{u,v} = \beta\kappa\widehat{C}^{(1)}_{u,v'}$ by Lemma 33(2), applied at v' (where $u(1)$ sits at a position ≥ 3 and $v'(2) = u(2)$), makes their weights cancel: $M_{u,v} = A_{u,v} = 0$.

If $v = ut_{12}$, the composites are $((1,2), \varnothing)$, $(\varnothing, (1,2))$ at level 1, and $(\varnothing, (1,2))$ at level 0, with weights $\beta\kappa t^{-2}_{u(1)}$, $-\beta\kappa t^{-1}_{u(1)}(t^{-1}_{u(1)} - a)$ and $-a\beta^3 q_1 \kappa t^{-1}_{u(1)}$; thus $A_{u,v} = a\beta\kappa t^{-1}_{u(1)}$ and $M_{u,v} = (1-\beta^2 q_1)A_{u,v}$.

Finally, if $v \notin E(u,1)$ but $v' \in E(u,1)$, then $\gamma_{v'}$ contains $(1,2)$, for otherwise $\gamma_{v'}(1,2)$ would be a chain to v; so $v' = \mathrm{end}(\gamma'(1,2))$ and $v = \mathrm{end}(\gamma') \in E(u,1)$, a contradiction. This proves (i), and the last assertion follows from Theorem 45 with Proposition 17. □

Lemma 47 (Repeated labels)**.** *Let (γ, λ) be a level-$(k-1)$ left composite. Every label occurring twice in its word occurs once in γ and once in λ, and if some label occurs twice then $(k-1, k)$ occurs in the word.*

Proof. The labels of γ are distinct, and so are those of λ. Suppose the word has a repeated label, and let (a, k) be the first label of λ that already occurs in γ, at block index r of γ. The labels preceding this edge are pairwise distinct, so by Lemma 10 (with $k-1$ in place of k and $\vartheta = u(k)$) all preceding edges are Bruhat in the rotated order. Along a Bruhat edge (a', k) the value at position k is replaced by the smaller value at a', and no value at a position strictly between a' and k lies between the two. Let e be the value at k just before the chain label (a, k) and $e' < e$ the value just after it; then e sits at position a from then on, since only the label (a, k) touches position a. If $a = k-1$ we are done, so let $a < k-1$. All labels between the chain label (a, k) and the repeated

edge have upper index k, so if one of them touches position $k-1$ it is $(k-1,k)$ and we are done. Otherwise let c be the value at $k-1$, constant along this stretch. The chain label (a,k) is a Bruhat edge with $k-1$ strictly between a and k, so $c \notin (e',e)$. Every subsequent edge before the repeated one has lower index $a' < k-1$ and lowers the value at k from some f to some $f' < f$ with $c \notin (f',f)$; hence, if y denotes the value at k just before the repeated edge, $c \notin (y,e')$, and so $c \notin (y,e)$. But the repeated edge is a quantum edge in the rotated order, since the value e at a exceeds the value y at k, and a quantum edge (a,k) requires every value at a position strictly between a and k, in particular c, to lie in (y,e). This contradiction shows that position $k-1$ is touched, so $(k-1,k)$ occurs. □

Lemma 48 (The first family)**.** *Let (γ,λ) be a distinct level-$(k-1)$ left composite whose word does not contain $(k-1,k)$. Then $(\gamma\,(k-1,k),\ (k-1,k)\,\lambda)$ is a left composite, and:*

1. *if γ has a label with lower index $k-1$, its weight is $-\beta^2 q_{k-1}\,\mathrm{wt}(\gamma,\lambda)$;*
2. *otherwise γ is a $(k-2)$-chain, (γ,λ) is also a level-$(k-2)$ left composite, and the weights of the three composites (γ,λ) at both levels and $(\gamma(k-1,k),(k-1,k)\lambda)$ add up to $(1-\beta^2 q_{k-1})\,\mathrm{wt}(\gamma,\lambda)$.*

Every left composite with a repeated label whose chain ends with $(k-1,k)$, whose left path begins with $(k-1,k)$, and which has no other repeated label arises in this way, and so does every level-$(k-2)$ left composite whose word contains neither $(k-1,k)$ nor a label of upper index $k-1$. Consequently the backtrack identity (34) *holds if and only if it holds after discarding these composites together with the distinct composites whose word does not contain $(k-1,k)$; that is, if and only if the remaining left composites—the distinct ones containing $(k-1,k)$, the ones with a repeated label not of the form just described, and the level-$(k-2)$ ones containing $(k-1,k)$ or a label of upper index $k-1$—satisfy* (34) *among themselves.*

Proof. Let x be the endpoint of γ, $x' = xt_{k-1,k}$, and let κ,κ' be the weights of the edges $x \to x'$ and $x' \to x$, so $\kappa\kappa' = q_{k-1}$. By Lemma 33(1), $\gamma(k-1,k)$ is an admissible $(k-1)$-chain, and $(k-1,k)\lambda$ is a left path from x' because $(k-1,k) \notin \lambda$ and $k-1$ is the largest lower index. The new composite has the same endpoint v, one more chain label and one more line label, and its weight is $-\beta\kappa'\,(-\beta)^{|\lambda|}q^{D(\lambda)}t^{-1}_{v(k)}\widehat{C}^{(k-1)}_{u,x'}$. Since $(k-1,k) \notin \gamma$, the value $u(k-1)$ is not at position k in x. In case (1), $x(k-1) \neq u(k-1)$ and Lemma 33(2) gives $\widehat{C}^{(k-1)}_{u,x'} = \beta\kappa\,\widehat{C}^{(k-1)}_{u,x}$, whence the claim. In case (2), γ has no label with lower index $k-1$, so it is an admissible $(k-2)$-chain with the same data, and Lemma 33(3) gives $\widehat{C}^{(k-1)}_{u,x} = (t^{-1}_{u(k-1)} - a)\widehat{C}^{(k-2)}_{u,x}$ and $\widehat{C}^{(k-1)}_{u,x'} = \beta\kappa\,t^{-1}_{u(k-1)}\widehat{C}^{(k-2)}_{u,x}$. With $X = (-\beta)^{|\lambda|}q^{D(\lambda)}t^{-1}_{v(k)}\widehat{C}^{(k-2)}_{u,x}$ the three weights are $X(t^{-1}_{u(k-1)} - a)$, $-\beta^2 q_{k-1}X\,t^{-1}_{u(k-1)}$ and $a\beta^2 q_{k-1}X$, whose sum is $(1-\beta^2 q_{k-1})X(t^{-1}_{u(k-1)} - a)$.

Conversely, removing the two labels $(k-1,k)$ from a composite of the first kind gives a distinct composite with the same endpoint whose word avoids $(k-1,k)$, and a level-$(k-2)$ composite of the second kind is a distinct level-$(k-1)$ composite by Lemma 47, with a $(k-2)$-chain as chain part. The last assertion follows, since the composites discarded contribute $(1-\beta^2 q_{k-1})$ times the discarded part of $A_{u,v}$. □

Remark 49 (Evidence and structure for $k \geq 3$)**.** *We have verified* (34) *for all $u \in S_5$ and $2 \leq k \leq 5$, by evaluating both sides over a prime field at random values of β, q, y, a; by Theorem 45 this reproves Theorem 3 in that range. The same computations show that $M_{u,v}$ is rigid at every v, that $A_{u,v}$ agrees with the rotated classical sum of Lemma 44, and that* (34) *holds separately for each prefix: grouping the left composites by the labels of their chain with upper index greater than k, each group satisfies* (34) *on its own. Thus the identity is a statement about the positions $1,\dots,k$ alone; Section 4.13*

proves this (Proposition 53). The composites left over by Lemma 48 all involve the label $(k{-}1,k)$, and a second exact cancellation accounts for many of them: a composite $(\gamma'(k-1,k)\mu,\ (k-1,k)\nu)$ with $\mu \neq \varnothing$ and the level-$(k{-}2)$ composite $(\gamma'\mu',\nu)$, where μ' is μ with its upper indices k replaced by $k{-}1$, have opposite weights whenever both are directed paths, by the braid relation $t_{k-1,k}\,t_{ak} = t_{a,k-1}\,t_{k-1,k}$ applied along μ and $\kappa\kappa' = q_{k-1}$. Section 4.13 turns this cancellation into a bijection (Lemma 55) and isolates what remains for $k \geq 3$ as the identity (37)*, in which the directedness exceptions of the braid relation appear as the wild chains. For $k = 2$ neither phenomenon occurs, which is why the identity is immediate there. The reduction replaces the difficulties noted after Corollary 11 and in Section 4.10—that line paths do not rotate to Bruhat paths, the backtracks, and the diagonal case with q—by this one identity: left paths of distinct composites rotate together with the chain (Lemma 44), the right paths and the diagonal are handled by Lemma 42 and Proposition 43, and the backtracks are the content of* (34).

4.13 The block model and the braid reduction

This subsection reduces the backtrack identity (34) to a statement about permutations of k letters, proves it in one of the two resulting cases, and in the other case rewrites it as an identity, (37), that involves neither a nor the fixed value at position $k-1$ and in which no repeated label occurs. Three facts drive this: non-fresh window values never change status inside a chain (Lemma 52); the factor $1-\beta^2 q_{k-1}$ is $(1-\beta E)(1+\beta E)$ for the edge $E=(k-1,k)$, whose double traversal has weight q_{k-1} (Proposition 51); and the $(k-1)$-chains containing $(k-1,k)$ are in bijection with a class of $(k-2)$-chains by the braid relation of Lemma 34 (Lemma 55). Section 4.14 then recasts the identity in terms of edge operators and proves it.

The closed form of A. We first make Lemma 44 explicit. For $v \in E(u,k-1)$ let γ_v be its $(k-1)$-chain and call its labels with upper index k its *block*; the positions $i \leq k-1$ that are lower indices of no label of γ_v with upper index greater than k are *fresh*. Say that γ_v has a *fresh suffix* if every entry of the longest strictly decreasing suffix of its block (read by lower indices) is fresh.

Lemma 50. *For every v,*

$$A_{u,v} = [\,v \in E(u,k-1) \text{ and } \gamma_v \text{ has a fresh suffix}\,]\ \widehat{C}^{(k-1)}_{u,v}\ \psi_k(v), \qquad \psi_k(v) = \begin{cases} a & v(k) \neq u(k), \\ t^{-1}_{u(k)} & v(k) = u(k). \end{cases}$$

Proof. By Lemma 44, $A_{u,v} = q^{\sigma(v)-\sigma(u)}\beta^{\ell(\rho v)-\ell(\rho u)} n_{\rho u,\rho v}\Psi(u,v)$, and by Proposition 36(3) a word with empty right part survives exactly when $J=m$, that is when the whole word is an admissible chain, and its decreasing suffix is fresh. Since there is at most one chain from ρu to ρv (Corollary 11), $n_{\rho u,\rho v} \in \{0,1\}$, and it is 1 exactly when $v \in E(u,k-1)$ and γ_v, which has the same labels as its rotation, has a fresh suffix. In that case the rotated chain is a Bruhat chain, so $\beta^{\ell(\rho v)-\ell(\rho u)} = \beta^{s(u,v;k-1)}$ and, by Lemma 26, $q^{\sigma(v)-\sigma(u)} = q^{D(u,v;k-1)}$; finally $\Psi(u,v) = \prod_{i<k}\phi_i(u,v;k-1)\,\psi_k(v)$, because $u(k) \in \mathrm{Left}(u,v;k)$ when $v(k) \neq u(k)$ (proof of Proposition 43). □

Where the factor $1-\beta^2 q_{k-1}$ comes from. For a formal combination $\Theta = \sum_v \theta_v \mathfrak{G}^q_v$ with x-free coefficients write

$$\mathcal{U}_k\Theta = \sum_v \theta_v \sum_\lambda \frac{(-\beta)^{|\lambda|}q^{D(\lambda)}}{1+\beta y_{\mathrm{end}(\lambda)(k)}}\mathfrak{G}^q_{\mathrm{end}(\lambda)}, \qquad \mathcal{R}_k\Theta = \sum_v \theta_v \sum_\varrho \beta^{|\varrho|}q^{D(\varrho)}\mathfrak{G}^q_{\mathrm{end}(\varrho)},$$

the sums over left paths λ and right paths ϱ at k from v, so that $\mathcal{T}_k = \mathcal{R}_k\mathcal{U}_k$ and $M_{u,v}$ is the coefficient of $\mathfrak{G}^q_v$ in $\mathcal{U}_k\widehat{H}_{k-1}$, where $\widehat{H}_{k-1} = \widehat{P}_{k-1} + a\beta^2 q_{k-1}\widehat{P}_{k-2}$ and $\widehat{P}_j = \sum_x \widehat{C}^{(j)}_{u,x}\mathfrak{G}^q_x$. Let $\mathcal{R}'_{k-1}$ be defined like $\mathcal{R}_{k-1}$ but with the right paths at $k-1$ restricted to upper indices $b > k$, let $\mathcal{U}'_k$ be defined like $\mathcal{U}_k$ but with the left paths at k restricted to lower indices $a \le k-2$, and put $\mathcal{V}_k = \mathcal{U}'_k\mathcal{R}'_{k-1}$. Finally let $A^{(j)}$ denote the vector of Lemma 50 at level j, that is $A^{(k)} = \sum_v A_{u,v}\mathfrak{G}^q_v$ and $A^{(k-1)}$ the same with $k-1$ in place of k.

Proposition 51. $\mathcal{U}_k\mathcal{T}_{k-1} = (1-\beta^2 q_{k-1})\,\mathcal{V}_k\mathcal{U}_{k-1}$. *Consequently, if Theorem 3 holds at levels $\le k-1$ for u and* (34) *holds for $(u, k-1)$, then* (34) *holds for (u,k) if and only if*

$$\mathcal{V}_k A^{(k-1)} = A^{(k)} + a\,\mathcal{U}_k\widehat{P}_{k-2}. \tag{35}$$

Proof. A right path at $k-1$ ends with $(k-1,k)$ or not, and a left path at k begins with $(k-1,k)$ or not; the edge $(k-1,k)$ exists from every permutation, and its two traversals have weights with product q_{k-1}. Writing E for the operator $\mathfrak{G}^q_x \mapsto \kappa_x\mathfrak{G}^q_{xt_{k-1,k}}$, we have $\mathcal{R}_{k-1} = (1+\beta E)\mathcal{R}'_{k-1}$, $\mathcal{U}_k = \mathcal{U}'_k(1-\beta E)$ and $E^2 = q_{k-1}$, whence $\mathcal{U}_k\mathcal{T}_{k-1} = \mathcal{U}'_k(1-\beta^2E^2)\mathcal{R}'_{k-1}\mathcal{U}_{k-1}$. For the second statement, the two-step identities of Lemma 21 give $\beta H_{k-1} = T_{k-1}H_{k-2}/(1-\beta^2 q_{k-1}) - aB^q_{k-2}$, so with $\mathbf{h}_j := \beta^j H_j\mathfrak{G}^q_u = \widehat{H}_j/(1-\beta^2 q_j)$ (by (18) and the Theorem at levels $j, j-1$) and Lemma 16, $\mathbf{h}_{k-1} = \mathcal{T}_{k-1}\mathbf{h}_{k-2}/(1-\beta^2 q_{k-1}) - a\widehat{P}_{k-2}$. Now (34) for (u,k) says $\mathcal{U}_k\mathbf{h}_{k-1} = A^{(k)}$, and (34) for $(u,k-1)$ says $\mathcal{U}_{k-1}\mathbf{h}_{k-2} = A^{(k-1)}$; by the operator identity, $\mathcal{U}_k\mathbf{h}_{k-1} = \mathcal{V}_k\mathcal{U}_{k-1}\mathbf{h}_{k-2} - a\,\mathcal{U}_k\widehat{P}_{k-2} = \mathcal{V}_kA^{(k-1)} - a\,\mathcal{U}_k\widehat{P}_{k-2}$. □

Identity (35) contains no factor $1-\beta^2 q_{k-1}$ and no repeated edge $(k-1,k)$; all three terms are explicit by Lemma 50. We have verified it for all $u \in S_5$ and $2 \le k \le 5$. We do not use it below, but it explains the shape of (34): the two traversals of the junction edge that are hidden in the product $\mathcal{R}_{k-1}\mathcal{U}_k$ account for the factor $1-\beta^2 q_{k-1}$.

Non-fresh values are inert. Fix u and k, and an admissible $(k-1)$-chain η from u all of whose labels have upper index greater than k; every left composite at level $k-1$ or $k-2$ whose chain begins with the labels of upper index $> k$ equal to η will be said to have *prefix* η. Put $x_0 = u\eta$ and let $F = \{p \le k-1 :\ x_0(p) = u(p)\}$ be the set of fresh positions; by Corollary 11(2) these are the positions that are lower indices of no label of η. Position k is never touched by η. Note that level-$(k-2)$ composites with prefix η exist only if η is a $(k-2)$-chain, that is if $k-1 \in F$.

Lemma 52 (Inertness)**.** *Let (γ,λ) be a left composite at level $j \in \{k-1, k-2\}$ with prefix η, and let $i \le j$ with $x_0(i) \ne u(i)$. Then the status of $u(i)$ (Left or Out) is the same in x_0 and in the endpoint of γ. More precisely, if $u(i)$ is at a position $p < i$ in x_0, then no label of γ after η has lower index p; if it is at a position $p > i$, $p \le k-1$, the labels of γ after η move it only to positions greater than p; and if it is at a position greater than k, it is not moved.*

Proof. The last case is clear, since the labels after η have upper index k or $k-1$. Rotate by ρ_ϑ with $\vartheta = u(j+1)$; by Corollary 11, γ becomes a saturated chain in the j-Bruhat order satisfying (P0) and (P1), with the same labels, and statuses are determined by positions, so we may argue in the rotated chain. Let $p_0 = i, b_1, p_1, b_2, \dots, b_r, p_r = p$ be the successive positions of $u(i)$ along η, so that η contains the labels (p_{s-1}, b_s) and (p_s, b_s) in this order, with $b_1 > b_2 > \cdots > b_r > k$ by condition 3 and the distinctness of labels. Suppose $p < i$. Then some s has $p_{s-1} > p_s$; take the largest. The prohibition lemma [6, Lemma 2.9] states that a chain satisfying (P0) and (P1) contains no subsequence $(j',m), \dots, (i',m), \dots, (i',l)$ with $i' < j' \le j < l < m$. Applied to $(p_{s-1}, b_s), (p_s, b_s)$ it shows that no later label has lower index p_s and upper index $< b_s$; every later label has upper

index $\le b_s$, and (p_s, b_s) itself is not repeated. If $s < r$ the label (p_s, b_{s+1}) contradicts this, so $s = r$, and no label after (p_r, b_r), in particular none after η, has lower index p. Suppose $p > i$. Position p is not fresh, so p is a repeated lower index at every later label with lower index p, and by condition 4 the label following such a label (p, b) is (p', b) with $p' > p$ or has a smaller upper index. If (p, k) occurs after η, it moves $u(i)$ to position k; the next label is (p', k) with $p' > p$, which moves $u(i)$ to p', or has upper index $k-1$, after which position k is not touched. Position p' can be touched again only by $(p', k-1)$, which moves $u(i)$ to $k-1$; then p' is repeated, so the next label is $(p'', k-1)$ with $p'' > p'$, which moves $u(i)$ to p'', a position touched by no later label. If instead $(p, k-1)$ occurs, the same argument applies from position $k-1$. In all cases $u(i)$ only moves to positions greater than $p > i$, so it remains Out at the endpoint of γ. □

The block model. Let ξ be a sequence of k distinct positive integers, regarded as the values at positions $1, \dots, k$, and let $F \subseteq \{1, \dots, k-1\}$. The quantum Bruhat graph on these k positions is the one of S_k, with the edge condition of Lemma 8 applied to the values of ξ. A *block chain at level* $j \in \{k-1, k-2\}$ is a directed path from ξ with labels $(a_1, b_1), \dots, (a_m, b_m)$, $a_i \le j < b_i \le k$, that are pairwise distinct, have weakly decreasing upper indices, and satisfy: if $i < m$ and $a_i \notin F$ or $a_i = a_h$ for some $h < i$, then $(a_i, b_i) \prec (a_{i+1}, b_{i+1})$. Its weight is

$$\widehat{C}_F(\mu) = \beta^m q^{D(\mu)} \prod_{p \in F,\ p \le j} \phi_p(\xi, x), \qquad \phi_p(\xi, x) = \begin{cases} t_{\xi(p)}^{-1} - a & x(p) = \xi(p), \\ a & \xi(p) \text{ is at a position } < p \text{ in } x, \\ t_{\xi(p)}^{-1} & \text{otherwise,} \end{cases}$$

where x is the endpoint of μ; level $k-2$ is used only when $k-1 \in F$. A *block composite* is a pair (μ, λ) of a block chain and a left path at k from its endpoint, with weight

$$(a\beta^2 q_{k-1})^{[j=k-2]}\, (-\beta)^{|\lambda|} q^{D(\lambda)}\, t_{v(k)}^{-1}\, \widehat{C}_F(\mu),$$

v the endpoint of λ; it is *distinct* if it is at level $k-1$ and no label of λ occurs in μ. Let $M_{\xi,v}^F$ and $A_{\xi,v}^F$ be the sums of the weights of all, respectively of the distinct, block composites with endpoint v. The *block identity* is

$$M_{\xi,v}^F = (1 - \beta^2 q_{k-1})\, A_{\xi,v}^F \qquad \text{for all } v \in S_k. \tag{36}$$

Proposition 53 (Reduction to the block model)**.** *Let η, x_0 and F be as above and let ξ be the restriction of x_0 to positions $1, \dots, k$. The block chains for (ξ, F) are exactly the parts after η of the admissible chains from u with prefix η, and for every v the sum of the weights of the left composites with prefix η and endpoint v equals $c_\eta\, M_{\xi,v'}^F$, where v' is the restriction of v to positions $1, \dots, k$ and*

$$c_\eta = \beta^{|\eta|} q^{D(\eta)} \prod_{i \le k-1,\ i \notin F} \phi_i(u, x_0; k-1);$$

the same holds for the distinct composites, with A in place of M. Consequently, if (36) *holds for all (ξ, F), then the backtrack identity* (34) *holds for all u and k, and Theorem 3 follows. Conversely, every pair (ξ, F) arises from some (u, η) in this way.*

Proof. The labels after η have upper index at most k, so they are distinct from those of η, and the conditions of Definition 2 for them reduce to the conditions on block chains: a lower index a_i is repeated in the whole chain exactly when $a_i \notin F$ or a_i occurs earlier after η. Directedness of a label (a, b) with $a < b \le k$ depends only on the values at positions $a, \dots, b$. The weight of a left composite with prefix η is $\beta^{|\eta|} q^{D(\eta)}$ times the block weight, except for the status factors $\phi_i(u, x; j)$ of the indices

$i \le j$ with $i \notin F$, which by Lemma 52 equal $\phi_i(u, x_0; k-1)$ at every chain endpoint x, and which for $j = k-2$ run over the same set of indices because $k-1 \in F$; the factor $t_{v(k)}^{-1}$ is the same on both sides, and so is distinctness. Every left composite (γ, λ) has exactly one prefix, namely the initial segment of γ formed by its labels of upper index greater than k (an initial segment by condition 3), so the left composites with endpoint v are partitioned according to their prefix η. Those with prefix η exist only if $u\eta$ agrees with v outside positions $1, \dots, k$, since block chains and left paths do not move the other positions, and for such η they correspond bijectively, by what was just shown, to the block composites for (ξ_η, F_η) with endpoint v', with weights related by the factor c_η. Summing over η gives $M_{u,v} = \sum_\eta c_\eta M^{F_\eta}_{\xi_\eta, v'}$ and likewise for A, so (36) for all (ξ_η, F_η) implies (34), and Theorem 45 applies. For the converse, given (ξ, F) with $\xi \in S_k$ after standardization, let $u(i) = 2\xi(i) - 1$ for $i \le k$, enumerate the positions $p \notin F$ as $p_1, \dots, p_r$, put $u(k + r + 1 - s) = 2\xi(p_s)$, and fill the remaining positions with the remaining values. Then $\eta = (p_1, k+r), (p_2, k+r-1), \dots, (p_r, k+1)$ is a chain of Bruhat covers (each label exchanges the adjacent values $2\xi(p_s) - 1$ and $2\xi(p_s)$) with distinct lower indices, hence admissible, and $u\eta$ restricted to positions $\le k$ has the relative order of ξ, with fresh set F. □

As a consistency check (Remark 6), we have verified (36) for every $\xi \in S_k$ and every F for $k \le 5$, and for a random sample of configurations with $k = 6$; by Proposition 53 this covers all $u \in S_\infty$ for $k \le 5$. The proof for all k occupies the rest of this section and the next.

Proposition 54 (The block identity when position $k-1$ is not fresh)**.** *If $k-1 \notin F$, then* (36) *holds.*

Proof. There are no level-$(k-2)$ composites. If $(k-1, k)$ occurs in a block chain it is the last label, since $k-1$ is a repeated lower index and no label (a, k) with $a > k-1$ exists. By Lemma 33(1), $\mu \mapsto \mu(k-1, k)$ is a bijection from the block chains not containing $(k-1, k)$ onto those containing it, and $\widehat{C}_F(\mu(k-1, k)) = \beta\kappa \widehat{C}_F(\mu)$ with κ the weight of the edge $(k-1, k)$ from the endpoint of μ: no fresh value sits at position $k-1$, and a fresh value at position k moves to $k-1$, where it is still Out. Write x for the endpoint of μ and $x' = x t_{k-1,k}$, with κ' the weight of $x' \to x$, $\kappa\kappa' = q_{k-1}$. The block composites are of four kinds: (i) (μ, λ) with $(k-1, k) \notin \mu\lambda$; (ii) $(\mu, (k-1, k)\lambda')$ with $(k-1, k) \notin \mu$; (iii) $(\mu(k-1, k), \lambda')$ with λ' a left path from x' avoiding $k-1$; (iv) $(\mu(k-1, k), (k-1, k)\lambda)$. In (ii) and (iii), λ' is a left path from x' with lower indices $\le k-2$, and the two weights are $-\beta\kappa$ and $\beta\kappa$ times the same quantity, so (ii) and (iii) cancel in pairs, also after restricting to distinct composites. In (iv), λ is a left path from x, and the weight is $\beta\kappa \cdot (-\beta)\kappa' = -\beta^2 q_{k-1}$ times the weight of the composite (μ, λ) of kind (i). The composites of kind (i) are distinct by Lemma 47, those of kind (iv) are not, so $M^F_{\xi,v} = (1 - \beta^2 q_{k-1}) \sum_{(\mathrm{i})}$ and $A^F_{\xi,v} = \sum_{(\mathrm{i})}$. □

In the language of Proposition 53, the backtrack identity therefore holds prefix by prefix whenever the prefix η moves position $k-1$. From now on $k-1 \in F$, and we write $t^* = t_{\xi(k-1)}$.

The braid bijection. A block chain at level $k-2$ is $\mu\nu$ with μ the labels of upper index k and ν those of upper index $k-1$. Let $\rho = \rho_\vartheta$ with $\vartheta = \xi(k)$. Call $\mu\nu$ *tame* if no lower index of ν occurs in μ and every edge of ν is a Bruhat edge after rotation by ρ (in particular if $\nu = \varnothing$), and *wild* otherwise. For a sequence ν of labels of upper index $k-1$ let $\tilde{\nu}$ be the same sequence with upper index k.

Lemma 55 (Braid bijection)**.** *Let $k-1 \in F$. The map $\mu(k-1, k)\tilde{\nu} \mapsto \mu\nu$ is a bijection from the block chains at level $k-1$ containing $(k-1, k)$ onto the tame block chains at level $k-2$. If*

$\gamma = \mu(k-1,k)\tilde{\nu}$ corresponds to $c = \mu\nu$ with endpoint x, then γ ends at $xt_{k-1,k}$, every edge of γ is Bruhat after rotation by ρ, and

$$\widehat{C}_F(\gamma) = \beta\kappa_x\, \widehat{C}_F(c) \cdot \begin{cases} a & \nu \neq \varnothing, \\ t^* & \nu = \varnothing, \end{cases}$$

where κ_x is the weight of the edge $x \to xt_{k-1,k}$.

Proof. A level-$(k-1)$ block chain containing $(k-1,k)$ has the form $\mu(k-1,k)\tilde{\nu}$ with all lower indices of μ and $\tilde{\nu}$ at most $k-2$, since labels are distinct. Admissibility of $\mu(k-1,k)\tilde{\nu}$ and of $\mu\nu$ are equivalent: labels are distinct in both; upper indices are weakly decreasing in both; the label after the last label (a,k) of μ is $\succ (a,k)$ in both (it is $(k-1,k)$, respectively a label of upper index $k-1$); at $(k-1,k)$ no condition is imposed because $k-1 \in F$; and inside $\tilde{\nu}$ and ν the repeated lower indices and the required inequalities between consecutive lower indices are the same. The two label sequences have the same product of transpositions, by the braid relation $t_{k-1,k}t_{bk} = t_{b,k-1}t_{k-1,k}$ applied to each label of ν in turn.

Directedness. Let x_μ be the endpoint of μ and $c^* = \xi(k-1)$, $c_\mu = x_\mu(k)$. The path $\mu(k-1,k)$ is directed, has distinct labels of upper index k, so by Lemma 10 (with $k-1$ in place of k) its edges are Bruhat after rotation; hence $c^* < c_\mu$ in the rotated order. Suppose γ is directed; then all its edges are rotated-Bruhat, again by Lemma 10. Along $\tilde{\nu}$ the value at position k decreases from c^*, and the cover condition for a label (b,k) involves the positions $b+1,\dots,k-1$, where position $k-1$ holds c_μ, which exceeds every value that position k takes along $\tilde{\nu}$. Therefore the same label $(b,k-1)$, applied in $\mu\nu$ at the corresponding moment, exchanges the same two values, with the same intermediate values except that c_μ is no longer among them; it is a rotated-Bruhat edge, and $\mu\nu$ is tame. Conversely, if $\mu\nu$ is tame the same comparison shows that each label (b,k) of $\tilde{\nu}$ is a Bruhat cover after rotation, the only additional intermediate value being c_μ, which lies above the interval in question. By Corollary 9 the two paths are directed in the quantum Bruhat graph, and by Lemma 34, applied at each step, they have the same quantum weight up to the weight κ_x of the final edge $x \to xt_{k-1,k}$ of the braided form $\mu\nu(k-1,k)$; the endpoints agree.

Weights. The chain γ has one more label than c, which gives the factor β, and $q^{D(\gamma)} = \kappa_x q^{D(c)}$ was just shown. For $p \in F$, $p \le k-2$, the value $\xi(p)$ occupies the same position in the endpoints of γ and of c unless it is at position $k-1$ or k, where it is Out in both. The value c^* is moved by $(k-1,k)$ to position k and, if $\tilde{\nu} \neq \varnothing$, by its first label (b_1,k) to position $b_1 < k-1$, where it stays; so ϕ_{k-1} is a if $\nu \neq \varnothing$ and t^* otherwise. □

The reduction. For a block chain c at level $k-2$ and a left path λ at k from its endpoint put

$$\varpi(c,\lambda) = (-\beta)^{|\lambda|} q^{D(\lambda)}\, t_{v(k)}^{-1}\, \widehat{C}_F(c),$$

the weight of the level-$(k-2)$ composite without its prefactor $a\beta^2 q_{k-1}$. Let

$$X_v = \sum_{c \text{ wild}} \sum_{\lambda} \varpi(c,\lambda),$$

$$Y_v = \sum_{c \text{ tame}} \sum_{\substack{\lambda = (k-1,k)\lambda' \\ \lambda' \text{ shares a lower index with } c}} \varpi(c,\lambda),$$

both over composites with endpoint v.

Theorem 56 (Braid reduction). *Let $k-1 \in F$. For every $v \in S_k$,*

$$M^F_{\xi,v} - (1-\beta^2 q_{k-1}) A^F_{\xi,v} = a\big(\beta^2 q_{k-1} X_v - (1-\beta^2 q_{k-1}) Y_v\big).$$

Hence the block identity (36) *holds for (ξ, F) if and only if*

$$\beta^2 q_{k-1}\, X_v = (1-\beta^2 q_{k-1})\, Y_v \qquad \text{for all } v \in S_k. \tag{37}$$

Proof. We list all block composites with endpoint v according to the level-$(k-2)$ chain $c = \mu\nu$ they are built from. Write x for the endpoint of c, $x' = x t_{k-1,k}$, $\kappa = \kappa_x$, κ' for the weight of $x' \to x$, so $\kappa\kappa' = q_{k-1}$, and note that a left path from x' beginning with $(k-1,k)$ is $(k-1,k)\lambda$ with λ a left path from x of lower indices $\le k-2$, and conversely.

(a) Level $k-2$: the composite (c,λ), of weight $a\beta^2 q_{k-1}\varpi(c,\lambda)$, for every c and λ.

(b) Level $k-1$, chain without $(k-1,k)$: such a chain is a level-$(k-2)$ chain c with $\nu = \varnothing$, and $\widehat{C}_F(c)$ at level $k-1$ equals $(t^*-a)\widehat{C}_F(c)$ at level $k-2$. The composite (c,λ) has weight $(t^*-a)\varpi(c,\lambda)$ for every left path λ from x.

(c) Level $k-1$, chain $\gamma = \mu(k-1,k)\tilde{\nu}$ containing $(k-1,k)$: by Lemma 55, γ corresponds to a tame c and ends at x'. If the left path avoids $k-1$, it is a left path λ from x' with lower indices $\le k-2$, and the weight is $\beta\kappa\,\theta\,\widehat{C}_F(c)(-\beta)^{|\lambda|}q^{D(\lambda)}t^{-1}_{v(k)} = -\theta\,\varpi(c,(k-1,k)\lambda)$, where $\theta = a$ if $\nu \neq \varnothing$ and $\theta = t^*$ otherwise. If the left path is $(k-1,k)\lambda$, then λ is a left path from x and the weight is $\beta\kappa\theta\widehat{C}_F(c)\cdot(-\beta)\kappa'\cdot(-\beta)^{|\lambda|}q^{D(\lambda)}t^{-1}_{v(k)} = -\beta^2 q_{k-1}\theta\,\varpi(c,\lambda)$.

Now fix c and a left path λ from x and collect the terms. If c is wild, only (a) contributes: $a\beta^2 q_{k-1}\varpi(c,\lambda)$. If c is tame with $\nu \neq \varnothing$ and λ avoids $k-1$, (a) and the second part of (c) give $a\beta^2 q_{k-1}\varpi - a\beta^2 q_{k-1}\varpi = 0$; if $\lambda = (k-1,k)\lambda'$, (a) and the first part of (c) give $a\beta^2 q_{k-1}\varpi - a\varpi = -a(1-\beta^2 q_{k-1})\varpi(c,\lambda)$. If $\nu = \varnothing$ and λ avoids $k-1$, (a), (b) and the second part of (c) give $(a\beta^2 q_{k-1} + (t^*-a) - \beta^2 q_{k-1}t^*)\varpi = (1-\beta^2 q_{k-1})(t^*-a)\varpi(c,\lambda)$; if $\lambda = (k-1,k)\lambda'$, they give $(a\beta^2 q_{k-1} + (t^*-a) - t^*)\varpi = -a(1-\beta^2 q_{k-1})\varpi(c,\lambda)$.

The distinct composites are: those of (b) whose left path avoids $k-1$, all of which are distinct by Lemma 47; those of (b) with $\lambda = (k-1,k)\lambda'$ such that λ' shares no lower index with c; and those of the first part of (c) whose λ shares no lower index with μ or ν. The second part of (c) repeats $(k-1,k)$. Hence $(1-\beta^2 q_{k-1})A^F_{\xi,v}$ is, for $\nu = \varnothing$ and λ avoiding $k-1$, $(1-\beta^2 q_{k-1})(t^*-a)\varpi$; for $\nu = \varnothing$ and $\lambda = (k-1,k)\lambda'$ with λ' sharing no index with c, $(1-\beta^2 q_{k-1})(t^*-a-t^*)\varpi = -a(1-\beta^2 q_{k-1})\varpi$; for c tame with $\nu \neq \varnothing$ and $\lambda = (k-1,k)\lambda'$ sharing no index, $-a(1-\beta^2 q_{k-1})\varpi$; and 0 otherwise. Subtracting, every term cancels except $-a(1-\beta^2 q_{k-1})\varpi(c,\lambda)$ for tame c and $\lambda = (k-1,k)\lambda'$ with λ' sharing a lower index with c, and $a\beta^2 q_{k-1}\varpi(c,\lambda)$ for wild c. □

Remark 57 (The reduced identity). *Identity* (37) *is the remaining content of the backtrack identity, hence of Theorem 3, for $k \ge 3$. Its left side runs over level-$(k-2)$ chains that are wild, that is whose $(k-1)$-labels either reuse a lower index of μ or contain an edge that is quantum after rotation by $\xi(k)$, followed by arbitrary left paths; its right side runs over tame chains followed by $(k-1,k)$ and a left path that returns to a lower index of the chain. Neither a nor t^* appears. For $k=2$ there are no labels of upper index $k-1$ and no wild chains, both sides vanish, and Proposition 46(i) is recovered. We have verified* (37) *for all $\xi \in S_k$ and all $F \ni k-1$ with $k \le 5$. The same computations show that X_v and Y_v are each rigid at v in the sense of Section 4.12—each is a y-free multiple of $\Psi(\xi,v)$—although neither is rigid word by word; the rigidity is produced by cancellations between words related by the braid relation $(k-1,k)(a,k) = (a,k-1)(k-1,k)$, the three forms $(a,k)(a,k-1) = (a,k-1)(k-1,k) = (k-1,k)(a,k)$ of a 3-cycle, and the commutation $(a,k)(b,k-1) = (b,k-1)(a,k)$ for $a \neq b$. For instance, for $\xi = 213$, $F = \{1,2\}$, $k=3$ and $v = 123$, the wild chains are $(1,2)$ and $(1,3)(1,2)$, with $X_v = \beta q_1(1-\beta^2 q_2)t_2^{-1}t_3^{-1}$ coming from the*

two composites $((1,2),\varnothing)$ *and* $((1,3)(1,2),(2,3))$*, the second being the first with a backtrack inserted through the 3-cycle, and* $Y_v = \beta^3 q_1 q_2 t_2^{-1} t_3^{-1}$ *comes from the single composite* $((1,3),(2,3)(1,3))$*. Identity* (37) *is proved in Section 4.14 (Theorem 77), which completes the proof of Theorem 3.*

The same computations ($k \le 5$) *exhibit a finer structure. Both sides of* (37) *are polynomials in* q*, so* (37) *splits according to the* q*-degree* D *of the word* $c\lambda$ *(with the convention that the factor* $\beta^2 q_{k-1}$ *is counted in* D*): writing* $X_{v,D}$*,* $Y_{v,D}$ *for the parts of degree* D*, it reads* $Y_{v,D} = \beta^2(X_{v,D-e_{k-1}} + Y_{v,D-e_{k-1}})$*. Numerically,* $X_{v,D}$ *and* $Y_{v,D}$ *are each* y*-free multiples of* $\Psi(\xi, v)$*, and for every* v *there are a sign* $\epsilon_v \in \{0, \pm 1\}$ *and a degree* D_v *such that* $X_v = \epsilon_v(1 - \beta^2 q_{k-1})\beta^m q^{D_v}\Psi(\xi, v)/a$ *and* $Y_v = \epsilon_v \beta^2 q_{k-1}\beta^m q^{D_v}\Psi(\xi, v)/a$*, with* $m = \ell(v) - \ell(\xi) + 2|D_v|$*; that is,* $X_v + Y_v$ *is a single monomial in* q*. When* $v(k) = \xi(k)$ *the monomial is the weight of the unique level-*$(k-2)$ *chain to* v *(and* $\epsilon_v = 1$ *exactly when that chain is wild); when* $v(k) \ne \xi(k)$ *both signs occur.*

4.14 Operator form, Yang–Baxter relations, and the proof of the block identity

In this subsection we rewrite the block identity as an identity between operators built from the edges of the quantum Bruhat graph. The edge operators satisfy the Yang–Baxter relations of Brenti, Fomin and Postnikov [1], which make the braid move of Lemma 34 and the 3-cycle relations of Remark 57 exact; the wild chains are identified by a single inequality; the left paths are eliminated altogether, leaving an identity between chains (Proposition 62). When no position below $k-1$ is fresh the block identity is then proved (Theorem 68) by an explicit bijection between two families of wild chains described through right-to-left maxima. For general F the chain identity splits into components indexed by the subsets of the fresh positions below $k-1$, of which the top one is the case just proved; the other components involve chains with descents, whose structure is worked out in Lemmas 71–74, and the same bijection, extended to these chains, proves them all (Theorem 77). This completes the proof of the backtrack identity, hence of Theorem 3 (Corollary 78).

Conventions. Throughout, $k \ge 3$, ξ and $F \ni k-1$ are as in the block model, and $F' = F \cap \{1, \dots, k-2\}$. Rotating all values by $\rho_{\xi(k)}$ (Section 4.2) changes the quantum weight of every directed path from ξ to v in the block by the same factor $q^{\sigma(v)-\sigma(\xi)}$, $\sigma_j(x) = \#\{i \le j : x(i) \le \xi(k)\}$, because an edge (a,b) with $a \le j < b$ changes σ_j by $+1$ exactly when it is quantum before and Bruhat after rotation, by -1 in the opposite case, and otherwise not at all (Corollary 9); the status factors depend only on positions, and tameness is defined through the rotated graph. Hence both sides of (36) at v are multiplied by the same unit, and we may and do assume that $\xi(k) = k$ is the largest value (the *clean frame*). Then by Lemma 10 every directed path from ξ with pairwise distinct labels of upper index k is a chain of Bruhat covers. We write $c^* = \xi(k-1)$ and $t^* = t_{c^*}^{-1}$ as before.

Let V be the free module on the symbols $\mathfrak{G}_x$, $x \in S_k$, over the coefficient ring. For a label (a,b), $a < b \le k$, let E_{ab} be the operator with $E_{ab}\mathfrak{G}_x = \kappa\,\mathfrak{G}_{xt_{ab}}$ if $x \to xt_{ab}$ is an edge of the quantum Bruhat graph of weight κ, and $E_{ab}\mathfrak{G}_x = 0$ otherwise. For a word $w = \ell_1\ell_2\cdots\ell_r$ of labels put $P[w] = E_{\ell_r}\cdots E_{\ell_1}$, so that $P[w]\mathfrak{G}_x$ is the weight of w times $\mathfrak{G}_{\mathrm{end}}$ if w is a directed path from x and 0 otherwise; products of operators are composed from right to left. We abbreviate

$$E = E_{k-1,k}, \qquad K_a = E_{ak}, \qquad L_a = E_{a,k-1} \qquad (1 \le a \le k-2).$$

Lemma 58 (Yang–Baxter relations). *1.* $E_{ab}E_{cd} = E_{cd}E_{ab}$ *whenever* $\{a,b\} \cap \{c,d\} = \varnothing$*.*

2. For $a < b < c$ *and a formal variable* ε*,*

$$(1 + \varepsilon E_{ab})(1 + \varepsilon E_{ac})(1 + \varepsilon E_{bc}) = (1 + \varepsilon E_{bc})(1 + \varepsilon E_{ac})(1 + \varepsilon E_{ab}).$$

Explicitly, in degree two,

$$P[(a,b)(b,c)] = P[(b,c)(a,c)] + P[(a,c)(a,b)], \tag{38}$$
$$P[(b,c)(a,b)] = P[(a,b)(a,c)] + P[(a,c)(b,c)], \tag{39}$$

and in degree three $P[(a,b)(a,c)(b,c)] = P[(b,c)(a,c)(a,b)]$.

3. $E^2_{a,a+1} = q_a$, *and* $E^2_{ab} = 0$ *if* $b - a \geq 2$.

Proof. (1) and (2) are Corollary 4.1 of [1], equations (2.3)–(2.4) there, for the quantum Bruhat operators Q_τ with the multiplicative function $E(\epsilon_a - \epsilon_b) = q_a \cdots q_{b-1}$; the operators of [1] act by left multiplication $w \mapsto \tau w$, and inversion $w \mapsto w^{-1}$, which preserves lengths, identifies them with our edge operators, which act by right multiplication. In (2) the two sides agree in degrees 0 and 1, and in degree two each side is a sum of three terms; the group elements $t_{ab}t_{bc} = t_{bc}t_{ac} = t_{ac}t_{ab}$ and $t_{bc}t_{ab} = t_{ab}t_{ac} = t_{ac}t_{bc}$ are the two 3-cycles, and collecting the terms according to the 3-cycle they represent gives (38) and (39). (3) The adjacent transposition is an edge from every permutation, and its two traversals have weights with product q_a. For $b - a \geq 2$, a Bruhat edge (a, b) requires the values at the positions strictly between a and b to lie outside the interval spanned by the two exchanged values, while the reverse quantum edge would require them to lie inside it. □

Each $P[w]\mathfrak{G}_x$ is either 0 or a monomial in the q_i with coefficient 1 times a single basis vector, and the three words in (38) have the same product of transpositions, hence the same endpoint. Applied to $\mathfrak{G}_x$, the right side of (38) is therefore a sum of two terms with nonnegative coefficients, which cannot cancel against each other, while the left side is 0 or a single monomial. Hence (38) with $(a, b, c) = (a, k-1, k)$ says: for every x, the path $(a, k-1)(k-1, k)$ is directed from x if and only if exactly one of $(k-1, k)(a, k)$ and $(a, k)(a, k-1)$ is, and then the weights agree. This sharpens Lemma 34: the braid move fails exactly when the wild pattern $(a, k)(a, k-1)$ is directed. The same cancellation-free bookkeeping, valid because every nonzero coefficient is a monomial with coefficient 1, is used again in Lemma 63 and Proposition 75.

Lemma 59 (Wild chains)**.** *Let $c = \mu\nu$ be a level-$(k-2)$ block chain with endpoint x. Then c is wild if and only if $x(k-1)$ lies in* $\mathrm{arc}(\xi(k-1), \xi(k)) \cup \{\xi(k)\}$*; in the clean frame, if and only if $x(k-1) > c^*$.*

Proof. Work in the clean frame. Rotate by ρ_{c^*}; by Lemma 10 with $k-2$ in place of k (so $\vartheta = \xi(k-1) = c^*$), every edge of c is Bruhat after this rotation. A Bruhat edge $(c, k-1)$ replaces the value at $k-1$ by the smaller value at c, and no other label of c touches position $k-1$; so the value at position $k-1$, which starts at the rotated maximum c^*, strictly decreases in the rotated order $c^*+1 < \cdots < k < 1 < \cdots < c^*$ at each label of ν. Thus it runs first through values $\leq c^*$ and, once it has become larger than c^*, stays larger. By Corollary 9 an edge $(c, k-1)$ of ν applied at y is quantum in the clean frame exactly when $y(k-1) \leq c^* < y(c)$, that is, exactly when it carries the value at $k-1$ from $\leq c^*$ to $> c^*$. Hence ν has a clean-frame quantum edge if and only if $x(k-1) > c^*$. Finally, if ν reuses a lower index a_i of μ, the value entering position $k-1$ at that label is the value e_{i-1} that position k held before the label (a_i, k) of μ, and $e_{i-1} > c^*$: along μ the value at k decreases from $e_0 = k$ through $e_1 > \cdots > e_m$, position $k-1$ lies strictly between every a_i and k and holds c^*, so the cover condition gives $c^* \notin (e_i, e_{i-1})$ for all i, whence $c^* < e_m$. So in this case too $x(k-1) > c^*$, and conversely $x(k-1) > c^*$ forces a quantum edge of ν. □

The block identity as an operator identity. Put

$$\mathcal{U}' = (1-\beta K_1)(1-\beta K_2)\cdots(1-\beta K_{k-2}), \qquad \mathcal{K} = (1+\beta K_{k-2})\cdots(1+\beta K_1),$$

so that $\mathcal{U}'$ is the sum over left paths at k with lower indices $\le k-2$ of $(-\beta)^{|\lambda|}P[\lambda]$, the operator $\mathcal{U}_k$ of Section 4.13 is $\mathcal{U}'(1-\beta E)$, and $\mathcal{K}$ is the sum over strictly increasing sequences $a_1 < \cdots < a_m \le k-2$ of $\beta^m P[(a_1,k)\cdots(a_m,k)]$, the Left-free chains with indices below $k-1$. Let $\mathcal{U}'_0$ be defined like $\mathcal{U}'$ with each K_a replaced by its Bruhat part. By Lemma 58(3), $K_a^2 = 0$, so

$$\mathcal{U}'\mathcal{K} = \mathcal{K}\mathcal{U}' = 1, \tag{40}$$

the factors cancelling from the middle outwards. Let Θ be the diagonal operator $\mathfrak{G}_v \mapsto t_{v(k)}^{-1}\mathfrak{G}_v$, let Π_w and $\Pi_t = 1-\Pi_w$ be the projections onto the span of the $\mathfrak{G}_x$ with $x(k-1) > c^*$, respectively $x(k-1) \le c^*$, and put

$$\mathcal{P} = \sum_c \widehat{C}_F(c)\,\mathfrak{G}_{\mathrm{end}(c)}, \qquad \mathcal{Q}_\varnothing = \sum_{c:\ \nu=\varnothing} \widehat{C}_F(c)\,\mathfrak{G}_{\mathrm{end}(c)},$$

the sums over the level-$(k-2)$ block chains $c = \mu\nu$. By Lemma 59, $\Pi_w\mathcal{P}$ and $\Pi_t\mathcal{P}$ are the wild and the tame parts of $\mathcal{P}$.

Proposition 60 (Operator form)**.** *With $B = \beta^2 q_{k-1}$,*

$$\sum_v M^F_{\xi,v}\mathfrak{G}_v = \Theta\,\mathcal{U}'\Big[(1-B)(t^*-a)\mathcal{Q}_\varnothing + a(1-B)\beta E\,\Pi_t\mathcal{P} + aB(1-\beta E)\Pi_w\mathcal{P}\Big], \tag{41}$$

$$\sum_v A^F_{\xi,v}\mathfrak{G}_v = \Theta\Big[\mathcal{U}'(t^*-a)\mathcal{Q}_\varnothing + a\,\mathcal{U}'_0\,\beta E\,\Pi_t\mathcal{P}\Big], \tag{42}$$

and (37) *is equivalent to*

$$\Omega\,\mathcal{P} = 0, \qquad \Omega := (1-B)(\mathcal{U}'-\mathcal{U}'_0)\,\beta E\,\Pi_t + B\,\mathcal{U}'(1-\beta E)\,\Pi_w. \tag{43}$$

Moreover, writing $\mathcal{P}^0 = \sum_c \beta^{|c|}q^{D(c)}\mathfrak{G}_{\mathrm{end}(c)}$ for the chain sum without status factors, (43) *holds if and only if $\Omega\,\Pi_S\mathcal{P}^0 = 0$ for every $S \subseteq F'$, where Π_S projects onto the span of the $\mathfrak{G}_x$ in which no value $\xi(p)$, $p \in S$, is Left and no value $\xi(p)$, $p \in F'\setminus S$, is fixed.*

Proof. Level-$(k-1)$ block chains without $(k-1,k)$ are the chains $c = \mu$ with $\nu = \varnothing$, with weight $(t^*-a)\widehat{C}_F(c)$, and those containing $(k-1,k)$ correspond by Lemma 55 to the tame chains c, with weight $\beta\kappa\theta_c\widehat{C}_F(c)$ at the endpoint $\mathrm{end}(c)t_{k-1,k}$, that is, their sum is $\beta E\big(t^*\mathcal{Q}_\varnothing + a(\Pi_t\mathcal{P} - \mathcal{Q}_\varnothing)\big)$. Hence the sum $\widehat{H}^F$ of the weights of the level-$(k-1)$ block chains plus aB times the sum of the weights of the level-$(k-2)$ block chains, each placed at its endpoint, is $\widehat{H}^F = (t^*-a)\mathcal{Q}_\varnothing + \beta E(t^*\mathcal{Q}_\varnothing + a\Pi_t\mathcal{P} - a\mathcal{Q}_\varnothing) + aB\mathcal{P}$, and by definition of the block composites $\sum_v M^F_{\xi,v}\mathfrak{G}_v = \Theta\,\mathcal{U}'(1-\beta E)\widehat{H}^F$. Expanding with $(1-\beta E)\beta E = \beta E - B$ gives (41). For (42), the proof of Theorem 56 lists the distinct composites: those of kind (b) with left path avoiding $k-1$, with total $\Theta\mathcal{U}'(t^*-a)\mathcal{Q}_\varnothing$; those of kind (b) with left path $(k-1,k)\lambda'$ and those of kind (c) with left path λ' avoiding $k-1$, in both cases with λ' sharing no lower index with c, whose weights add up to $a\,(-\beta)^{|\lambda'|}q^D t_{v(k)}^{-1}\,\beta\kappa\,\widehat{C}_F(c)$ for each tame c and each such λ', by the computation in that proof. The word $\mu(k-1,k)\tilde{\nu}\lambda'$ has pairwise distinct labels of upper index k exactly when λ' shares no lower index with c; by Lemma 10 all its edges are then Bruhat, and conversely a repeated label is a quantum edge, since the position it touches holds a value exceeding the value at k (as in the proof of Lemma 47). Therefore the left paths λ' sharing no index with c are exactly the left paths from $\mathrm{end}(c)t_{k-1,k}$ consisting of Bruhat

edges, and the sum is $\Theta\, a\,\mathcal{U}_0'\beta E\Pi_t\mathcal{P}$. Subtracting $(1-B)$ times (42) from (41) and dividing by a gives $\Theta\Omega\mathcal{P}$, which by Theorem 56 is $\sum_v(\beta^2 q_{k-1}X_v-(1-\beta^2 q_{k-1})Y_v)\mathfrak{G}_v$; as Θ is invertible, (37) is equivalent to (43).

For the last assertion write $s_p=t_p^{-1}-a$ for $p\in F'$. The status factor of $\xi(p)$ at a chain endpoint x is s_p, a or s_p+a according as $\xi(p)$ is fixed, Left or Out in x, that is, $s_p(1-[\xi(p)\text{ Left}])+a(1-[\xi(p)\text{ fixed}])$. Expanding the product over $p\in F'$, $\mathcal{P}=\sum_{S\subseteq F'}s^S a^{|F'\setminus S|}\,\Pi_S\mathcal{P}^0$, where $s^S=\prod_{p\in S}s_p$. The operator Ω involves neither a nor the equivariant parameters, and the monomials s^S are linearly independent over $\mathbb{Z}[\beta,q,a]$; hence $\Omega\mathcal{P}=0$ if and only if $\Omega\Pi_S\mathcal{P}^0=0$ for every S. □

The last assertion says that (37) is equivalent to a family of identities that involve neither a nor y: for each $S\subseteq F'$, the signed sum of the wild composites and the composites counted by Y_v whose chain endpoint has no Left value among $\{\xi(p):p\in S\}$ and no fixed value among $\{\xi(p):p\in F'\setminus S\}$ vanishes. For $S=F'$ these are the composites whose chain has no Left value below $k-1$ at its endpoint; for $F'=\varnothing$ there is a single identity, treated next.

Remark 61 (The case $a=0$ in one line)**.** *At $a=0$ only Left-free chains contribute to $\widehat{P}_{k-1}$, each with status factor $\prod_{p\in F}t_p^{-1}$, and the level-$(k-2)$ part of $\widehat{H}^F$ vanishes. The Left-free level-$(k-1)$ chains are the increasing sequences in $\{1,\dots,k-1\}$, with operator $(1+\beta E)\mathcal{K}$, so by* (40)

$$\mathcal{U}_k(1+\beta E)\mathcal{K}=\mathcal{U}'(1-\beta E)(1+\beta E)\mathcal{K}=(1-B)\,\mathcal{U}'\mathcal{K}=1-B,$$

which is the block identity at $a=0$: $M^F|_{a=0}=(1-B)A^F|_{a=0}$, both sides being supported at $v=\xi$. This is the block-model form of Proposition 46(ii).

Eliminating the left paths. The operator $\mathcal{U}'$ can be removed from the block identity altogether, at the price of a closed form for A^F. Let $c=\mu\nu$ be a level-$(k-2)$ block chain with $\nu=(c_1,k-1)\cdots(c_n,k-1)$, $n\geq 1$, and $c_n\in F'$. The value $\xi(c_n)$ sits at position $k-1$ in $\mathrm{end}(c)$, where it is Out, so $\widehat{C}_F(c)$ contains the factor $t_{\xi(c_n)}^{-1}$; let $\widehat{C}_F^{\flat}(c)$ denote $\widehat{C}_F(c)$ with this factor replaced by a, and put

$$\mathcal{P}_t^{\flat}=\sum_{\substack{c=\mu\nu\ \text{tame}\\ \nu\neq\varnothing,\ c_n\in F'}}\widehat{C}_F^{\flat}(c)\,\mathfrak{G}_{\mathrm{end}(c)},\qquad \mathcal{Q}=\mathcal{Q}_{\varnothing}+\mathcal{P}_t^{\flat}.$$

For $F=\{k-1\}$ we have $F'=\varnothing$ and $\mathcal{Q}=\mathcal{Q}_{\varnothing}$.

Proposition 62 (Chain form of the block identity)**.** *Let $k-1\in F$. The block identity* (36) *for (ξ,F) is equivalent to*

$$E\,\Pi_t\mathcal{P}+\beta q_{k-1}\,\mathcal{P}=(1+\beta E)\,\mathcal{K}E\,\mathcal{Q}. \tag{44}$$

Proof. We first compute A^F. The distinct block composites are those whose word has pairwise distinct labels, and the distinct composites with a given word ω are the valid cuts of ω in the sense of Proposition 36. That proposition and its proof concern the composites with one fixed word and apply verbatim in the block model, where the fresh positions are those in F and the non-fresh positions carry no status factor; in the clean frame every edge of a word with distinct labels is Bruhat (Lemma 10), so no power of q occurs. Hence, exactly as in Lemma 50, $A^F_{\xi,v}$ is the sum of $\Sigma(\gamma)$ over the level-$(k-1)$ block chains γ with endpoint v and a fresh suffix, where

$$\Sigma(\gamma)=\beta^{|\gamma|}\,t_{v(k)}^{-1}\,a^S\prod_{p\in F\text{ not in }\gamma}(t_{\xi(p)}^{-1}-a)\prod_{r\leq|\gamma|-S}\psi_r,$$

S is the length of the longest strictly decreasing suffix of the sequence of lower indices of γ, and ψ_r is a if the rth lower index is fresh and exceeds the next one, $t_{\xi(a)}^{-1}$ if it is a fresh index a that is smaller than the next one or is the last index before the suffix, and 1 if it is not fresh.

The words of the composites counted by $\Theta(t^*-a)\mathcal{U}'\mathcal{Q}_\varnothing$ are the words $\mu\lambda$ without the label $(k{-}1,k)$; these composites are all distinct (Lemma 47), and grouping them by their words shows that $\Theta(t^*{-}a)\mathcal{U}'\mathcal{Q}_\varnothing$ is the part of A^F coming from the chains γ without $(k{-}1,k)$. The remaining chains are $\gamma=\mu(k{-}1,k)\tilde\nu$ with $c=\mu\nu$ tame (Lemma 55). The lower indices of γ are $a_1,\dots,a_m,k{-}1,c_1,\dots,c_n$; if $n=0$ the decreasing suffix is $(k-1)$, and if $n\ge1$ it ends with c_n and its other entries are descents of an admissible chain, hence fresh. So γ has a fresh suffix if and only if $n=0$ or $c_n\in F'$. For such γ the factors a^S and ψ_r assign a to the index $k-1$ (it is the suffix if $n=0$, and otherwise it is fresh and exceeds c_1), a to c_n if $n\ge1$, and to every other fresh index p of γ the factor a or $t_{\xi(p)}^{-1}$ according as the next index is smaller or larger than p. The latter is the status factor of $\xi(p)$ in $\mathrm{end}(c)$: the label with lower index p moves $\xi(p)$ to position k or $k-1$, the next label moves it to the position of the next index, and it stays there because no index is repeated in a tame chain; for $p=a_m$ the next index is $k-1>a_m$ and $\xi(a_m)$ ends at position k, Out. The indices of F' outside γ are fixed, and $k-1$ is in γ. Therefore $\Sigma(\gamma)=t_{v(k)}^{-1}\,a\beta\,\widehat{C}_F(\mu)$ if $n=0$ and $\Sigma(\gamma)=t_{v(k)}^{-1}\,a\beta\,\widehat{C}_F^{\flat}(c)$ if $n\ge1$, where $v=\mathrm{end}(\gamma)=\mathrm{end}(c)t_{k-1,k}$, and since the edge $\mathrm{end}(c)\to v$ is Bruhat in the clean frame,

$$\sum_v A^F_{\xi,v}\mathfrak{G}_v=\Theta\big[(t^*-a)\,\mathcal{U}'\mathcal{Q}_\varnothing+a\beta\,E\,\mathcal{Q}\big].$$

Comparing with (41), the block identity reads

$$\mathcal{U}'\big[(1-B)(t^*-a)\mathcal{Q}_\varnothing+a(1-B)\beta E\Pi_t\mathcal{P}+aB(1-\beta E)\Pi_w\mathcal{P}\big]=(1-B)\big[(t^*-a)\mathcal{U}'\mathcal{Q}_\varnothing+a\beta E\mathcal{Q}\big].$$

Cancelling the terms with $\mathcal{Q}_\varnothing$, dividing by $a\beta$ and applying $\mathcal{K}=\mathcal{U}'^{-1}$ (40) gives $(1-B)E\Pi_t\mathcal{P}+\beta q_{k-1}(1-\beta E)\Pi_w\mathcal{P}=(1-B)\mathcal{K}E\mathcal{Q}$, as $B/\beta=\beta q_{k-1}$. Now $1-B=(1-\beta E)(1+\beta E)$ and $1-\beta E$ is invertible, with inverse $(1+\beta E)/(1-B)$, because $E^2=q_{k-1}$. Cancelling $1-\beta E$ leaves $(1+\beta E)E\Pi_t\mathcal{P}+\beta q_{k-1}\Pi_w\mathcal{P}=(1+\beta E)\mathcal{K}E\mathcal{Q}$, and $(1+\beta E)E=E+\beta q_{k-1}$ turns this into (44). □

Identity (44) involves chains only: on the left the tame chains followed by the edge $(k-1,k)$ and all chains with the factor βq_{k-1}, on the right the chains of $\mathcal{Q}$ followed by $(k{-}1,k)$, by a strictly increasing sequence of labels (c,k), $c\le k-2$, and optionally by $(k-1,k)$ again.

No fresh position below $k-1$. Assume now $F=\{k-1\}$, so that $F'=\varnothing$ and no status factor appears below $k-1$. Condition 4 forbids every descent among lower indices $\le k-2$, so the level-$(k-2)$ block chains are the directed words $\mu\nu$ with $\mu=(a_1,k)\cdots(a_m,k)$, $a_1<\cdots<a_m$, and $\nu=(c_1,k-1)\cdots(c_n,k-1)$, $c_1<\cdots<c_n$, where μ and ν may share lower indices; we identify such a chain with the pair (μ,ν) of increasing sequences, call it *shared* if $\mu\cap\nu\neq\varnothing$, and note that shared chains are wild. With $\mathcal{L}=(1+\beta L_{k-2})\cdots(1+\beta L_1)$ we have $\mathcal{Q}_\varnothing=\mathcal{K}\mathfrak{G}_\xi$ and $\mathcal{P}=\mathcal{L}\mathcal{K}\mathfrak{G}_\xi$. The level-$(k-1)$ block chains are the words μ and the words $\mu(k-1,k)\tilde\nu$ with $\mu\cap\nu=\varnothing$, and by Lemma 50 the only ones with a fresh suffix are $\varnothing$ and the words $\mu(k-1,k)$; hence $\sum_v A^F_{\xi,v}\mathfrak{G}_v=\Theta\big[(t^*-a)\mathfrak{G}_\xi+a\beta E\mathcal{K}\mathfrak{G}_\xi\big]$. For a pair (μ,ν) with $|\nu|=n$ and $0\le l\le n$ put

$$Z_l(\mu,\nu)=P[\mu\,\nu_{\le l}\,(k-1,k)\,\tilde\nu_{>l}]\,\mathfrak{G}_\xi,\qquad W_l(\mu,\nu)=P[\mu\,(c_l,k)\,\nu_{\le l}\,\tilde\nu_{>l}]\,\mathfrak{G}_\xi\quad(l\ge1),$$

where $\nu_{\le l}$ consists of the first l labels of ν and $\tilde\nu_{>l}$ of the remaining ones with upper index k. Thus $Z_0(\mu,\nu)=P[\mu(k-1,k)\tilde\nu]\mathfrak{G}_\xi$ is the braided word and $Z_n(\mu,\nu)=E\,P[\mu\nu]\mathfrak{G}_\xi$ is the chain followed by $(k-1,k)$.

Lemma 63 (Telescoping). *For every pair (μ,ν) and $1 \le l \le n$, $Z_l(\mu,\nu) - Z_{l-1}(\mu,\nu) = W_l(\mu,\nu)$. Consequently exactly one of the following holds: $Z_0 = Z_n \neq 0$ and all W_l vanish; or $Z_0 = 0$, $Z_n \neq 0$, and exactly one W_l is nonzero, equal to Z_n; or $Z_0 = Z_n = 0$ and all W_l vanish. In particular $Z_0 \neq 0$ implies $Z_0 = Z_n$.*

Proof. Write $c = c_l$. By (38) with $(a,b,c) = (c, k-1, k)$, $EL_c = K_cE + L_cK_c$, and K_c commutes with $L_{c'}$ for $c' \neq c$ by Lemma 58(1). Hence

$$\begin{aligned} Z_l &= P[\tilde{\nu}_{>l}]\, E\, L_c\, P[\nu_{<l}]P[\mu]\mathfrak{G}_\xi \\ &= P[\tilde{\nu}_{>l}]K_cE\, P[\nu_{<l}]P[\mu]\mathfrak{G}_\xi + P[\tilde{\nu}_{>l}]L_c\, P[\nu_{<l}]\, K_c\, P[\mu]\mathfrak{G}_\xi = Z_{l-1} + W_l. \end{aligned}$$

Each of Z_0, Z_n, W_l is zero or a single monomial in q with coefficient 1 times a basis vector, and all of them have the same endpoint, the words having the same product of transpositions. Hence in $Z_n - Z_0 = \sum_l W_l$ the right side has nonnegative coefficients. If $Z_0 \neq 0$, this forces $Z_n = Z_0$ (a nonzero $Z_n \neq Z_0$, or $Z_n = 0$, would give a negative coefficient on the left), and then all W_l vanish; if $Z_0 = 0$ and $Z_n \neq 0$, it forces exactly one W_l to be nonzero, equal to Z_n; and if $Z_0 = Z_n = 0$ all W_l vanish. □

The proof uses only the Yang–Baxter relations, so the lemma and its proof apply verbatim to arbitrary words $\mu = (a_1,k)\cdots(a_m,k)$ and $\nu = (c_1,k-1)\cdots(c_n,k-1)$ with $c_1,\dots,c_n$ pairwise distinct, with Z_l and W_l defined by the same formulas; we shall use it in this generality for general F.

For a pair that is not shared the lemma recovers Lemma 55: $Z_0 = Z_n$ if the chain is tame and $Z_0 = 0$ if it is wild. For shared pairs it gives the new information that a directed braided word has the same weight as the chain followed by $(k-1,k)$. Let $\mathcal{W}$ be the set of directed wild pairs, and split it as $\mathcal{W} = \mathcal{W}_A \sqcup \mathcal{W}_B$, where $\mathcal{W}_B$ consists of the shared pairs with $Z_0 \neq 0$.

Proposition 64. *Let $F = \{k-1\}$. The block identity* (36) *for (ξ, F) is equivalent to*

$$\sum_{(\mu,\nu)\in\mathcal{W}_B} \beta^{|\mu|+|\nu|} P[\mu\nu]\, \mathfrak{G}_\xi = \beta E \sum_{(\mu',\nu')\in\mathcal{W}_A} \beta^{|\mu'|+|\nu'|} P[\mu'\nu']\, \mathfrak{G}_\xi. \tag{45}$$

Proof. Here $\mathcal{Q} = \mathcal{Q}_\varnothing = \mathcal{K}\mathfrak{G}_\xi$ and $\mathcal{P} = \mathcal{L}\mathcal{K}\mathfrak{G}_\xi$, so by Proposition 62 the block identity is equivalent to

$$E\mathcal{P}_t + \beta q_{k-1}(\mathcal{P}_t + \mathcal{P}_w) = (1+\beta E)\,\mathcal{K}E\mathcal{K}\mathfrak{G}_\xi,$$

where $\mathcal{P}_t = \Pi_t\mathcal{P}$ and $\mathcal{P}_w = \Pi_w\mathcal{P}$ are the tame and wild parts. Let $\mathcal{R} = \sum \beta^{|\mu|+|\nu|} Z_0(\mu,\nu)$ over the shared pairs. Since $\mathcal{K}E\mathcal{K}\mathfrak{G}_\xi = \sum_{\mu,\nu} \beta^{|\mu|+|\nu|} Z_0(\mu,\nu)$ over all pairs, and the non-shared pairs contribute $E\mathcal{P}_t$ by Lemma 55, we have $\mathcal{K}E\mathcal{K}\mathfrak{G}_\xi = E\mathcal{P}_t + \mathcal{R}$. Substituting, the right side becomes $E\mathcal{P}_t + \mathcal{R} + \beta E(E\mathcal{P}_t + \mathcal{R}) = E\mathcal{P}_t + \beta q_{k-1}\mathcal{P}_t + (1+\beta E)\mathcal{R}$, so the block identity is equivalent to $(1+\beta E)\mathcal{R} = \beta q_{k-1}\mathcal{P}_w$. Now $\mathcal{R} = \sum_{\mathcal{W}_B} \beta^{|\mu|+|\nu|} Z_0(\mu,\nu)$, and by Lemma 63 $Z_0 = Z_n = E\,P[\mu\nu]\mathfrak{G}_\xi$ on $\mathcal{W}_B$, so $\beta E\mathcal{R} = \beta q_{k-1} \sum_{\mathcal{W}_B} \beta^{|\mu|+|\nu|} P[\mu\nu]\mathfrak{G}_\xi$ and $\mathcal{R} = E \sum_{\mathcal{W}_B} \beta^{|\mu|+|\nu|} P[\mu\nu]\mathfrak{G}_\xi$. Since $\mathcal{P}_w$ is the sum of $\beta^{|\mu|+|\nu|} P[\mu\nu]\mathfrak{G}_\xi$ over $\mathcal{W}_A \sqcup \mathcal{W}_B$, cancelling the $\mathcal{W}_B$ part and applying the invertible operator $E^{-1} = q_{k-1}^{-1}E$ gives (45). □

Identity (45) says that the wild chains with a directed braided word are, up to the edge $(k-1,k)$ and a factor β, exactly the remaining wild chains. We now describe both families explicitly and match them. The description rests on the following elementary observation.

Right-to-left maxima.

Lemma 65 (Increasing chains are right-to-left maxima). *Let z be an arrangement of distinct values at the positions $1,\dots,b$ in which $z(b)$ is the largest value, and let $a_1 < \cdots < a_m < b$. The word $(a_1,b)(a_2,b)\cdots(a_m,b)$ is a directed path from z in the quantum Bruhat graph if and only if every a_i is a right-to-left maximum of z on the positions $1,\dots,b-1$, that is, $z(a_i) > z(p)$ for all $a_i < p < b$. In that case all its edges are Bruhat, position b carries the values $z(a_1) > z(a_2) > \cdots > z(a_m)$ in turn, and position a_i receives $z(a_{i-1})$, where $z(a_0) := z(b)$.*

Proof. Induction on m, the case $m = 0$ being empty. Suppose the first $m-1$ labels form a directed path, that $a_1,\dots,a_{m-1}$ are right-to-left maxima and that the edges are as described; so just before the label (a_m,b) position b carries $z(a_{m-1})$, position a_m still carries $z(a_m)$, and the positions strictly between a_m and b, none of which is an a_i, carry their original values. If $m \geq 2$, these values are smaller than $z(a_{m-1})$, because $a_{m-1} < a_m$ is a right-to-left maximum, and so is $z(a_m)$; a quantum edge (a_m,b) would require $z(a_m) > z(a_{m-1})$. If $m = 1$, position b carries the largest value. So the edge (a_m,b), if it exists, is Bruhat, and by Lemma 8 it exists if and only if no value at a position strictly between a_m and b lies in $(z(a_m), z(a_{m-1}))$; as these values are below $z(a_{m-1})$, this says that they are below $z(a_m)$, that is, a_m is a right-to-left maximum. The edge then moves $z(a_m)$ to b and $z(a_{m-1})$ to a_m. □

We apply this to the pairs (μ,ν). Let

$$R_\xi = \{p \leq k-2 :\ \xi(p) > \xi(p') \text{ for all } p < p' \leq k-1\}$$

be the set of right-to-left maxima of ξ on $\{1,\dots,k-1\}$ other than $k-1$ itself; their values exceed c^*. Let $H = \{p \leq k-2 : \xi(p) < c^*\}$, let $H_{\max}$ be the set of right-to-left maxima of ξ restricted to H, that is, the $p \in H$ with $\xi(p) > \xi(p')$ for all $p' \in H$ with $p' > p$, and let $p_H = \max H$, with $p_H = 0$ if $H = \varnothing$. For a word μ with lower indices $a_1 < \cdots < a_m$ put $a_m = 0$ if $m = 0$ and

$$R_\mu = H_{\max} \ \cup\ \{a \in \mu :\ a > p_H\} \ \cup\ \{r \in R_\xi :\ r > a_m,\ r > p_H\}.$$

We identify μ and ν with their sets of lower indices, and split $\nu = \nu^H\nu^L$, where ν^H consists of the labels of ν with lower index $\leq p_H$ and ν^L of those with lower index $> p_H$; as ν is increasing, ν^H precedes ν^L.

Lemma 66 (Directed pairs and their braided words). *Let $F = \{k-1\}$ and work in the clean frame.*

1. *The chain $\mu\nu$ is directed if and only if $\mu \subseteq R_\xi$ and $\nu \subseteq R_\mu$.*
2. *Let $\mu\nu$ be directed. Along μ the value at position k runs through $e_0 = k > e_1 > \cdots > e_m$, $e_i = \xi(a_i)$, position a_i receives e_{i-1}, and $e_m > c^*$. Along ν the value at position $k-1$ starts at c^*, then runs through the values $\xi(c)$, $c \in \nu^H$, which are smaller than c^* and decrease, and then through the values of ν^L, which exceed c^* and decrease: the value of $c = a_i \in \nu^L \cap \mu$ is $e_{i-1} > e_m$, and the value of $c \in \nu^L \setminus \mu$ is $\xi(c) \in (c^*, e_m)$. In particular $\mu\nu$ is wild if and only if $\nu^L \neq \varnothing$, and it is shared if and only if $\nu^L \cap \mu \neq \varnothing$.*
3. *$Z_0(\mu,\nu) \neq 0$ if and only if $\mu\nu$ is directed and $\nu^L \subseteq \mu$.*

Proof. (1) The word μ is a directed path from ξ if and only if $\mu \subseteq R_\xi$, by Lemma 65 with $z = \xi$ and $b = k$. Let x_μ be its endpoint and $y = y_\mu$ the restriction of x_μ to the positions $1,\dots,k-2$; thus $y(a_i) = e_{i-1}$ and $y(p) = \xi(p)$ for $p \notin \mu$, and $x_\mu(k-1) = c^*$. The labels of ν involve only the positions $1,\dots,k-1$. Rotate by ρ_{c^*}; by Corollary 9 ν is directed from x_μ if and only if it is

directed from $\rho_{c^*}x_\mu$, where c^* has become the largest value and sits at position $k-1$. By Lemma 65 with $b = k-1$, ν is directed if and only if its lower indices are right-to-left maxima of $\rho_{c^*}y$. In the rotated order every value below c^* exceeds every value above c^*, and $\mu \cap H = \varnothing$ since the values of μ exceed c^*. Hence a position $p \in H$ is a right-to-left maximum of $\rho_{c^*}y$ if and only if $\xi(p) > \xi(p')$ for all $p' \in H$ with $p' > p$, that is, $p \in H_{\max}$; and a position $p \notin H$ is one if and only if $p > p_H$ and $y(p) > y(p')$ for all $p < p' \le k-2$, that is, p is a right-to-left maximum of y in the usual order. The latter are the positions $a_i \in \mu$—the values to the right of a_i are $e_{i'-1} \le e_i$ at the $a_{i'}$ with $i' > i$ and $\xi(p') < \xi(a_i) = e_i$ elsewhere, since $a_i \in R_\xi$—and the positions $p \notin \mu$ with $p > a_m$ and $p \in R_\xi$: for $p > a_m$ the arrangement y agrees with ξ at all positions $\ge p$, so p is a right-to-left maximum of y if and only if $\xi(p) > \xi(p')$ for all $p < p' \le k-2$, which together with $\xi(p) > c^* = \xi(k-1)$ says $p \in R_\xi$; and for $p \notin \mu$ with $p < a_m$, let i be minimal with $a_i > p$: then $y(a_i) = e_{i-1}$ exceeds $\xi(p)$, because $e_0 = k$, and for $i \ge 2$ the position $a_{i-1} < p$ is a right-to-left maximum of ξ. Hence the right-to-left maxima of $\rho_{c^*}y$ are exactly the elements of R_μ.

(2) The statements about μ are Lemma 65; $e_m > c^*$ because position $k-1$ lies strictly between every a_i and k, so the cover conditions give $c^* \notin (e_i, e_{i-1})$ for all i, whence $c^* < e_m$ by induction. Along ν the value at $k-1$ is replaced by $y(c)$ at the label $(c, k-1)$, and by Lemma 65 in the rotated frame these values strictly decrease in the rotated order $c^*+1 < \cdots < k < 1 < \cdots < c^*$; the values of $\nu^H \subseteq H$ are $\xi(c) < c^*$ and the values of ν^L are $y(c) > c^*$, namely e_{i-1} for $c = a_i \in \mu$, and $\xi(c)$ for $c \in R_\xi \setminus \mu$ with $c > a_m$, where $c^* < \xi(c) < \xi(a_m) = e_m$ because $a_m < c$ are both right-to-left maxima of ξ. The final value at $k-1$ exceeds c^* exactly when $\nu^L \ne \varnothing$, so Lemma 59 gives the characterization of wild chains, and $\nu \cap \mu = \nu^L \cap \mu$ since $\mu \cap H = \varnothing$.

(3) Compare the words $\mu\nu$ and $\mu(k-1,k)\tilde{\nu}$. After $\mu(k-1,k)$ the positions $1, \ldots, k-2$ carry y, position $k-1$ carries e_m and position k carries c^*. Suppose both words are directed up to their jth label of ν, respectively $\tilde{\nu}$, exclusive, and write $f_0 = c^*$ and $f_i = y(c_i)$ for the values that position $k-1$ takes along ν. The jth label $(c_j, k-1)$ of ν exchanges the value f_j at c_j, a position not touched by the earlier labels, with the value f_{j-1} at $k-1$; the jth label (c_j, k) of $\tilde{\nu}$ exchanges the same two values between the positions c_j and k. The positions strictly between are the same, with the same values, except that $k-1$, carrying e_m, is intermediate for the second label only. By Lemma 8 the second label is an edge if and only if the first one is and $e_m \notin \operatorname{arc}(f_j, f_{j-1})$. Hence, by induction on j, $Z_0(\mu,\nu) \ne 0$ if and only if $\mu\nu$ is directed and $e_m \notin \operatorname{arc}(f_j, f_{j-1})$ for $j = 1, \ldots, n$. We evaluate these conditions using (2). For a label of ν^H we have $f_j < f_{j-1} \le c^* < e_m$, and the arc is (f_j, f_{j-1}), which does not contain e_m. For the first label of ν^L we have $f_{j-1} \le c^* < f_j$, and the arc $\{f_j+1, \ldots, k\} \cup \{1, \ldots, f_{j-1}-1\}$ avoids e_m if and only if $f_j > e_m$. For a later label of ν^L we have $c^* < f_j < f_{j-1}$ and, inductively, $f_{j-1} > e_m$, so the arc (f_j, f_{j-1}) avoids e_m if and only if $f_j > e_m$. Thus $Z_0(\mu,\nu) \ne 0$ if and only if every value of ν^L exceeds e_m, which by (2) means $\nu^L \subseteq \mu$. □

Proposition 67 (The two families). *Let $F = \{k-1\}$, in the clean frame.*

1. *$\mathcal{W}_B$ is the set of pairs with $\mu \subseteq R_\xi$, $\nu^H \subseteq H_{\max}$ and $\varnothing \ne \nu^L \subseteq \{a \in \mu : a > p_H\}$, and $\mathcal{W}_A$ is the set of pairs with $\mu \subseteq R_\xi$, $\nu \subseteq R_\mu$ and $\nu^L \not\subseteq \mu$.*

2. *For $(\mu,\nu) \in \mathcal{W}_B$ let $j = \max \nu$, which lies in μ, and put*

$$\Phi(\mu,\nu) = \big(\mu_{<j},\ \nu \cup \mu_{>j}\big),$$

where $\mu_{<j}$, $\mu_{>j}$ are the parts of μ below and above j. For $(\mu',\nu') \in \mathcal{W}_A$ let $U = \nu'^L \setminus \mu'$, which is nonempty, and put

$$\Psi(\mu',\nu') = \big(\mu' \cup U,\ \nu'^H \cup (\nu'^L \cap \mu') \cup \{\min U\}\big).$$

Then $\Phi\colon \mathcal{W}_B \to \mathcal{W}_A$ *and* $\Psi\colon \mathcal{W}_A \to \mathcal{W}_B$ *are mutually inverse bijections,* $|\Phi(\mu,\nu)| = |\mu| + |\nu| - 1$*, and*

$$P[\mu\nu]\,\mathfrak{G}_\xi = E\,P[\Phi(\mu,\nu)]\,\mathfrak{G}_\xi \qquad \text{for all } (\mu,\nu) \in \mathcal{W}_B.$$

Proof. (1) By Lemma 66, a pair is in $\mathcal{W}_B$ (directed, shared, $Z_0 \neq 0$) if and only if $\mu \subseteq R_\xi$, $\nu \subseteq R_\mu$, $\nu^L \subseteq \mu$ and $\nu^L \cap \mu \neq \varnothing$; here $\nu^H \subseteq R_\mu \cap H = H_{\max}$, and $\nu^L \subseteq R_\mu \cap \mu = \{a \in \mu : a > p_H\}$. A pair is in $\mathcal{W}_A$ (directed, wild, $Z_0 = 0$) if and only if $\mu \subseteq R_\xi$, $\nu \subseteq R_\mu$, $\nu^L \neq \varnothing$ and $\nu^L \not\subseteq \mu$, and the third condition follows from the fourth.

(2) Let $(\mu,\nu) \in \mathcal{W}_B$ and $(\mu',\nu') = \Phi(\mu,\nu)$. Since $\nu^L \subseteq \mu$ is nonempty, $j = \max\nu = \max\nu^L \in \mu$ and $j > p_H$. We have $\mu' = \mu_{<j} \subseteq R_\xi$, $\nu'^H = \nu^H \subseteq H_{\max}$ and $\nu'^L = \nu^L_{<j} \cup \{j\} \cup \mu_{>j}$. Here $\nu^L_{<j} \subseteq \{a \in \mu' : a > p_H\} \subseteq R_{\mu'}$, and $\{j\} \cup \mu_{>j}$ consists of elements of R_ξ exceeding $\max\mu'$ and p_H, so it lies in $R_{\mu'}$ as well. Thus $\nu' \subseteq R_{\mu'}$, and $j \in \nu'^L \setminus \mu'$; by (1), $(\mu',\nu') \in \mathcal{W}_A$. Clearly $|\mu'| + |\nu'| = |\mu| + |\nu| - 1$.

Let $(\mu',\nu') \in \mathcal{W}_A$ and $(\mu,\nu) = \Psi(\mu',\nu')$. The elements of $U = \nu'^L \setminus \mu' \subseteq R_{\mu'} \setminus \mu'$ lie in R_ξ and exceed $\max\mu'$ and p_H, so $\mu = \mu' \cup U \subseteq R_\xi$, and $\mu' < U$ elementwise. Further $\nu^H = \nu'^H \subseteq H_{\max}$ and $\nu^L = (\nu'^L \cap \mu') \cup \{\min U\}$ is a nonempty subset of μ all of whose elements exceed p_H. By (1), $(\mu,\nu) \in \mathcal{W}_B$.

The two maps are inverse to each other. Starting from $(\mu,\nu) \in \mathcal{W}_B$ with $j = \max\nu$, the pair $(\mu',\nu') = \Phi(\mu,\nu)$ has $\nu'^L \setminus \mu' = \{j\} \cup \mu_{>j}$ with minimum j and $\nu'^L \cap \mu' = \nu^L_{<j}$, so $\Psi(\mu',\nu') = (\mu_{<j} \cup \{j\} \cup \mu_{>j},\ \nu^H \cup \nu^L_{<j} \cup \{j\}) = (\mu,\nu)$. Starting from $(\mu',\nu') \in \mathcal{W}_A$ with $c = \min U$, the pair $(\mu,\nu) = \Psi(\mu',\nu')$ has $\max\nu = c$, since $\nu'^L \cap \mu' \subseteq \mu' < U$, so $\mu_{<c} = \mu'$ and $\mu_{>c} = U \setminus \{c\}$, and $\Phi(\mu,\nu) = (\mu',\ \nu'^H \cup (\nu'^L \cap \mu') \cup U) = (\mu',\nu')$.

For the identity of weights, let $(\mu,\nu) \in \mathcal{W}_B$, $j = \max\nu$ and $(\mu',\nu') = \Phi(\mu,\nu)$. The label j is the l_0th label of $\nu' = \nu_{<j}\,(j, k-1)\,\overline{\mu_{>j}}$, where the bar converts upper indices k to $k-1$ and $l_0 = |\nu_{<j}| + 1$. Using Lemma 58(1) to move the operators K_a, $a > j$, of $\mu_{>j}$ past L_j and past the L_c, $c < j$, of $\nu_{<j}$, and to move K_j past the latter,

$$P[\mu\nu]\mathfrak{G}_\xi = P[\mu_{>j}]\,L_j\,P[\nu_{<j}]\,K_j\,P[\mu_{<j}]\mathfrak{G}_\xi = W_{l_0}(\mu',\nu').$$

As (μ,ν) is directed, $W_{l_0}(\mu',\nu') \neq 0$, so Lemma 63 gives $Z_0(\mu',\nu') = 0$ and $E\,P[\mu'\nu']\mathfrak{G}_\xi = Z_{|\nu'|}(\mu',\nu') = W_{l_0}(\mu',\nu') = P[\mu\nu]\mathfrak{G}_\xi$. □

The proof shows, in passing, that for $(\mu',\nu') \in \mathcal{W}_A$ the unique nonzero defect $W_l(\mu',\nu')$ of Lemma 63 occurs at the first label of ν' whose lower index exceeds p_H and is not in μ'.

Theorem 68 (The block identity when only position $k-1$ is fresh)**.** *The block identity* (36) *holds for* $F = \{k-1\}$ *and every* ξ*.*

Proof. By the conventions of this subsection we may assume the clean frame. By Proposition 67, summing $\beta^{|\mu|+|\nu|}P[\mu\nu]\mathfrak{G}_\xi = \beta E\,\beta^{|\Phi(\mu,\nu)|}P[\Phi(\mu,\nu)]\mathfrak{G}_\xi$ over $(\mu,\nu) \in \mathcal{W}_B$ and using that Φ is a bijection onto $\mathcal{W}_A$ gives (45), and Proposition 64 gives (36). □

General F: the components of the chain form. We return to arbitrary $F \ni k-1$ and record what the chain form (44) looks like component by component. Write $s_p = t^{-1}_{\xi(p)} - a$ for $p \in F'$. The status factor of $\xi(p)$ at a chain endpoint is s_p, a or $s_p + a$ according as $\xi(p)$ is fixed, Left or Out, and in $\widehat{C}^\flat_F$ the factor of $\xi(c_n)$ is a. Expanding the products over F', both sides of (44) become sums of monomials $s^S a^{|F'\setminus S|}$, $S \subseteq F'$, with coefficients in the span of the $q^D\beta^m\mathfrak{G}_v$; the operators E and $\mathcal{K}$ do not involve s or a. Hence (44) is equivalent to the family of identities

$$E\,\Pi_l\mathcal{P}^0_S + \beta q_{k-1}\mathcal{P}^0_S = (1+\beta E)\,\mathcal{K}E\big(\mathcal{Q}^0_{\varnothing,S} + \mathcal{P}^{\flat,0}_{t,S}\big), \qquad S \subseteq F', \tag{46}$$

in which $\mathcal{P}^0_S$ and $\mathcal{Q}^0_{\varnothing,S}$ are the sums of $\beta^{|c|}q^{D(c)}\mathfrak{G}_{\mathrm{end}(c)}$ over the chains c, respectively the chains $c=\mu$, whose endpoint has no Left value $\xi(p)$ with $p\in S$ and no fixed value $\xi(p)$ with $p\in F'\setminus S$, and $\mathcal{P}^{\flat,0}_{t,S}$ is the same sum over the tame chains with $c_n\in F'\setminus S$ for which the same holds when $\xi(c_n)$ is counted as Left. We call such chains *S-admissible*. These identities contain neither a nor the equivariant parameters.

Lemma 69 (Fresh descents leave a value Left)**.** *Let $c=\mu\nu$ be a level-$(k-2)$ block chain with a descent, that is, two consecutive labels $(a_i,k)(a_{i+1},k)$ with $a_i>a_{i+1}$ or $(c_j,k-1)(c_{j+1},k-1)$ with $c_j>c_{j+1}$. Then some fresh value $\xi(p)$, $p\in F'$, is Left in* end(c)*. More precisely, a descent of ν at c_j leaves $\xi(c_j)$ Left, and a descent of μ at a_i leaves $\xi(a_i)$ Left unless ν contains the label $(a_{i+1},k-1)$ preceded by a label with a larger lower index.*

Proof. Work in the clean frame. A descent $c_j>c_{j+1}$ of ν is allowed only if $c_j\in F'\setminus\mu$; the label $(c_j,k-1)$ moves $\xi(c_j)$ to position $k-1$ and the next label moves it to position $c_{j+1}<c_j$, which no later label touches. A descent $a_i>a_{i+1}$ of μ is allowed only if $a_i\in F'$; the value $e_i=\xi(a_i)$ ends at position $a_{i+1}<a_i$ after the label (a_{i+1},k) and is moved again only if ν contains $(a_{i+1},k-1)$, say as its jth label. Suppose that all labels of ν before this one have lower indices smaller than a_{i+1}. Then at the jth label the positions a_{i+1} and a_i still carry e_i and e_{i-1}, and a_i lies strictly between a_{i+1} and $k-1$. The label exchanges e_i with the value f_{j-1} at position $k-1$; a quantum edge would need $e_{i-1}\in(f_{j-1},e_i)$, which is absurd, so the edge is Bruhat with $e_i<f_{j-1}$ and $e_{i-1}\notin(e_i,f_{j-1})$, that is, $f_{j-1}\in(e_i,e_{i-1})$. We show by descending induction that $f_{j'}\in(e_i,e_{i-1})$ for all $j'<j$, which is absurd for $j'=0$ because $f_0=c^*<e_m\le e_i$. Let $1\le j'<j$ with $f_{j'}\in(e_i,e_{i-1})$. The value $f_{j'}$ is the value at position $c_{j'}$ after μ; it is not one of the e_h, so $c_{j'}\notin\mu$ and $f_{j'}=\xi(c_{j'})$. The cover condition of the ith label of μ says that no position in $(a_i,k-1]$ other than $a_1,\dots,a_{i-1}$ carries a value in (e_i,e_{i-1}), so $c_{j'}<a_i$; also $c_{j'}<a_{i+1}$ by assumption. Hence the positions a_{i+1} and a_i, carrying e_i and e_{i-1}, lie strictly between $c_{j'}$ and $k-1$ at the j'th label. A quantum edge would need $e_{i-1}\in(f_{j'-1},f_{j'})$, contradicting $f_{j'}<e_{i-1}$; so the edge is Bruhat, $f_{j'}<f_{j'-1}$, and $e_{i-1}\notin(f_{j'},f_{j'-1})$ forces $f_{j'-1}<e_{i-1}$, that is, $f_{j'-1}\in(e_i,e_{i-1})$. □

Corollary 70. *For every $F\ni k-1$ and every ξ, the component $S=F'$ of (46) holds.*

Proof. For $S=F'$ the S-admissible chains are those without a Left fresh value, and $\mathcal{P}^{\flat,0}_{t,F'}=0$. By Lemma 69 the S-admissible chains have no descent, that is, μ and ν are increasing; conversely an increasing chain never moves a value to a smaller position. Hence $\mathcal{P}^0_{F'}=\mathcal{LKG}_\xi$ and $\mathcal{Q}^0_{\varnothing,F'}=\mathcal{KG}_\xi$, and the component is the identity (44) for $(\xi,\{k-1\})$, which holds by Theorem 68 and Proposition 62. □

Chains with descents. We now drop the assumption $F=\{k-1\}$. For the rest of this subsection $c=\mu\nu$ denotes a level-$(k-2)$ block chain for (ξ,F) in the clean frame, with $\mu=(a_1,k)\cdots(a_m,k)$ and $\nu=(c_1,k-1)\cdots(c_n,k-1)$. Since the labels are distinct, the lower indices $a_1,\dots,a_m$ are pairwise distinct, and so are $c_1,\dots,c_n$; condition 4 allows a descent $a_i>a_{i+1}$ only when $a_i\in F'$, and a descent $c_j>c_{j+1}$ only when $c_j\in F'$ and $c_j\notin\mu$. We write $e_0=k$ and $e_i=\xi(a_i)$, let y be the endpoint of μ, and put $f_0=c^*$ and $f_j=y(c_j)$; as before we also write μ and ν for the sets of lower indices. The vectors $Z_l(\mu,\nu)$ and $W_l(\mu,\nu)$ are defined by the same formulas as before, and Lemma 63 applies to them.

Lemma 71 (Shape of a block chain)**.** *1. Every edge of μ is Bruhat. Along μ the value at position k runs through $e_0>e_1>\cdots>e_m$; we have $y(a_i)=e_{i-1}$, $y(p)=\xi(p)$ for $p\notin\mu$, $y(k-1)=c^*$ and $y(k)=e_m$. For every i, no position p with $a_i<p\le k-1$ other than $a_1,\dots,a_{i-1}$ satisfies $\xi(p)\in(e_i,e_{i-1})$; in particular $c^*<e_m$.*

2. *There is $j_0 \in \{1, \dots, n+1\}$ such that $c^* = f_0 > f_1 > \cdots > f_{j_0-1}$ and $f_{j_0} > f_{j_0+1} > \cdots > f_n > c^*$. The j_0th edge of ν is quantum, all other edges of ν are Bruhat, and c is wild if and only if $j_0 \le n$. If $c_j \in \mu$, say $c_j = a_i$, then $f_j = e_{i-1} > e_m$ and $j \ge j_0$; if $c_j \notin \mu$, then $f_j = \xi(c_j)$. In particular $c_j \notin \mu$ for $j < j_0$.*

3. *The jth label of ν exchanges the value f_j at position c_j with the value f_{j-1} at position $k-1$. If $j \ne j_0$, no position strictly between c_j and $k-1$ carries a value in (f_j, f_{j-1}); if $j = j_0$, every position strictly between c_{j_0} and $k-1$ carries a value in (f_{j_0-1}, f_{j_0}).*

Proof. (1) Lemma 10, applied to μ with $k-1$ in place of k, rotates by $\vartheta = \xi(k) = k$, which is the identity; so every edge of μ is Bruhat. A Bruhat edge (a_i, k) replaces the value at k by the smaller value at a_i, which is $\xi(a_i)$ because no earlier label of μ has lower index a_i, and it leaves the old value e_{i-1} at a_i, where it stays. By Lemma 8 the edge requires that no value at a position strictly between a_i and k lies in (e_i, e_{i-1}); the positions $a_1, \dots, a_{i-1}$ carry $e_0, \dots, e_{i-2}$, which exceed e_{i-1}, and every other position still carries its value in ξ. Position $k-1$ is never a lower index of μ, so $c^* \notin (e_i, e_{i-1})$ for all i, and $c^* < e_0$ gives $c^* < e_m$ by induction on i.

(2) By Lemma 10, applied to c with $k-2$ in place of k and $\vartheta = \xi(k-1) = c^*$, every edge of the rotated chain is Bruhat, so along ν the value at position $k-1$ strictly decreases in the rotated order $c^*+1 < \cdots < k < 1 < \cdots < c^*$; this gives the two decreasing runs, and by the last assertion of that lemma the jth edge is quantum exactly when $f_{j-1} \le c^* < f_j$, that is for $j = j_0$. The chain is wild if and only if $f_n > c^*$ (Lemma 59), that is if and only if $j_0 \le n$. The value at position c_j just before the jth label of ν is $y(c_j)$, because no earlier label of ν has lower index c_j; by (1) it is $e_{i-1} > e_m > c^*$ if $c_j = a_i$, which forces $j \ge j_0$, and $\xi(c_j)$ if $c_j \notin \mu$.

(3) is Lemma 8: for $j \ne j_0$ the edge is Bruhat with $f_j < f_{j-1}$, and $\mathrm{arc}(f_j, f_{j-1}) = (f_j, f_{j-1})$; for $j = j_0$ it is quantum with $f_{j_0} > f_{j_0-1}$, and $\mathrm{arc}(f_{j_0}, f_{j_0-1})$ is the complement of $[f_{j_0-1}, f_{j_0}]$. □

For an increasing chain the labels of ν with $j < j_0$ form ν^H and those with $j \ge j_0$ form ν^L. The next lemma extends to arbitrary chains the fact, contained in Lemma 66(2), that the labels of ν^L with lower index in μ carry values above e_m and the others values below e_m, and that the latter come last.

Lemma 72 (The labels after the quantum edge)**.** *Let $J = \{j : c_j \notin \mu \text{ and } c^* < \xi(c_j) < e_m\}$. Then $J = \{j \ge j_0 : c_j \notin \mu\} = \{j \ge j_0 : f_j < e_m\}$, and J is a final segment of $\{1, \dots, n\}$. Thus, if $J = \varnothing$, then $c_j \in \mu$ for every $j \ge j_0$; and if $J \ne \varnothing$ and $l^* = \min J$, then $c_j \in \mu$ for $j_0 \le j < l^*$ and $c_j \notin \mu$ for $j \ge l^*$. Moreover, writing $c_j = a_{i(j)}$ for the labels of ν with lower index in μ, the indices $i(j)$ strictly increase with j.*

Proof. If $c_j \in \mu$ then $j \ge j_0$ and $f_j = e_{i(j)-1}$ (Lemma 71(2)); as the f_j with $j \ge j_0$ strictly decrease, the $i(j)$ strictly increase. For $j \ge j_0$ the value f_j differs from $e_m = y(k)$, and $f_j < e_m$ forces $c_j \notin \mu$ and then $f_j = \xi(c_j) \in (c^*, e_m)$, so $j \in J$; conversely $j \in J$ gives $f_j = \xi(c_j) > c^*$, hence $j \ge j_0$, and $f_j < e_m$. So $J = \{j \ge j_0 : f_j < e_m\}$, which is a final segment because the f_j, $j \ge j_0$, decrease. It remains to show that $c_j \in \mu$ whenever $j \ge j_0$ and $f_j > e_m$; together with the above this gives $J = \{j \ge j_0 : c_j \notin \mu\}$ and the description of the labels c_j, $j \ge j_0$. Suppose not, and let j be the least index with $j \ge j_0$, $f_j > e_m$ and $c_j \notin \mu$; then $f_j = \xi(c_j)$, and $e_m < \xi(c_j) < k = e_0$ gives a unique $h \in \{1, \dots, m\}$ with $e_h < \xi(c_j) < e_{h-1}$. By Lemma 71(1), applied to the hth label of μ and the position $c_j \notin \mu$, we get $c_j < a_h$; so a_h lies strictly between c_j and $k-1$ at the jth label of ν. We use Lemma 71(3).

If $j = j_0$, every position strictly between c_{j_0} and $k-1$ carries a value smaller than $f_{j_0} = \xi(c_j)$. Position a_h is not among $c_1, \dots, c_{j_0-1}$, which are not in μ, so it still carries $y(a_h) = e_{h-1} > \xi(c_j)$: a contradiction.

If $j > j_0$, then $c_{j-1} \in \mu$ by the minimality of j (as $f_{j-1} > f_j > e_m$), say $c_{j-1} = a_i$, and $f_{j-1} = e_{i-1}$. No position strictly between c_j and $k-1$ carries a value in $(\xi(c_j), e_{i-1})$. From $e_{i-1} > \xi(c_j) > e_h$ we get $i \le h$. If $h = i$, then $c_{j-1} = a_h > c_j$ is a descent of ν at a lower index of μ, which condition 4 forbids. So $h > i$. If a_h is not among $c_1, \dots, c_{j-1}$, it carries e_{h-1}, and $\xi(c_j) < e_{h-1} < e_{i-1}$: a contradiction. If $a_h = c_{j'}$ with $j' \le j-1$, then $j' \ne j-1$ because $a_h \ne a_i$, so $j' < j-1$; and $f_{j'} = e_{h-1} > c^*$ gives $j' \ge j_0$, whence $f_{j'} > f_{j-1}$, that is $e_{h-1} > e_{i-1}$ and $h < i$: a contradiction. □

Lemma 73 (Types). *For $0 \le l \le n$ we have $Z_l(\mu,\nu) \ne 0$ if and only if $J \subseteq \{1,\dots,l\}$. Hence $Z_0(\mu,\nu) \ne 0$ if and only if $J = \varnothing$; and if $J \ne \varnothing$ and $l^* = \min J$, then $Z_l(\mu,\nu) = 0$ for $l < l^*$, $Z_l(\mu,\nu) = E\,P[\mu\nu]\mathfrak{G}_\xi$ for $l \ge l^*$, and $W_{l^*}(\mu,\nu) = E\,P[\mu\nu]\mathfrak{G}_\xi$ is the only nonzero $W_l(\mu,\nu)$.*

Proof. We compare the words $\mu\nu_{\le l}(k-1,k)\tilde{\nu}_{>l}$ and $\mu\nu$ as in the proof of Lemma 66(3). After $\mu\nu_{\le l}(k-1,k)$, which is directed, the positions $1,\dots,k-2$ carry the same values as after $\mu\nu_{\le l}$, while position $k-1$ carries e_m and position k carries f_l. Suppose that both words agree in this way just before their jth labels (c_j,k), respectively $(c_j,k-1)$, for some $j > l$. Both labels exchange the value f_j at position c_j with the value f_{j-1}, which sits at k in the first word and at $k-1$ in the second; the positions strictly between are the same, with the same values, except that $k-1$, carrying e_m, is intermediate for the first label only. By Lemma 8 the first label is an edge if and only if the second one is and $e_m \notin \operatorname{arc}(f_j, f_{j-1})$, and then the two words agree again in the same way. Hence $Z_l(\mu,\nu) \ne 0$ if and only if $e_m \notin \operatorname{arc}(f_j, f_{j-1})$ for all $j > l$.

We evaluate this condition with Lemma 71(2) and Lemma 72. For $j < j_0$ the arc is $(f_j, f_{j-1}) \subseteq \{1,\dots,c^*-1\}$, which does not contain $e_m > c^*$. For $j = j_0$ the arc is $\{f_{j_0}+1,\dots,k\} \cup \{1,\dots,f_{j_0-1}-1\}$, and since $e_m > c^* \ge f_{j_0-1}$ it contains e_m if and only if $f_{j_0} < e_m$, that is $j_0 \in J$. For $j > j_0$ the arc is (f_j, f_{j-1}), which contains e_m if and only if $f_j < e_m < f_{j-1}$, that is $j \in J$ and $j-1 \notin J$. As J is a final segment of $\{j_0,\dots,n\}$, the arc contains e_m exactly for $j = \min J$. Hence $Z_l \ne 0$ if and only if $J = \varnothing$ or $\min J \le l$. The remaining assertions follow from Lemma 63: $Z_l = Z_0 + \sum_{h \le l} W_h$ with at most one nonzero W_h, which must then be W_{l^*}, and $Z_n = E\,P[\mu\nu]\mathfrak{G}_\xi$. □

We call c of *type B* if $J = \varnothing$, that is if its braided word $Z_0(\mu,\nu)$ is directed, and of *type A* otherwise. A tame chain has $j_0 = n+1$, hence is of type B; a chain of type A has $J \ne \varnothing$, hence $j_0 \le n$, and is wild. For increasing chains, type B means $\nu^L \subseteq \mu$, and Lemma 73 reduces to Lemma 66(3).

Lemma 74 (Chains of type B). *Let c be of type B.*

1. *If c is wild, then $c_j \in \mu$ for every $j \ge j_0$. Thus the lower indices that ν shares with μ are $a_{i(j_0)}, \dots, a_{i(n)}$ with $i(j_0) < \dots < i(n)$; in particular $c_n \in \mu$, and no label of ν has a lower index a_i with $i > i(n)$.*

2. *If $a_i > a_{i+1}$ is a descent of μ, then $a_{i+1} \notin \nu$, and $\xi(a_i)$ is Left in the endpoint of every prefix $\mu\nu_{\le l}$, $0 \le l \le n$.*

Proof. (1) is Lemma 72 with $J = \varnothing$, together with Lemma 71(2) for the labels $j < j_0$.

(2) Suppose $a_{i+1} = c_j$. Then $f_j = y(a_{i+1}) = e_i > c^*$, so $j \ge j_0$, the chain is wild, and by (1) every $c_{j'}$ with $j' \ge j_0$ lies in μ, with $i(j')$ increasing. Position a_i lies strictly between $c_j = a_{i+1}$ and $k-1$. If $j = j_0$, every such position carries a value smaller than $f_{j_0} = e_i$ (Lemma 71(3)); but a_i is not among $c_1, \dots, c_{j_0-1}$, which are not in μ, so it carries $e_{i-1} > e_i$: a contradiction. If $j > j_0$, let $c_{j-1} = a_{i'}$; then $i' < i(j) = i+1$, so $i' \le i$, and $f_{j-1} = e_{i'-1}$. If $i' = i$, then $c_{j-1} = a_i > a_{i+1} = c_j$ is a descent of ν at a lower index of μ, which condition 4 forbids; so $i' < i$. No position strictly

between a_{i+1} and $k-1$ carries a value in $(e_i, e_{i'-1})$. If a_i is not among $c_1, \dots, c_{j-1}$, it carries e_{i-1}, and $e_i < e_{i-1} < e_{i'-1}$: a contradiction. If $a_i = c_{j''}$ with $j'' < j$, then $i = i(j'') \le i(j-1) = i' < i$: a contradiction.

Finally, the value $\xi(a_i) = e_i$ is carried to position k by the label (a_i, k) and to position a_{i+1} by the label (a_{i+1}, k), and no later label of μ and no label of ν has lower index a_{i+1}; so in the endpoint of every prefix $\mu\nu_{\le l}$ it sits at $a_{i+1} < a_i$, where it is Left. □

The right side of the component identity. Fix $S \subseteq F'$. Let $\mathcal{C}(S)$ be the set of S-admissible chains, $\mathcal{T}(S)$ the set of tame chains in $\mathcal{C}(S)$, and $\mathcal{W}_A(S)$, $\mathcal{W}_B(S)$ the sets of wild chains in $\mathcal{C}(S)$ of type A, respectively of type B; so $\mathcal{C}(S) = \mathcal{T}(S) \sqcup \mathcal{W}_A(S) \sqcup \mathcal{W}_B(S)$, and $\mathcal{T}(S) \sqcup \mathcal{W}_B(S)$ is the set of S-admissible chains of type B. By an *S-prefix* we mean a chain counted in $\mathcal{Q}^0_{\varnothing,S} + \mathcal{P}^{\flat,0}_{t,S}$: either a chain $\gamma = \mu$ in $\mathcal{C}(S)$, or a tame chain $\gamma = \mu\nu$ with $\nu \neq \varnothing$ and $c_n \in F' \setminus S$ that is S-admissible when $\xi(c_n)$ is counted as Left. Expanding $\mathcal{K}$,

$$\mathcal{K}E\big(\mathcal{Q}^0_{\varnothing,S} + \mathcal{P}^{\flat,0}_{t,S}\big) = \sum_{\gamma,\nu'} \beta^{|\gamma|+|\nu'|}\, P[\gamma\,(k-1,k)\,\tilde{\nu'}]\,\mathfrak{G}_\xi,$$

the sum over the S-prefixes γ and the strictly increasing sequences ν' in $\{1, \dots, k-2\}$, where $\tilde{\nu'}$ is the word of the labels (c', k), $c' \in \nu'$, in increasing order.

Proposition 75 (The right side)**.** *Let $S \subseteq F'$. For a chain $c = \mu\nu$ of type B in $\mathcal{C}(S)$ let $l_0(c)$ be the largest j such that $c_j \in F' \setminus S$ and $c_j \notin \mu$, or $l_0(c) = 0$ if there is no such j. Then*

$$c \longmapsto \big(\mu\nu_{\le l_0(c)},\ \nu_{>l_0(c)}\big)$$

is a bijection from $\mathcal{T}(S) \sqcup \mathcal{W}_B(S)$ onto the set of pairs (γ, ν') of an S-prefix γ and a strictly increasing sequence ν' for which $\gamma(k-1,k)\tilde{\nu'}$ is directed, and for these pairs $P[\gamma(k-1,k)\tilde{\nu'}]\,\mathfrak{G}_\xi = E\,P[c]\,\mathfrak{G}_\xi$. Consequently

$$\mathcal{K}E\big(\mathcal{Q}^0_{\varnothing,S} + \mathcal{P}^{\flat,0}_{t,S}\big) = E \sum_{c \in \mathcal{T}(S) \sqcup \mathcal{W}_B(S)} \beta^{|c|} P[c]\,\mathfrak{G}_\xi.$$

Proof. Step 1: the tail avoids the prefix. Let $\gamma = \mu\nu_{\le l}$ be an S-prefix ($l \ge 0$) and $\nu' = (c'_1 < \cdots < c'_r)$ a strictly increasing sequence such that $\omega = \gamma(k-1,k)\tilde{\nu'}$ is directed. We claim that no c'_h equals a c_j with $j \le l$. Suppose it does, and let i be the least index with $c'_i = c_j$ for some $j \le l$; so $l \ge 1$. Since γ is tame, $c^* = f_0 > f_1 > \cdots > f_l$. After $\gamma(k-1,k)$ position $k-1$ carries e_m, which no label of $\tilde{\nu'}$ moves, and position k carries $g_0 = f_l$; let g_h be the value brought to position k by the hth label of $\tilde{\nu'}$. Position $k-1$ lies strictly between c'_h and k for every h. By Lemma 8, if the hth label is a quantum edge, then $g_h > g_{h-1}$ and every intermediate value, in particular e_m, lies in (g_{h-1}, g_h), so $g_h > e_m$; if it is a Bruhat edge, then $g_h < g_{h-1}$ and $e_m \notin (g_h, g_{h-1})$, so $g_h > e_m$ whenever $g_{h-1} > e_m$. Hence once some g_h exceeds e_m, all later ones do. Now consider the ith label (c_j, k). Position c_j carries f_{j-1}: the jth label of ν put it there, and no later label of γ or earlier label of $\tilde{\nu'}$ has lower index c_j. Since $f_{j-1} \le c^* < e_m$ and e_m is intermediate, the label cannot be a quantum edge, so it is a Bruhat edge with $f_{j-1} < g_{i-1}$ and $e_m \notin (f_{j-1}, g_{i-1})$, that is $g_{i-1} < e_m$. Then no g_h with $h < i$ exceeds e_m, so the first $i-1$ labels of $\tilde{\nu'}$ are Bruhat edges and $g_{i-1} \le g_0 = f_l$. But $j \le l$ gives $f_l < f_{j-1} < g_{i-1}$: a contradiction.

Step 2: the word is a chain of type B in $\mathcal{C}(S)$, cut at l_0. Keep γ, ν', ω as in Step 1 and let $c = \mu\nu$ with $\nu = \nu_{\le l}\nu'$, the labels of ν' taken with upper index $k-1$. The word c has pairwise distinct labels by Step 1, and it satisfies condition 4: inside γ and inside the increasing ν' there is nothing to check; at the junction, if $l = 0$ the upper index drops from k to $k-1$, and if $l \ge 1$

and $c_l > c'_1$, then $c_l \in F'$ by the definition of an S-prefix and $c_l \notin \mu$ because γ is tame (by the definition of tame chains, $\nu_{\le l}$ shares no lower index with μ), so the descent is allowed. By Lemma 63, $Z_n(\mu,\nu) = Z_l(\mu,\nu) + \sum_{h>l} W_h(\mu,\nu)$, where $Z_l(\mu,\nu) = P[\omega]\mathfrak{G}_\xi \neq 0$ and every term is zero or a monomial with coefficient 1 times a basis vector; no cancellation can occur among such terms, so $Z_n(\mu,\nu) = E\,P[c]\mathfrak{G}_\xi \neq 0$, and c is directed, hence a level-$(k-2)$ block chain. By Lemma 73, $J(c) \subseteq \{1,\dots,l\}$; if $J(c) \neq \varnothing$ this gives $l \ge \min J(c) \ge j_0$, so the prefix γ would contain the quantum edge of ν, contradicting its tameness. Hence $J(c) = \varnothing$, c is of type B, and $P[\omega]\mathfrak{G}_\xi = Z_l(\mu,\nu) = Z_n(\mu,\nu) = E\,P[c]\mathfrak{G}_\xi$.

We compare the statuses of the fresh values $\xi(p)$, $p \in F'$, in the endpoints of γ and of c. The labels of ν' are the only difference.

- If $p \notin \mu \cup \nu$, then $\xi(p)$ is fixed in both.
- If $p = c_l$ (with $l \ge 1$), then $p \in F' \setminus S$, and $\xi(p)$ is not fixed in either endpoint.
- If $p = c_j$ with $j < l$, then $p \notin \mu$ by Lemma 71(2), since $j < j_0$; the value $\xi(p)$ is moved to $k-1$ by the jth label and to c_{j+1} by the $(j+1)$th, and no label of ν' has lower index c_{j+1}: the same position in both.
- If $p \in \nu'$ and $p \notin \mu$, then $\xi(p)$ is fixed in the endpoint of γ, since no label of γ has lower index p; and in the endpoint of c it is Out, because ν' is increasing: the label $(p, k-1)$ moves it to $k-1$, and the next label, if any, moves it to a position larger than p.
- If $p = a_i$, the value e_i is carried to position k and then, if $i < m$, to a_{i+1} by μ. If $i = m$ or $a_{i+1} \notin \nu$, it stays there in both endpoints. Otherwise $a_{i+1} \in \nu'$, because a lower index of μ cannot occur in $\nu_{\le l}$ (it would have $j \ge j_0 > l$); then $a_i < a_{i+1}$ by Lemma 74(2), the value e_i is Out at a_{i+1} in the endpoint of γ, and in the endpoint of c it is Out as well, at position $k-1$ or at a position larger than a_{i+1}, again because ν' is increasing.

Since γ is an S-prefix, every $p \in F' \cap \nu'$ with $p \notin \mu$, whose value is fixed in the endpoint of γ, belongs to S. Hence the status of every $\xi(p)$ in the endpoint of c is admissible for S: it is unchanged, or $p \in F' \setminus S$ and the value is not fixed, or $p \in S$ and the value is Out. So $c \in \mathcal{C}(S)$, and $c \in \mathcal{T}(S) \sqcup \mathcal{W}_B(S)$. Finally $l_0(c) = l$: if $l \ge 1$ then $c_l \in F' \setminus S$ and $c_l \notin \mu$, and no $j > l$ qualifies, since $c_j \in F' \cap \nu'$ with $c_j \notin \mu$ lies in S.

Step 3: every chain of type B in $\mathcal{C}(S)$ arises. Let $c = \mu\nu \in \mathcal{T}(S) \sqcup \mathcal{W}_B(S)$, $l = l_0(c)$, $\gamma = \mu\nu_{\le l}$ and $\nu' = \nu_{>l}$. The sequence ν' is increasing: a descent $c_j > c_{j+1}$ with $j > l$ has $c_j \in F'$ and $c_j \notin \mu$ by condition 4, and $\xi(c_j)$ is Left in the endpoint of c by Lemma 69, so $c_j \notin S$ by S-admissibility; thus $c_j \in F' \setminus S$, $c_j \notin \mu$ and $j > l_0(c)$, which is absurd. The word $\gamma(k-1,k)\tilde{\nu}'$ is directed, because $Z_l(\mu,\nu) \neq 0$ by Lemma 73. It remains to see that γ is an S-prefix. If $l \ge 1$, then $c_l \in F' \setminus S$ and $c_l \notin \mu$ by definition, and γ is tame: this is clear if c is tame, and if c is wild then $c_j \in \mu$ for all $j \ge j_0$ (Lemma 74(1)), so $c_l \notin \mu$ forces $l < j_0$, whence the first l edges of ν are Bruhat and their lower indices lie outside μ (Lemma 71(2)). The comparison of statuses in Step 2 applies verbatim (c is of type B, ν' is increasing and avoids $\nu_{\le l}$, and $c_l \notin \mu$) and shows that the endpoint of γ satisfies the conditions for an S-prefix: the only values whose status differs from that in the endpoint of c are $\xi(c_l)$, which is counted as Left and has $c_l \in F' \setminus S$, and the values $\xi(p)$ with $p \in F' \cap \nu'$, $p \notin \mu$, which are fixed in the endpoint of γ; these p lie in S, for otherwise the index of p in ν would exceed $l_0(c)$.

Step 4. The maps of Steps 2 and 3 are inverse to each other: Step 2 recovers $l = l_0(c)$ from c, and Step 3 produces the pair $(\mu\nu_{\le l_0(c)}, \nu_{>l_0(c)})$ whose word is c. The weights were computed in Step 2, and $|\gamma| + |\nu'| = |c|$. □

Proposition 76 (The two families for general F). *Let $S \subseteq F'$. For $c = \mu\nu \in \mathcal{W}_B(S)$ let i_1 be the index with $c_n = a_{i_1}$ (Lemma 74(1)) and put*

$$\Phi(c) = \big((a_1,k)\cdots(a_{i_1-1},k),\ \nu\,(a_{i_1+1},k-1)\cdots(a_m,k-1)\big).$$

For $c' = \mu'\nu' \in \mathcal{W}_A(S)$, with $\mu' = (a'_1,k)\cdots(a'_{m'},k)$, $\nu' = (c'_1,k-1)\cdots(c'_{n'},k-1)$ and $l^* = \min J(c')$, put

$$\Psi(c') = \big(\mu'\,(c'_{l^*},k)(c'_{l^*+1},k)\cdots(c'_{n'},k),\ \nu'_{\le l^*}\big).$$

Then $\Phi\colon \mathcal{W}_B(S) \to \mathcal{W}_A(S)$ and $\Psi\colon \mathcal{W}_A(S) \to \mathcal{W}_B(S)$ are mutually inverse bijections, $|\Phi(c)| = |c| - 1$, and

$$P[c]\,\mathfrak{G}_\xi = E\,P[\Phi(c)]\,\mathfrak{G}_\xi \qquad \text{for all } c \in \mathcal{W}_B(S).$$

For increasing chains these are the maps of Proposition 67.

Proof. Let $c \in \mathcal{W}_B(S)$ and $(\mu',\nu') = \Phi(c)$, so that $\mu' = (a_1,k)\cdots(a_{i_1-1},k)$ and ν' is ν followed by the labels $(a_i,k-1)$, $i > i_1$, in the order of μ. The labels of ν' are distinct, because no a_i with $i > i_1$ occurs in ν (Lemma 74(1)). Condition 4 holds for the word $\mu'\nu'$: inside μ' and inside ν it holds as in c, where only the weaker requirement $c_j \notin \mu'$ is now needed for a descent of ν; a descent $a_i > a_{i+1}$ with $i > i_1$ has $a_i \in F'$, and a_i is neither in μ' nor an earlier lower index of ν'; and at the junction, $c_n = a_{i_1}$ is neither in μ' nor an earlier lower index of ν', and if $a_{i_1} > a_{i_1+1}$ then $a_{i_1} \in F'$, since this is a descent of μ. Now

$$P[c] = P[\nu]\,K_{a_m}\cdots K_{a_{i_1+1}}\,K_{a_{i_1}}\,P[\mu'],$$

and the operators K_{a_i}, $i > i_1$, commute with every L_{c_j} (Lemma 58(1), as $a_i \ne c_j$), so

$$P[c]\,\mathfrak{G}_\xi = K_{a_m}\cdots K_{a_{i_1+1}}\,P[\nu]\,K_{a_{i_1}}\,P[\mu']\,\mathfrak{G}_\xi = W_n(\mu',\nu'),$$

the word $\mu'(a_{i_1},k)\nu'_{\le n}\tilde\nu'_{>n}$ of $W_n(\mu',\nu')$ being μ', then (c_n,k), then ν, then the labels (a_i,k), $i > i_1$. Since c is directed, $W_n(\mu',\nu') \ne 0$, and Lemma 63 gives $Z_0(\mu',\nu') = 0$ and $E\,P[\mu'\nu']\mathfrak{G}_\xi = Z_{|\nu'|}(\mu',\nu') = W_n(\mu',\nu') = P[c]\mathfrak{G}_\xi$. In particular $\Phi(c)$ is directed, hence a level-$(k-2)$ block chain, and it is of type A, hence wild. Its endpoint is $\mathrm{end}(c)\,t_{k-1,k}$, and exchanging the values at the positions $k-1$ and k does not change the status of any value $\xi(p)$ with $p \le k-2$; so $\Phi(c)$ is S-admissible, and $\Phi(c) \in \mathcal{W}_A(S)$. Clearly $|\Phi(c)| = |c| - 1$.

Let $c' = \mu'\nu' \in \mathcal{W}_A(S)$, $l^* = \min J(c')$ and $(\mu'',\nu'') = \Psi(c')$. By Lemma 72, $c'_j \notin \mu'$ for $j \ge l^*$, with $\xi(c'_j) \in (c^*, e'_{m'})$, while $c'_j \in \mu'$ for $j_0 \le j < l^*$ and $c'_j \notin \mu'$ for $j < j_0$, where j_0 and $e'_{m'} = \xi(a'_{m'})$ refer to c' (and $e'_0 = k$). We first show that $c'_{l^*} > a'_{m'}$ if $m' \ge 1$. Suppose $c'_{l^*} < a'_{m'}$, so that $a'_{m'}$ lies strictly between c'_{l^*} and $k-1$ at the l^*th label of ν', and apply Lemma 71(3) to c'. If $l^* = j_0$, every such position carries a value smaller than $f'_{l^*} = \xi(c'_{l^*}) < e'_{m'}$; but $a'_{m'}$ is not among $c'_1,\dots,c'_{j_0-1}$, so it carries $e'_{m'-1} > e'_{m'}$: a contradiction. If $l^* > j_0$, then $c'_{l^*-1} = a'_i \in \mu'$ and $f'_{l^*-1} = e'_{i-1}$, and no such position carries a value in $(\xi(c'_{l^*}), e'_{i-1})$. If $i = m'$, then $c'_{l^*-1} = a'_{m'} > c'_{l^*}$ is a descent of ν' at a lower index of μ', which condition 4 forbids; so $i < m'$. If $a'_{m'}$ is not among $c'_1,\dots,c'_{l^*-1}$, it carries $e'_{m'-1}$, and $\xi(c'_{l^*}) < e'_{m'} < e'_{m'-1} < e'_{i-1}$: a contradiction. If $a'_{m'} = c'_{j'}$ with $j' < l^*$, then $j' \ge j_0$ and, the indices $i(\cdot)$ being increasing, $m' = i(j') \le i(l^*-1) = i < m'$: a contradiction.

Hence $\mu''\nu''$ satisfies condition 4: inside μ' there is nothing new; the junction $a'_{m'} < c'_{l^*}$ is an ascent; a descent $c'_j > c'_{j+1}$ with $j \ge l^*$ has $c'_j \in F'$, by condition 4 for c', as $c'_j \notin \mu'$ is not a repeated index there; and inside $\nu'' = \nu'_{\le l^*}$ a descent $c'_j > c'_{j+1}$, $j < l^*$, has $c'_j \in F'$ and $c'_j \notin \mu'$, hence $c'_j \notin \mu''$ since c'_j is not among $c'_{l^*},\dots,c'_{n'}$. The labels of $\mu''\nu''$ are distinct. Now

$$P[\mu''\nu''] = P[\nu'_{\le l^*}]\,K_{c'_{n'}}\cdots K_{c'_{l^*+1}}\,K_{c'_{l^*}}\,P[\mu'] = K_{c'_{n'}}\cdots K_{c'_{l^*+1}}\,P[\nu'_{\le l^*}]\,K_{c'_{l^*}}\,P[\mu'],$$

by Lemma 58(1), and the right side applied to $\mathfrak{G}_\xi$ is $W_{l^*}(\mu',\nu')$, which equals $E\,P[c']\mathfrak{G}_\xi \neq 0$ by Lemma 73. So $\Psi(c')$ is directed, hence a level-$(k-2)$ block chain, with $P[\Psi(c')]\mathfrak{G}_\xi = E\,P[c']\mathfrak{G}_\xi$; it is wild, because ν'' reuses the lower index c'_{l^*} of μ''; it is S-admissible, its endpoint being $\mathrm{end}(c')\,t_{k-1,k}$; and it is of type B: the last value of μ'' is $e''_{m''} = \xi(c'_{n'})$, and a label c'_j of ν'' with $c'_j \notin \mu''$ has $j < l^*$ and $c'_j \notin \mu'$, hence $j < j_0$ and $\xi(c'_j) = f'_j < c^*$, so $j \notin J(\Psi(c'))$. Thus $\Psi(c') \in \mathcal{W}_B(S)$.

The two maps are inverse to each other. For $c' \in \mathcal{W}_A(S)$, the last label of ν'' is c'_{l^*}, which is the $(m'+1)$th lower index of μ'', so $\Phi(\Psi(c')) = (\mu', \nu'_{\le l^*}\nu'_{>l^*}) = c'$. For $c \in \mathcal{W}_B(S)$ with $(\mu',\nu') = \Phi(c)$, we determine $J(\Phi(c))$, which is computed with $e'_{m'} = e_{i_1-1}$: a label c'_j of ν' with $j \le n$ is c_j, which for $j < j_0$ has $\xi(c_j) = f_j < c^*$, and for $j \ge j_0$ is $a_{i(j)}$ with $i(j) \le i_1$, so that $c_j \notin \mu'$ only for $j = n$, where $\xi(a_{i_1}) = e_{i_1} \in (c^*, e_{i_1-1})$. Hence $\min J(\Phi(c)) = n$, and $\Psi(\Phi(c)) = \big(\mu'(a_{i_1},k)(a_{i_1+1},k)\cdots(a_m,k),\ \nu'_{\le n}\big) = (\mu,\nu) = c$. □

Theorem 77 (The block identity)**.** *The block identity* (36) *holds for every $k \ge 3$, every ξ and every F.*

Proof. If $k-1 \notin F$ this is Proposition 54. Let $k-1 \in F$. By Proposition 62 and the discussion preceding (46), it suffices to prove (46) for every $S \subseteq F'$, in the clean frame. Write $\Sigma(\mathcal{X}) = \sum_{c\in\mathcal{X}} \beta^{|c|} P[c]\mathfrak{G}_\xi$ for a set $\mathcal{X}$ of chains. Then $\mathcal{P}^0_S = \Sigma(\mathcal{C}(S))$ and $\Pi_t\mathcal{P}^0_S = \Sigma(\mathcal{T}(S))$, so the left side of (46) is

$$E\,\Sigma(\mathcal{T}(S)) + \beta q_{k-1}\big(\Sigma(\mathcal{T}(S)) + \Sigma(\mathcal{W}_A(S)) + \Sigma(\mathcal{W}_B(S))\big).$$

By Proposition 75 and $E^2 = q_{k-1}$, the right side is

$$(1+\beta E)E\big(\Sigma(\mathcal{T}(S)) + \Sigma(\mathcal{W}_B(S))\big) = (E + \beta q_{k-1})\big(\Sigma(\mathcal{T}(S)) + \Sigma(\mathcal{W}_B(S))\big).$$

The difference of the two sides is $E\,\Sigma(\mathcal{W}_B(S)) - \beta q_{k-1}\Sigma(\mathcal{W}_A(S))$. By Proposition 76,

$$\Sigma(\mathcal{W}_B(S)) = \sum_{c\in\mathcal{W}_B(S)} \beta^{|c|} E\,P[\Phi(c)]\mathfrak{G}_\xi = \beta E \sum_{c\in\mathcal{W}_B(S)} \beta^{|\Phi(c)|} P[\Phi(c)]\mathfrak{G}_\xi = \beta E\,\Sigma(\mathcal{W}_A(S)),$$

so $E\,\Sigma(\mathcal{W}_B(S)) = \beta E^2\Sigma(\mathcal{W}_A(S)) = \beta q_{k-1}\Sigma(\mathcal{W}_A(S))$, and the difference vanishes. □

Corollary 78. *The backtrack identity* (34) *holds for every $u \in S_\infty$ and every $k \ge 2$. Consequently Theorem 3 holds for all u and k.*

Proof. For $k = 2$ the backtrack identity is Proposition 46(i). For $k \ge 3$ it follows from Theorem 77 by Proposition 53. Theorem 45 then gives Theorem 3. □

Remark 79. *The proof matches the two sides of* (46) *term by term: the terms $E\,P[c]\mathfrak{G}_\xi$, $c \in \mathcal{T}(S)$, and $\beta q_{k-1}P[c]\mathfrak{G}_\xi$, $c \in \mathcal{T}(S)\sqcup\mathcal{W}_B(S)$, occur on both sides, and the terms $\beta q_{k-1}P[c']\mathfrak{G}_\xi$, $c' \in \mathcal{W}_A(S)$, of the left side are matched with the terms $E\,P[c]\mathfrak{G}_\xi$, $c \in \mathcal{W}_B(S)$, of the right side through Φ. As a consistency check (Remark 6), we have verified by computer the identities* (46)*, the statements of Lemmas 72–74 and Propositions 75–76, for every $\xi \in S_k$, every $F \ni k-1$ and every $S \subseteq F'$ with $k \le 7$, and the relations of Lemma 58 for $k \le 6$.*

5 A small example and specializations

Example 80. *Take $u = 21$ and $k = 1$. The endpoints are $21, 312, 132$, and the identity permutation e. Their chain data (s, D) are respectively*

$$(0,0), \qquad (1,0), \qquad (2,(1,0,\ldots)), \qquad (1,(1,0,\ldots)).$$

For instance, the path to 132 *is*

$$213 \xrightarrow{(1,3)} 312 \xrightarrow{(1,2)} 132,$$

whose second edge is quantum. Since $B_1^q = x_1 + z - \beta q_1(1+\beta x_1)$*, the formula reads*

$$\begin{aligned}\big(x_1 + z - \beta q_1(1+\beta x_1)\big)\mathfrak{G}_{21}^q = &\left(z - \frac{y_2}{1+\beta y_2}\right)\mathfrak{G}_{21}^q \\ &+ \frac{\mathfrak{G}_{312}^q + \beta q_1 \mathfrak{G}_{132}^q + q_1}{1+\beta y_2}.\end{aligned} \tag{47}$$

Here $\mathfrak{G}_e^q = 1$*, and all polynomials have alphabets* $(x; y)$*. This particular identity can be verified by expanding the four quantum double Grothendieck polynomials and clearing the denominator.*

Classical specialization. When all $q_i = 0$, only paths without quantum edges contribute. The multiplier becomes $\prod_{i=1}^{k}(x_i + z)$, giving the classical double Grothendieck top-block formula as a specialization of Theorem 3; this specialization is proved in Theorem 38. The equivariant classical formula is itself new, rather than an appeal to a previously published top-block formula.

Non-equivariant specialization. When all $y_a = 0$, the coefficient becomes

$$q^D \beta^{s-m} z^{|\,\mathrm{Fix}\,|}(1-\beta z)^{|\,\mathrm{Left}\,|}.$$

In addition setting $z = 0$ leaves only endpoints with Fix $= \varnothing$, and the surviving coefficient is $q^D\beta^{s-k}$. At $\beta = -1$ it is $(-1)^{s-k}q^D$. This specialization is Theorem 20, derived from the quantum Grothendieck Pieri formula of Naito–Sagaki [9] through the uniqueness of chains and Lemma 19.

Cohomological specialization. At $\beta = 0$ the polynomials in (5) are quantum double Schubert polynomials, not Grothendieck polynomials: $\mathfrak{G}_w^q$ has no negative powers of β, and

$$\mathfrak{G}_w^q(x; y)\big|_{\beta=0} = \mathfrak{S}_w^q(x; -y)$$

is the quantum double Schubert polynomial [3], the image of $\mathfrak{G}_w|_{\beta=0} = \mathfrak{S}_w(x; -y)$ under the cohomological quantization map $e_r(x_1, \ldots, x_j) \mapsto E_r^j(x; q)$ of Fomin–Gelfand–Postnikov [2], applied in the x-variables. For $y = 0$ this is [4, Proposition 3.22], read through the scaling (14). In general, expand $\mathfrak{G}_w(x; y) = \sum_v a_{v,w}(y; \beta)\,\mathfrak{G}_v(x)$ with $a_{v,w} \in \mathbb{Z}[\beta][y]$, which is possible because $\{\mathfrak{G}_v(x) : v \in S_N\}$ is a $\mathbb{Z}[\beta]$-basis of the polynomials with $\deg_{x_i} \le N - i$ (each $\mathfrak{G}_v(x)$ is $\mathfrak{S}_v(x)$ plus terms of higher x-degree). Both quantization maps are linear over y and β, so $\mathfrak{G}_w^q(x; y) = \sum_v a_{v,w}(y; \beta)\,\mathfrak{G}_v^q(x)$ has no negative powers of β, and at $\beta = 0$ it becomes $\sum_v a_{v,w}(y; 0)\,\mathfrak{S}_v^q(x)$, the cohomological quantization of $\sum_v a_{v,w}(y; 0)\,\mathfrak{S}_v(x) = \mathfrak{S}_w(x; -y)$. The block B_k^q is a polynomial in β by Proposition 13, and each coefficient (6) is regular at $\beta = 0$ because $s \ge m$ (Proposition 7). Hence (5) specializes termwise: terms with $s > m$ vanish, and those with $s = m$ have coefficient

$$q^D \prod_{a \in \mathrm{Fix}} (z - y_a).$$

The corresponding comparison is with the top quantum double Schubert block in coefficient alphabet $-y$.

Usual K-theory variables. At $\beta = -1$ the multiplicative equivariant variables in these conventions may be written

$$Y_a = \frac{1}{1 - y_a}.$$

Thus comparisons with formulas in the Y variables use $y_a = 1 - Y_a^{-1}$. This change of variables does not alter the permutation indices or the quantum parameters q_i.

References

[1] Brenti, F., Fomin, S., and Postnikov, A. Mixed Bruhat operators and Yang-Baxter equations for Weyl groups. *Internat. Math. Res. Notices*, 8 (1999), 419–441.

[2] Fomin, S., Gelfand, S., and Postnikov, A. Quantum Schubert polynomials. *J. Amer. Math. Soc. 10*, 3 (1997), 565–596.

[3] Kirillov, A. N., and Maeno, T. Quantum double Schubert polynomials, quantum Schubert polynomials and Vafa–Intriligator formula. *Discrete Math. 217* (2000), 191–223. arXiv:q-alg/9610022.

[4] Lenart, C., and Maeno, T. Quantum Grothendieck polynomials, 2006. arXiv:math/0608232.

[5] Lenart, C., Naito, S., and Sagaki, D. A general Chevalley formula for semi-infinite flag manifolds and quantum K-theory. *Selecta Math. (N.S.) 30* (2024). arXiv:2010.06143.

[6] Lenart, C., and Sottile, F. A Pieri-type formula for the K-theory of a flag manifold. *Trans. Amer. Math. Soc. 359*, 5 (2007), 2317–2342.

[7] Maeno, T., Naito, S., and Sagaki, D. A presentation of the torus-equivariant quantum K-theory ring of flag manifolds of type A, Part I: the defining ideal, 2023. arXiv:2302.09485.

[8] Maeno, T., Naito, S., and Sagaki, D. A presentation of the torus-equivariant quantum K-theory ring of flag manifolds of type A, Part II: quantum double Grothendieck polynomials. *Forum Math. Sigma 12* (2024). arXiv:2305.17685.

[9] Naito, S., and Sagaki, D. Pieri-type multiplication formula for quantum Grothendieck polynomials. *Adv. Math. 459* (2024). arXiv:2211.01578.